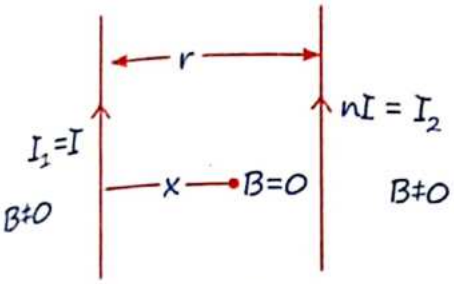


- At a point x from the wire carrying I and r − x from the wire carrying current nI, the magnetic field due to these currents:

$$B_1+B_2=0$$

$$\frac{\mu_0 I}{2\pi x} = \frac{\mu_0 I n}{2\pi(r-x)}$$

$$r-x = nx$$

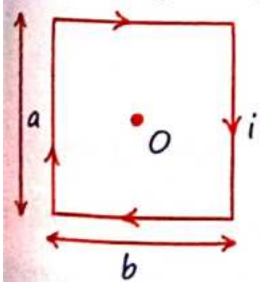
$$\boxed{x = \frac{r}{n+1}}$$

Similarities and difference between Biot Savart law and coloumbs law :-

(i) Electric field is produced by scalar source "charge"	(i) Magnetic field produced by vector source "current element $i\vec{dl}$"
(ii) Electric field along the position vector from source	(ii) Magnetic field perpendicular to the position vector from source
(iii) follow inverse square law	(iii) also follow same
(iv) linear inside source	(iv) linear inside source

MR wala sawal

The magnetic field at the centre O of a rectangular loop with sides a and b carrying a current i is given by:

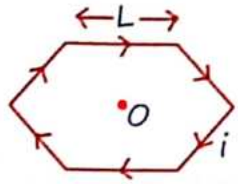


$$B = \frac{2\mu_0 i}{\pi ab}\sqrt{a^2+b^2}$$

a = b = L
"Square"

$$B_0 = \frac{2\sqrt{2}\,\mu_0 i}{\pi L}$$

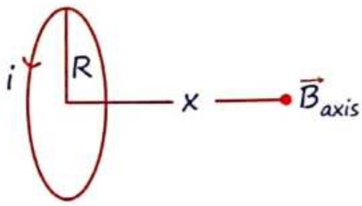
For a regular hexagonal loop with the side length L carrying a current i, the magnetic field at the centre O is:



$$B_0 = \frac{\sqrt{3}\,\mu_0 i}{\pi L}$$

4 Magnetic Field on the axis of a current carrying loop

For a circular loop of radius R carrying a current i the magnetic field at a distance x along the axis



$$\boxed{B_{axis} = \frac{\mu_0 i R^2}{2(R^2+x^2)^{3/2}}}$$

Special cases :-

B at the centre of circular loop:-

$$B_{centre} = \frac{\mu_0 i}{2R}$$

B at the centre of semi-circular loop:-

$$B_0 = \frac{\mu_0 i}{4R}$$

B at the centre of quarter circle:-

$$B_0 = \frac{\mu_0 i}{8R}$$

Generalised Formula for Circular Arc :-

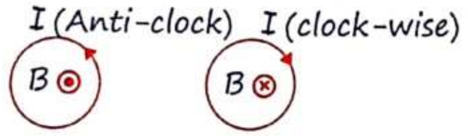
For an arc with a central angle θ

Vimp

$$\boxed{B_Q = \frac{\mu_0 i}{2R}\left[\frac{\theta}{2\pi}\right]}$$

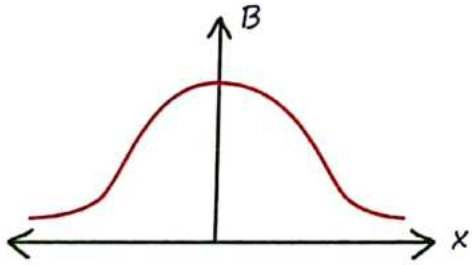
$\theta \rightarrow$ radian.

Direction of magnetic field



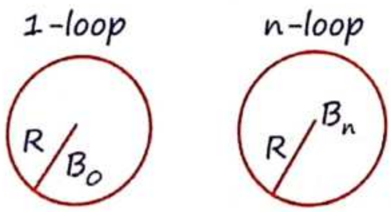
MR* **Anti-clockwise Current:** Produces an upward magnetic field.
Clockwise Current: Produces a downward magnetic field.

+ Graph of Mag. Field along the axis of Current Carrying circular loop :-



5 Magnetic Field due to looping of wire

"R = Constant." :-

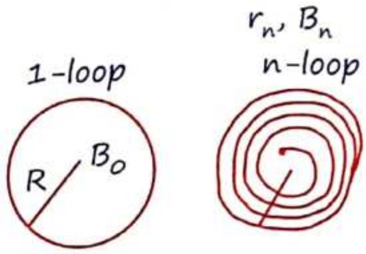


If the wire is wound into n loops, each with radius R, the total magnetic field at the center of the loops is:

$$B_n = n\,B_{1\,loop} = \frac{n\mu_0 I}{2R}$$

Same wire rewound with reduced radius:-

$L = 2\pi R = const^n.$



If the same wire is rewound into n loops, each with a reduced radius r_n:

$$r_n = \frac{R}{n}$$

The magnetic field

$$B_n = n^2 B_{1\text{-}loop} = \frac{n^2\mu_0 I}{2R}$$

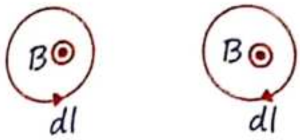
6 Amperes Circuital Law

+ $\vec{B}$ = due to all inside or outside current.
+ Always valid for all type of current.
+ only applicable when current distribution is symmetric.
+ Not a magnetic flux, because here is close line integral.

$$\oint \vec{B}\cdot\vec{dl} = \mu_0 i.$$

i = enclose current.

+ Assume direction in loop and if direction of loop same as magnetic field then current will be positive if opposite then it will be negative.



$\oint \vec{B}.\vec{dl} = +\mu_o I$ because B also Anti-Clock wise and loop

$\oint \vec{B}.\vec{dl} = -\mu_o I$ $\vec{B}$ is Anti-Clock but loop is clock

Steps to apply Ampere's circuital law

1. Draw close symmetric Amperian loop, that must be passes through the point where field have to calculate.

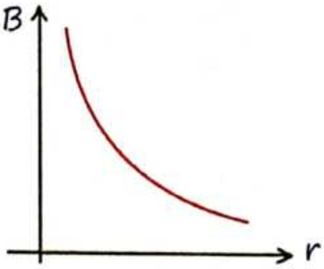
Ex circular, square loop etc

2. Angle between loop and magnetic field must be 0°, 90° or 180°

3. Value of B must be constant at all point of loop.

Magnetic Field Due to a straight infinite current carrying wire :-

$$\boxed{B = \frac{\mu_o i}{2\pi r}}$$

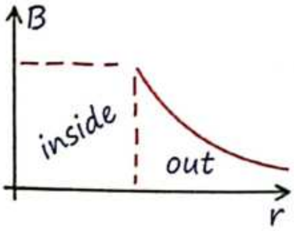


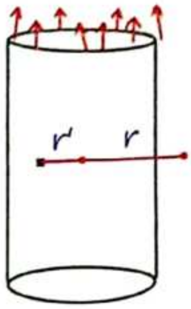
Magnetic Field Due to a infinite Current hollow Cylinder :-

Outside the cylinder, $B_o = \dfrac{\mu_o i}{2\pi r}$

Inside the cylinder, $B_{in} = 0$

$$B_o = \frac{\mu_o i}{2\pi r}, \quad B_{in} = 0$$





Magnetic Field Due to infinite Current Carrying solid Cylinder :-

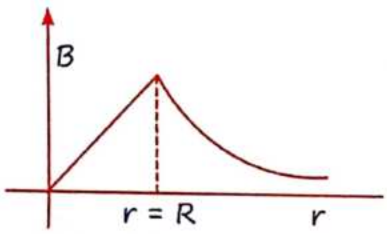
Outside $B_o = \dfrac{\mu_o i}{2\pi r}$

Inside $\boxed{B_{in} = \dfrac{\mu_o i r}{2\pi R^2}}$

$$B_{in} = \frac{\mu_o J r}{2}$$

$$\boxed{\vec{B}_{in} = \frac{\mu_o \vec{J} \times \vec{r}}{2}} \quad * \; J = \frac{I}{\pi R^2} \; *$$

On surface $B_{surface} = \dfrac{\mu_o i}{2\pi R}$



Q.1. Magnetic field Due to infinite current carrying solid cylinder where $J = J_o x$ find magnetic field outside the cylinder

Sol. Given $\hat{J} = j_o x$

Total current i_n enclosed within the cylinder up to radius R

$$i_{in} = \int_0^i di = \int_0^R J_0 x.(2\pi x).dx$$

$$\boxed{i = \frac{J_0 2\pi R^3}{3}}$$

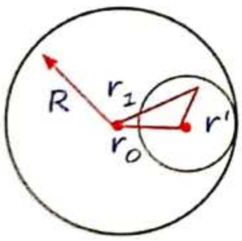
From ampere's circuital law

$$B(2\pi r) = \mu_0 i_{in}$$

$$B = \frac{\mu_0 J_0 R^3}{3r}$$

Magnetic Field Inside cavity of solid cylinder :-

A cylindrical cavity is located inside the cylinder. The center of this cavity is at a distance r_o from the center of the solid cylinder.



$$\vec{r}_o + \vec{r}' = \vec{r}_1$$

$$\vec{r}' = \vec{r}_1 - \vec{r}_o \;*$$

Where

$\vec{r}_o$ = vector from the center of the solid cylinder to the center of the cavity.

$\vec{r}_1$ = vector from the center of the solid cylinder to the point of interest with the cavity.

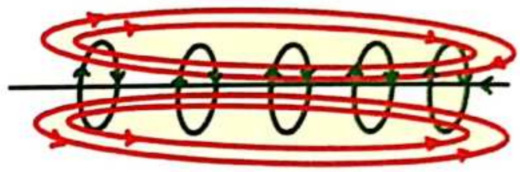
$\vec{r}'$ = Distance from the center of the cavity to the point of interst.

$$*B_{net} = \underbrace{\frac{\mu_0 J r_1}{2}}_{\text{Complete}} - \underbrace{\frac{\mu_0 J r'}{2}}_{\text{Cavity}} \;*$$

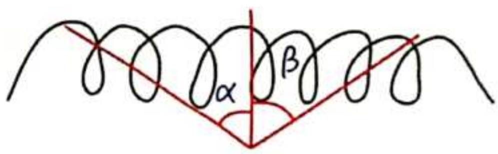
$$B = \frac{\mu_0 J r_0}{2}$$

+ Magnetic field wil be uniform inside cavity.

7 Solenoid



Finite Solenoid :-

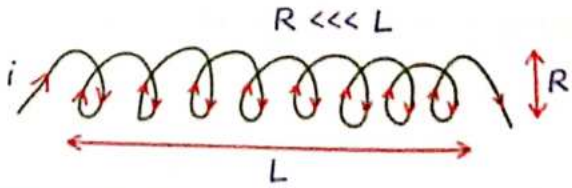


Magnetic field at a point inside a solenoid:

$$B = \frac{\mu_0 ni}{2}[\sin\alpha + \sin\beta]$$

At centre: $B_C = \mu_0 ni$, At ends: $B_{end} = \frac{\mu_0 ni}{2}$

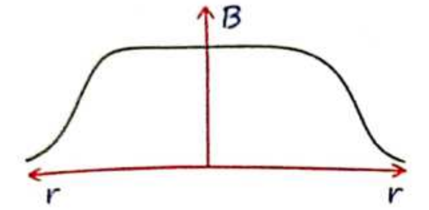
Infinite Solenoid :-



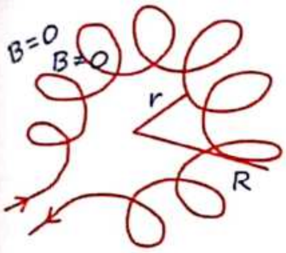
$$\boxed{B = \mu_0 ni = \frac{\mu_0 Ni}{L}}$$ N = Total turn
L = Total length

$n \Rightarrow$ Turns per unit length.

$$n = \frac{N}{l} = \frac{\text{Total turns}}{\text{length of solenoid}}$$



Toroid:-



$$\boxed{B = \frac{\mu_0 N i}{2\pi R_{avg}}}$$

$$R_{avg} = \frac{R + r}{2}$$

8 Magnetic Force

The magnetic force on a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{\beta}$ is

$$\vec{F} = qBv\sin\theta = q(\vec{V} \times \vec{B})$$

Magnetic field rest charge per force nahi lagata

*θ = Angle between $\vec{V}$ & $\vec{B}$.

*$\vec{F} \perp^{er} \vec{V}$ and $\vec{F} \perp^{er} \vec{B}$.

*$\vec{a} \perp^{er} \vec{v}$

*only direction will change, speed will remains constant.

*K.E = constant

*Work done = 0

*Power = $\vec{F} \cdot \vec{v} = 0$

$\therefore$ M.F. की औकात नही है "q" की speed change करे! # Garda visualize.

Lekin accl$^n \neq 0$

Kyuki dir^n change hokr $\vec{V}$ & $\vec{P}$. change hoga!

M.F. is like Centripetal Force ($F \perp V$)

9 Lorentz Force

$$\vec{F}_m = q[\vec{E} + \vec{V} \times \vec{B}]$$

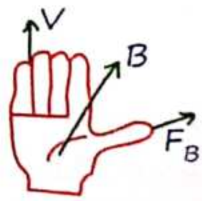
$$\vec{F}_m = \vec{F}_{elec.} + \vec{F}_{Mag.}$$

MR* Law for direction

Place your four finger of right hand along velocity and then slap magnetic field by palm then thumb will represent direction of force.

Ex.: $\xrightarrow{\quad F_B \quad}$

V ⊗ B=? (Upward)



10 Motion of Charge Particle in Magnetic Field

Charge is projected ||el to Mag. Field :-

$\vec{F}_m = 0 \qquad a = 0 \qquad V = \text{Const}^n$

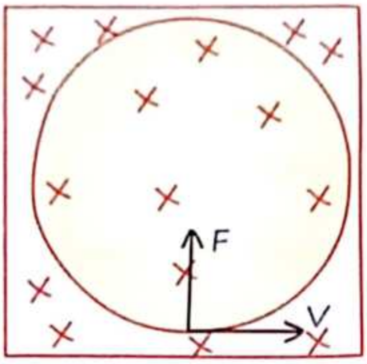
$$\boxed{\text{dist}^n = vt}$$

Charge is projected $\perp$er to Mag. Field :-

$\vec{V}$ = Variable

$F_m = qVB$

$$\boxed{a = \frac{qVB}{m}}$$



Radius of Circular path :-

$$\boxed{R = \frac{mv}{qB} = \frac{P}{qB} = \frac{\sqrt{2m(K.E)}}{qB} = \frac{\sqrt{2mqV}}{qB}}$$

Time Period & Frequency:-

Time taken to complete one rotation.

T does not depends upon speed

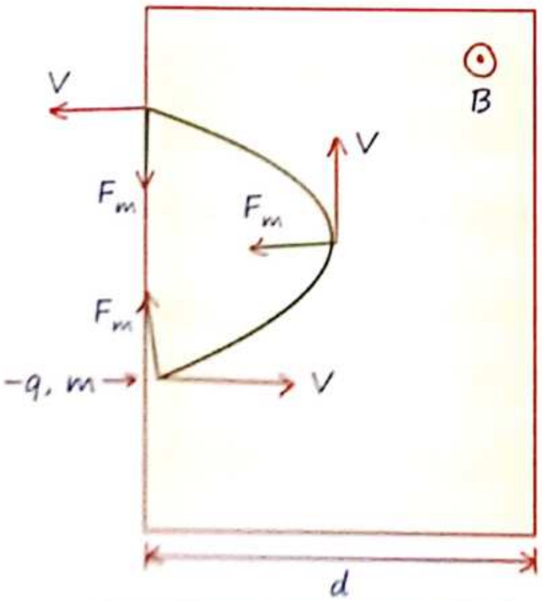
$$T = \frac{2\pi m}{qB} \qquad f = \frac{qB}{2\pi m}$$

time taken to Rotate through an θ angle

$$\boxed{T_\theta = \frac{m\theta}{qB}} \qquad f = \frac{qB}{m\theta} \ (\theta \text{ in radian})$$

Charge particle is projected from outside region of M.F. ⊥er to Field :-

$R \leq d$



$\therefore$ $\boxed{\text{Deviation} = 180^\circ = \pi}$

MR**

- Average Force

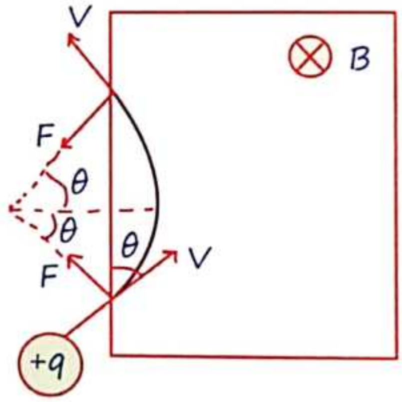
$$\boxed{F_{avg} = qVB \ \frac{\sin\frac{\theta}{2}}{\frac{\theta}{2}}}$$

for $\theta = 180^\circ \rightarrow F_{avg} = \frac{2qVB}{\pi}$

general formula for Time period $t_\theta = \frac{m\theta}{qB}$

for $\theta = 180^\circ \rightarrow T = \frac{m\pi}{qB}$

Charge particle is projected ⊥er to Mag. Field at some angle with boundary of Magnetic Field :- (M.F. :- inwards)

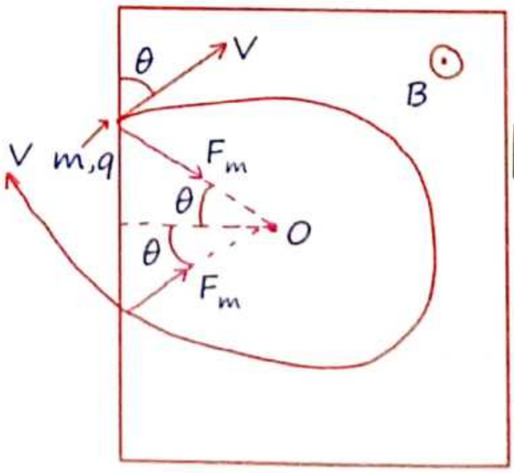


$$\boxed{\delta = 2\theta}$$

$$T_\theta = \frac{m(2\theta)}{qB}$$

$$F_{avg} = \frac{qVB\sin\theta}{\theta}$$

Charge particle is projected ⊥er to Mag. Field at some angle with boundary of outward magnetic field :-



$$\boxed{\delta = 2\pi - 2\theta}$$

$$T_Q = \frac{m\delta}{qB}$$

- θ = Angle between boundary & velocity

11 Important Questions

Q.2. Time required when "q" loose contact on smooth inclined plane

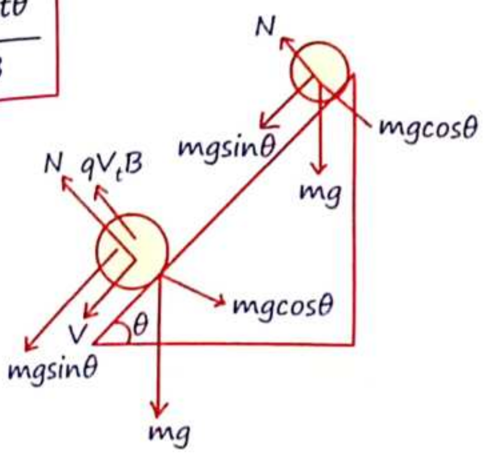
Magnetic field is outside the plane : B⊙

Sol. $N + qV_tB = mg\cos\theta$

$\because N = 0$ for losing contact

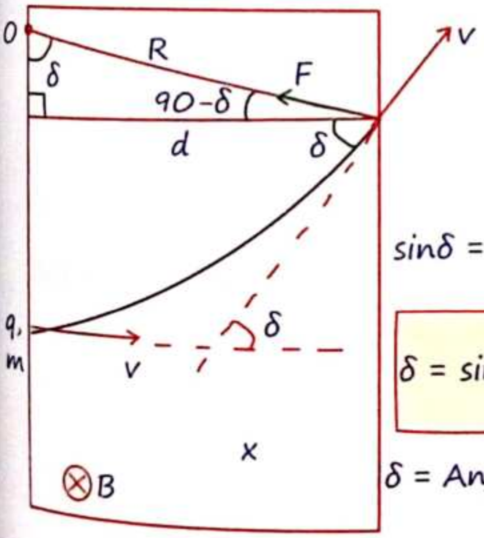
& $V_t = g\sin\theta . t$

$$\boxed{t = \frac{m\cot\theta}{qB}}$$



Q.3. Charge particle is projected ⊥er to Mag. Field where (d < R) then find δ.

Sol.



$\sin\delta = \frac{d}{R}$

$$\boxed{\delta = \sin^{-1}\left[\frac{d}{R}\right]}$$

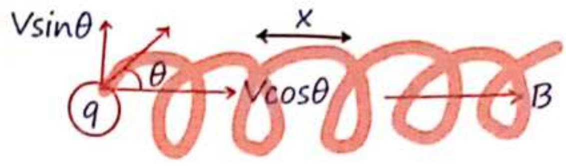
δ = Ang. deviat^n

Important Cases :-

- Object is projected with speed "v" at an angle "θ" from Mag. Field :-

  Path of the particle will be helical.

  $\theta \neq 0°, 90°$ & $180°$



- The perpendicular component $v\sin\theta$ causes the particle to move in a circular path with radius R and time period T:

$$R = \frac{mv\sin\theta}{qB} \qquad T = \frac{2\pi m}{qB}$$

- The parallel component $v\cos\theta$ causes the particle to move along the magnetic field, resulting in helical motion with pitch:

$$\boxed{\begin{aligned}\text{Pitch } (x) &= u_x T \\ &= v\cos\theta \cdot \frac{2\pi m}{qB}\end{aligned}}$$

12 Velocity Selector

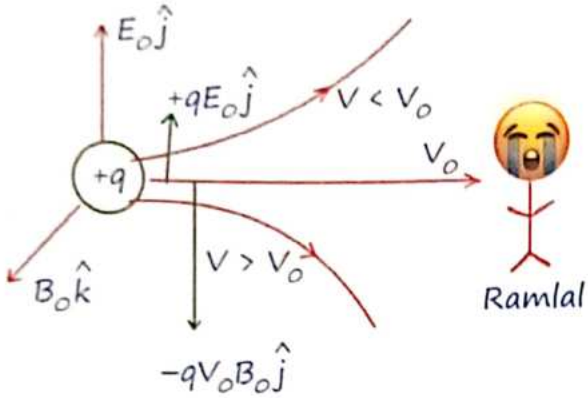
- A velocity selector is a device used to filter particles so that only those with a specific velocity pass through undeflected. This is achieved using
- Magnetic field, Electric field and velocity all are perpendicular to each other.

Condition for No Deviation:

- For a particle to pass through the velocity selector without deviation, the net force must be zero, i.e.,

$$\vec{F}_E + \vec{F}_B = 0$$

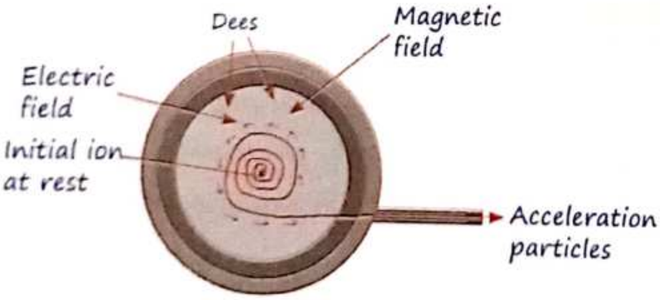
$$qE_0 = qV_0B_0 \qquad \boxed{V_0 = \frac{E_0}{B_0}}$$



- Particle which have velocity $V > V_0$ will experience large magnetic force and deviates downward
- Particle which have velocity $V < V_0$ will experience small magnetic force and deviates upward.

13 Cyclotron

- Device use to acceleration charge particle like proton deuteron, α -particle but not for electron, we use betatron for electron.



- Electric field used to acceleration and shift provide k.E. to the charge particle.
- Magnetic field used to keep the charge particle inside magnetic field.
- Freq. of oscillator = Freq. of charge particle

$$\boxed{KE_{1rot^n} = 2q\Delta V} \qquad \boxed{f_{req} = \frac{qB}{2\pi m}}$$

$$KE = \frac{1}{2}\frac{q^2B^2R^2}{m} \qquad \because \frac{mv^2}{r} = qvB$$

14 Magnetic Force on Current Carrying Wire

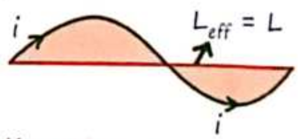
- Magnetic force always perpendicular to the plane of $I\vec{dl}$ and $\vec{B}$.
- Force on close loop of any shape is zero ($l_{eff} = 0$)

$$\boxed{\begin{array}{l} F = Bil_{eff}\sin\theta. \\ \vec{F} = i(\vec{l} \times \vec{B}) \end{array}}$$

$\vec{F}$ → Thumb, $\vec{l}$ → Fore Finger, $\vec{B}$ → Curl Finger — Right Hand

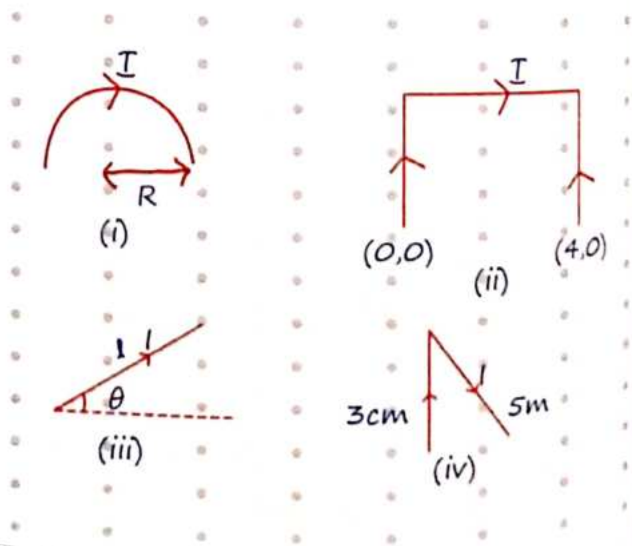
θ = Angle between i & B.

L_{eff} :-

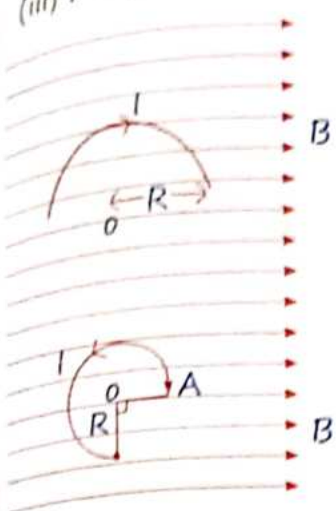


MR*for direction:- Place your four finger of right hand along $\vec{Il}$ and slap magnetic field thumb will represent force.

Q.4. Magnetic force on different wire.



Sol. (i) $F = 2BIR$ (ii) $F = 4IB$
(iii) $F = BIl$ (iv) $F = 4BI$

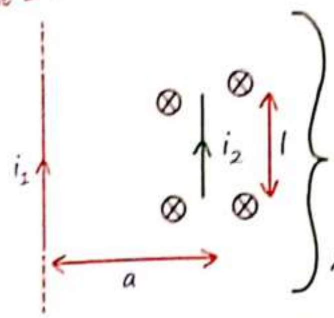


$F = 0$

$F = BIR$
effective length perpendicular to B is R.

Force on small current carrying wire due to infinite large current carrying wire :-

Case 1 :-



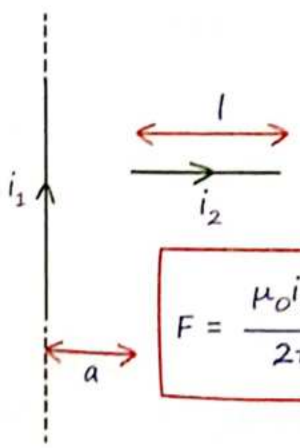
Both wire i dir^n
Same → Attract
Oppos. → Repel

Magnetic field due to infinite wire:

$$B_1 = \frac{\mu_0 i_1}{2\pi a}$$

Force on small wire $\boxed{F = B_1 i_2 l = \frac{\mu_0 i_1}{2\pi a} i_2 l}$

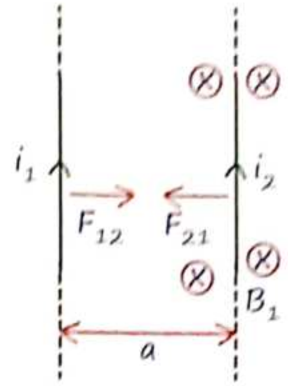
Case 2 :-



$$\boxed{F = \frac{\mu_0 i_1 i_2}{2\pi}\left[\ln\left(\frac{a+L}{a}\right)\right]}$$

#ye sab Mutual Force hai matlab jitna badi-choti wire pr Force lagayega utnahi Force choti-badi pr lagayegi.

Long prallel wires :-



$F = Bil$

$\frac{F}{l} = Bi$

$$\boxed{\frac{F}{l} = \frac{\mu_0 i_1 i_2}{2\pi a}}$$

"Force per unit length"

Biot-savart Law in Terms of Velocity of Particle :-

$$\boxed{B = \frac{\mu_0}{4\pi}\frac{qV\sin\theta}{r^2}}$$

θ = Angle between l and r.

Note :- For electron, $q = e^-$
Ratio of magnetic force to electric force

$$\frac{F_{MF}}{F_{EF}} = \frac{\mu_0 e^2 V^2}{4\pi d^2} \times \frac{4\pi\varepsilon_0 d^2}{e^2}$$

$$\boxed{\frac{F_{MF}}{F_{EF}} = \frac{V^2}{C^2}} \qquad C^2 = \frac{1}{\mu_0\varepsilon_0}$$

$F_{EF} >>>>>> F_{MF}$

15 Circular Current Loop as a Magnetic Dipole

A current-carrying circular loop generates a magnetic field similar to that of a bar magnet, acting as a magnetic dipole.

Magnetic Field on the Axis of the Loop:

$$B = \frac{\mu_0 i R^2}{2(R^2+X^2)^{3/2}}, \text{ where, } K = \frac{\mu_0}{4\pi}$$

M = magnetic moment

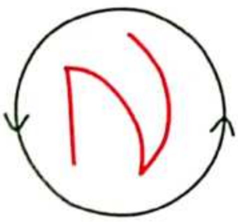
At $x >> R$

$$B = \frac{2\mu_0}{4\pi}\frac{R^2\pi i}{X^3} = \frac{2KM}{X^3}$$

$$\boxed{M = i(\pi R^2) = iA}$$

→ Vector → Amp m^2

→ Dir^n along Area vector

Clockwise
South-pole
M:- ⊗

Anticlockwise
North-pole
M:- ⊙

Direct^n of $\vec{M}$ is along $\vec{B}$.

$\boxed{M = NAI}$ → N = no. of turns.

Magnetic Moment of revolving electron :-

Frequency $f = \frac{V}{2\pi R}$,

Current $i = \frac{e}{T} = ef$

$$\boxed{M = IA = ef\pi R^2 = \frac{eVR}{2}}$$

Gyromagnetic Ratio:-

- Ratio of Magnetic Moment and Angular Momentum:-

$$\boxed{\frac{M}{L} = \frac{e}{2m_e} = \frac{8.8\times10^{10} C}{Kg}}$$

- Bohr Magneton:-

$$M_{e^-} = \frac{e}{2m_e}\left[\frac{h}{2\pi}\right]$$

$$\therefore M = \frac{e}{2m_{e^-}}[I\omega]$$

I = moment of inertia

Torque on a Current Carrying Loops, Magnetic Dipole:-

$$\boxed{\begin{aligned}\vec{\tau} &= \vec{M}\times\vec{B} = MB\sin\theta \\ \vec{\tau} &= BiNA\sin\theta\end{aligned}}$$

θ = Angle between M & B.

Torque perpendicular to magnetic moment and magnetic field

Net force = zero

- If angle given from plane of loop

$\tau = MB\sin(90-\theta) = MB\cos\theta$

Magnetic Potential Energy Stored in Magnetic Dipole:-

$$\boxed{\begin{aligned}U &= -MB\cos\theta \\ U &= -BiNA\cos\theta\end{aligned}}$$

Mag. Field Ka ekhi Udesh hai woh Mag. Dipole Ko Apne Along align Krna Chahta hai.

• Time Period :- $T = 2\pi\sqrt{\frac{I}{MB}}$

I = Moment of Inertia

M = Magnetic moment

• Work Done to Rotate dipole $W = \Delta U$.

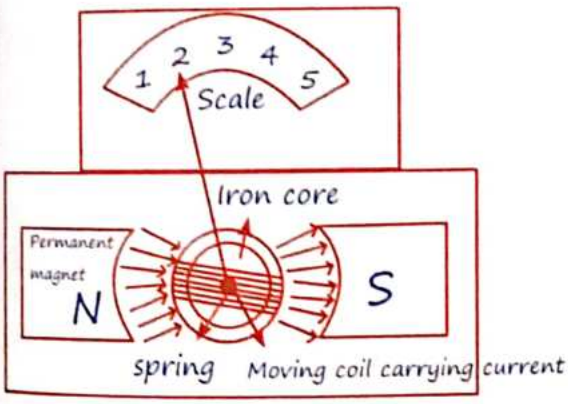
• Work done by M.F. to rotate dipole $= W = -\Delta U$

1.6 Moving Coil Galvanometer

Working Principle: Torque on current carrying loop

• If No. of turns in moving coil-galvanometer is increase then current sensitivity will increases but voltage sensitivity remains same because resistance will also increase.

$V_s \propto \frac{N}{R}$



Torque on the coil:

$$\tau = MB = NIAB = C\theta$$

Torsional Const^n

$\boxed{\tau_{spring} = C\theta}$ $\quad \theta = \frac{BiNA}{C}$

Area of Loop

$$\boxed{I_S = \frac{\theta}{I} = \frac{BAN}{C}}$$ Current Sensitivity.

Divided by R both sides

$$\boxed{V_S = \frac{I_S}{R} = \frac{BAN}{CR} = \frac{\theta}{V}}$$ Voltage Sensitivity.

Visualisation 1.

If an electron is not deflected in passing through a certain region of space, can we be sure that there is no magnetic field in that region?

No, electron would not be deflected if $\vec{v}$ and $\vec{B}$ are in the same direction.

Visualisation 2.

If a moving electron is deflected sideways on passing through a certain region of space, can we be sure that a magnetic field exists in that region?

No, the sideways deflection may be due to Electric field as well. In the absence of electric field, the sideways deflection shows the presence of magnetic field in the region.

Visualisation 3.

If a charged particle at rest experiences no electromagnetic force, then the electric field must be zero or

The magnetic field may or may not be zero

Visualisation 4.

If a charged particle kept at rest experiences an electromagnetic force, then The electric field must not be zero

or

The magnetic field may or may not be zero.

Visualisation 5.

If a charged particle projected in a gravity-free room deflects, then Both fields cannot be zero

or

Both fields can be non-zero

Visualisation 6.

A charged particle moves in a gravity-free space without change in velocity. Possible cases are

$E = 0\ B = 0$ or $E = 0\ B \neq 0$ or $E \neq 0\ B \neq 0$

Visualisation 7.

A charged particle moves along a circle under the action of possible constant electric and magnetic fields. Possible case is

$E = 0, B \neq 0$

Visualisation 8.

A charged particle goes undeflected in a region containing electric and magnetic field. It is possible that

$\vec{v} \parallel \vec{E}, \vec{E} \parallel \vec{B}$ or $\vec{E}$ is not parallel to $\vec{B}$

Visualisation 9.

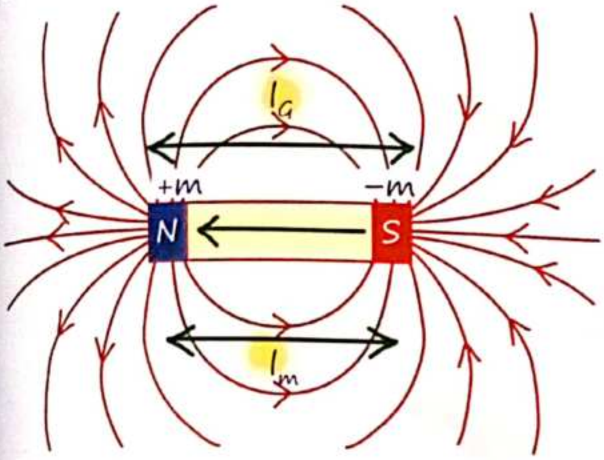
If a charged particle goes unaccelerated in a region containing electric and magnetic fields, then

$\vec{E}$ must be perperdicular to $\vec{B}$ and $\vec{v}$ must be perpendicular to $\vec{E}$.

Magnetism & Matter

1 Bar Magnet

A bar magnet creates a magnetic field, which can be visualized using magnetic field lines.



$$\boxed{\frac{l_{Mag}}{l_{Geo}} = 0.84}$$

- Magnetic field lines are also called magnetic force line → false because force acts perpendicular to magnetic field.
- Magnetic field lines always from N to S → false, Inside Magnet it is S to N

Properties of Magnetic field lines :-

- They form closed loop
- They never intersect each other
- Magnetic field lines are crowded near the Pole where magnetic field is strong. and spread apart from each other where field is weak.
- They flow from the South Pole to the north Pole within a magnet and north pole to South Pole outside
- They Comes out and go in at any angle from magnet.

2 Magnetic Dipole Moment

The magnetic dipole moment represents the strength and orientation of a magnetic source, such as a bar magnet or a current loop.

$\vec{M} = m\vec{l} = NiA$ m:- Pole Strength

$m\vec{l}$ → For bar magnet

NiA → For current loop

Direction:

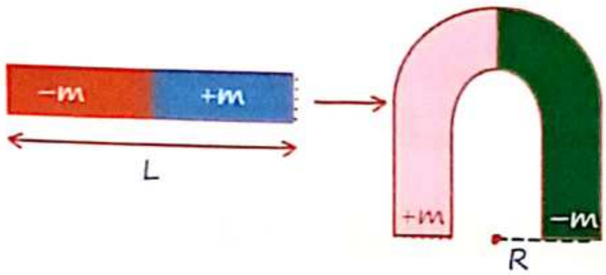
- The direction of $\vec{M}$ is from the south pole to the North pole for a bar magnet.
- For a current loop, the direction of $\vec{M}$ is determined using the right hand rule: curl your fingers in the direction of the currrent, and your thumb points in the direction of $\vec{M}$

Cutting of bar Magnet :-

*m ∝ Area, M ∝ Volume

Cute type	Resulting Length (l')	Pole Strength (m')	Magnetic Moment (M')
Cut along Length	$l' = l$	$m' = m/2$	$M' = m.l' = (m/2)l = M/2$
Cut Perpendicular to length	$l' = l/2$	$m' = m$	$M' = m.l' = m\,(l/2) = M/2$

Bending of bar Magnet :-



MR* **

Jab bhi mein Koi Circle dhekhu mera dil dewaana bole....

Magnetic moment circular Segment

$$\boxed{M' = \frac{M \sin(\theta/2)}{\theta/2}}$$

Complete Circle

$\theta = 2\pi$

$M' = 0$

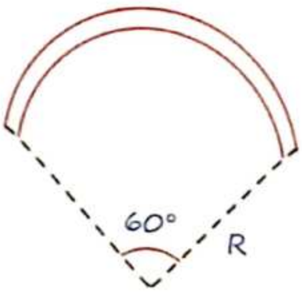
Semi-Circle.

$\theta = \pi$

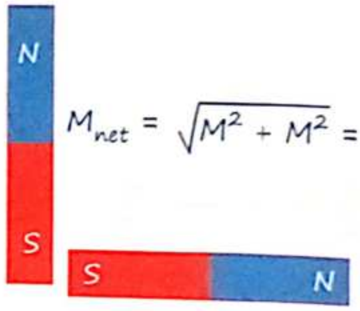
$M' = \frac{2M}{\pi}$

- Magnetic moment of 60° segment

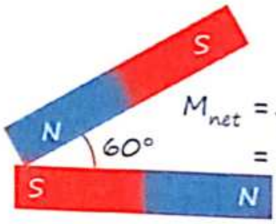
$$M' = \frac{M \sin\left(\frac{60^\circ}{2}\right)}{\frac{\pi}{3\times 2}} = \frac{3M}{\pi} \quad \left(\theta = 60^\circ = \frac{\pi}{3}\,\text{radian}\right)$$



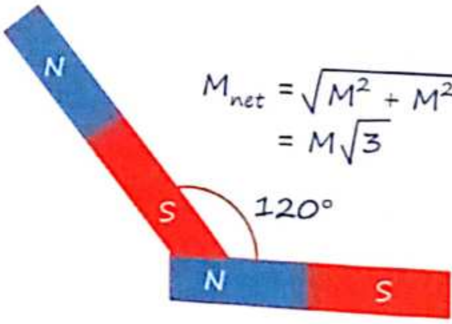
- Two Identical bar Magnet of magnetic Moment M



$$M_{net} = \sqrt{M^2 + M^2} = \sqrt{2}\,M$$



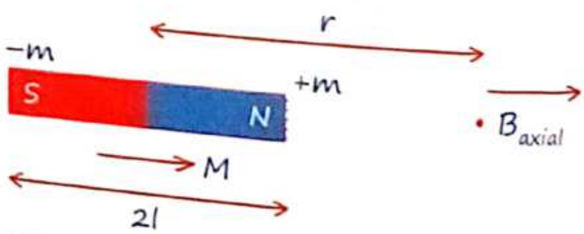
$$M_{net} = \sqrt{M^2 + M^2 + 2MM\cos(120^\circ)} = M$$



$$M_{net} = \sqrt{M^2 + M^2 + 2MM\cos 60^\circ} = M\sqrt{3}$$

3 Magnetic Field Due to Bar Magnet

On axial point :-



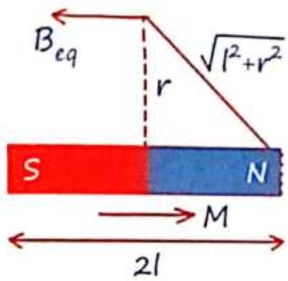
$$B_{axial} = \left(\frac{\mu_0}{4\pi}\right)\frac{2Mr}{(r^2-l^2)^2}$$

For $r \gg l$, $B_{axial} = \left(\frac{\mu_0}{4\pi}\right)\frac{2M}{r^3}$

- Magnetic Field along dipole

On Normal Bisector :-

$$B_{eq} = \left(\frac{\mu_0}{4\pi}\right)\frac{2ml}{(r^2+l^2)^{3/2}}$$

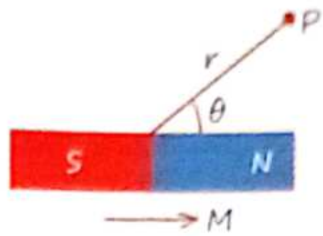


For $r \gg l$

$$B_{eq} = \left(\frac{\mu_0}{4\pi}\right)\frac{M}{r^3}$$

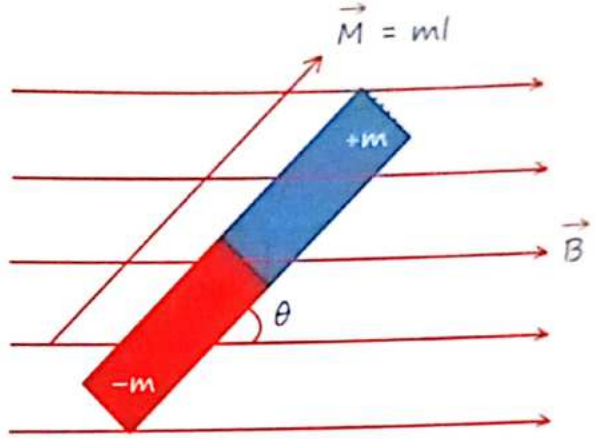
- Magnetic Field opposite to dipole.

Magnetic field due to Dipole at General Point :-



$$B_p = \left(\frac{\mu_0}{4\pi}\right)\frac{M}{r^3}\sqrt{3\cos^2\theta + 1}$$

Dipole in uniform Magnetic Field :-



Condition	Torque (t)	Potential Energy(U)	Explanation
General	$\tau = \vec{M}\times\vec{B} = MB\sin\theta$	$U = -\vec{M}.\vec{B} = -MB\cos\theta$	
$\theta = 0°$	$\tau = 0$	$U = -MB$ (Minimum energy)	Dipole aligned with the magnetic field
$\theta = 90°$	$\tau = MB$ (Maximum Torque)	$U = 0$	Dipole perpendicular to the magnetic field
$\theta = 180°$	$\tau = 0$	$U = MB$ (Maximum energy)	Dipole opposite to the magnetic field

- $W_{ext} = \Delta U$ $\qquad$ $W_B = -\Delta U$
- Stable equilibrium of $\theta = 0°$ (Diople opposite to field)
- Unstable equilibrium at $\theta = 180°$
- BAR magnet will oscillate in uniform magnetic field about stable equilibrium

$$T = 2\pi\sqrt{\frac{I}{MB}}$$

MR* feel

Magnetic Field Ka ekhi Udessh hai Ki woh Mag. dipole Ko apni taraf Khich Ke rakhna Chahta hai !

4 Analogy

Electrostatics	Magnetism
$\frac{1}{\varepsilon_0}$	μ_0
Charge q	Magnetic Pole Strength (m)
Dipole Moment $\vec{p} = q\vec{l}$	Magnetic Dipole Moment $\vec{M} = m\vec{l}$

$F = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2}$	$F = \frac{\mu_0}{4\pi}\frac{m_1 m_2}{r^2}$
$\vec{F} = q\vec{E}$	$\vec{F} = m\vec{B}$
Axial Field $\vec{E} = \frac{2\vec{p}}{4\pi\varepsilon_0 r^3}$	$\vec{B} = \frac{\mu_0}{4\pi}\frac{2\vec{M}}{r^3}$
Equatorial field $\vec{E} = \frac{-\vec{p}}{4\pi\varepsilon_0 r^3}$	$\vec{B} = -\frac{\mu_0}{4\pi}\frac{\vec{M}}{r^3}$
Torque $\vec{\tau} = \vec{p} \times \vec{E}$	$\vec{\tau} = \vec{M} \times \vec{B}$
Potential Energy U $= -\vec{p}\cdot\vec{E}$	$U = -\vec{M}\cdot\vec{B}$
Work, W $= pE(\cos\theta_1 - \cos\theta_2)$	Work, W $= MB(\cos\theta_1 - \cos\theta_2)$

MR Speical***

Force Between

- Two point Charge: $F \propto \frac{1}{r^2}$
- Dipole & Point Charge: $F \propto \frac{1}{r^3}$
- Dipole & Dipole: $F \propto \frac{1}{r^4}$

5 Gauss Law in Magnetism

- Gauss's law in magnetism states that the total magnetic flux through a closed surface is always zero, indicating that magnetic monopoles do not exist.

$$\int \vec{B}\cdot\vec{ds} = \phi_B$$

$$\oint \vec{B}\cdot\vec{ds} = 0 \text{ (Always)}$$

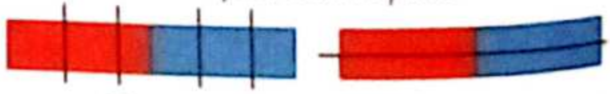
This means that the magnetic field lines are continuous and do not begin or end at any point, unlike electric field lines.

MR Speical***

1> T = Same $\Big\}$ When bar magnet cut along length.

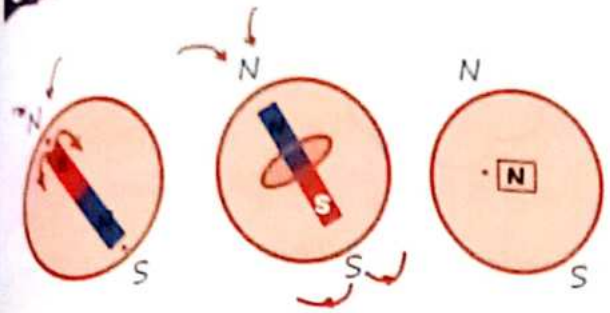
2> $T' = \frac{T}{n}$ $\Big\}$ When bar magnet cut $\perp^{er}$ to length.

n = no. of equal cutted part.



$T = \frac{T}{n}$ $\qquad$ $T' = T$

6 Neutral Point



2 = NP   ∞ = NP   1 = NP

7 Vibrational Magnetometer

(Oscillational Magnetometer)

$$T = 2\pi\sqrt{\frac{I}{MB_H}}$$

Application -1 :-

To Find Magnetic dipole Moment (M):-

$$M = \frac{4\pi^2 I}{T^2 B_H}$$

Application -2 :-

To Find ratio of Mag. dipole moment of two magnet of same size.

$$\frac{M_2}{M_1} = \left(\frac{T_1}{T_2}\right)^2$$

Application -3 :-

To Compare Mag. dipole moment of two magnet of diff. size.

$$\therefore \boxed{\frac{M_1}{M_2} = \frac{T_2^2 + T_1^2}{T_2^2 - T_1^2} = \frac{f_1^2 + f_2^2}{f_1^2 - f_2^2}}$$

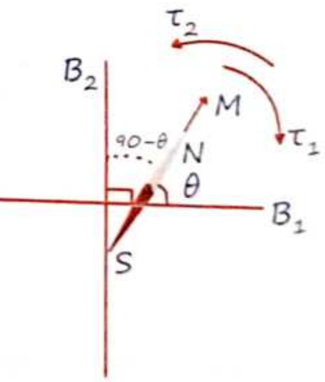
$$T_1 = 2\pi\sqrt{\frac{I_1 + I_2}{(M_1+M_2)B_H}}$$

$$T_2 = 2\pi\sqrt{\frac{I_1 + I_2}{(M_1-M_2)B_H}}$$

8 Tangent Law

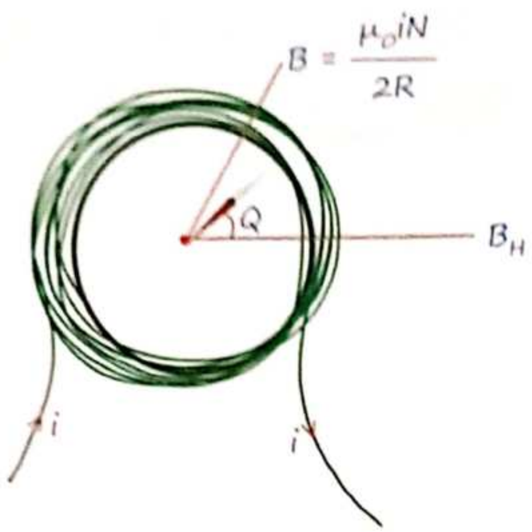
The tangent law in magnetism describes the relationship between two perpendicular magnetic fields acting on a magnetic needle, such as in a tangent galvanometer. When a magnetic needle is subjected to two perpendicular magnetic fields, B_1 and B_2, it aligns at an angle θ such that:

$$\tan\theta = \frac{B_2}{B_1}$$



<u>Tangent Galvanometer</u> :-

A tangent galvanometer is an instrument used to measure the horizontal component of the Earth's magnetic field.



When placed in the Earth's magnetic field.

$$M B_H \sin\theta = M\left(\frac{\mu_0 iN}{2R}\right)\cos\theta.$$

$$\boxed{\tan\theta = \frac{\mu_0 iN}{2R\,B_H}}$$

"K = Reduction Factor."

$$\underline{\underline{K\tan\theta = i}} \quad \left(K = \frac{2R\,B_H}{\mu_0 N}\right)$$

Magnetic Field Intensity:- (H)

Source की असली औकात

$$H = \frac{B_0}{\mu_0} = \frac{B_m}{\mu_m}$$

Magnetic Permeability :- (μ)

$$\mu = \frac{B}{H}$$

Scalar Unit :- Wb/Amp-m

Medium Source

Magnetisation & Magnetic Intensity:- (I)

$$\vec{I} = \frac{\vec{M}}{V}$$

→ Vector
Medium independent.

→ dir^n $\parallel^{el}$ to $\vec{M}$

Magnetic Susceptibility:- (χ)

$$\chi_m = \frac{I}{H}$$

→ Scalar
→ Unit & Dimensionless

$\chi_m \uparrow$ = Easily magnetised.

Relation Between μ_r & χ_m:-

$$\mu_r = 1 + \chi_m$$

9 Cause of Magnetism

Atom → (Nucleus + electron in rotational motion)

#revolving electron Produce magnetic field, (magnetic moment) along axis of Rotation.

#In Paired electron atom, two electrons are in Opposite spin.

$$\vec{M}_{atom} = 0$$

In unpaired electron atom.

$\vec{M}_{atom} \neq 0$ and $\vec{M}_{crystal} = 0$ due to random orientation of atoms.

10 Materials

Diamagnetic:-

- Diamagnetic have tendency to Move from region of Stronger to weaker part of external Magnetic Field

∵ They are Magnetised in opposite direction.

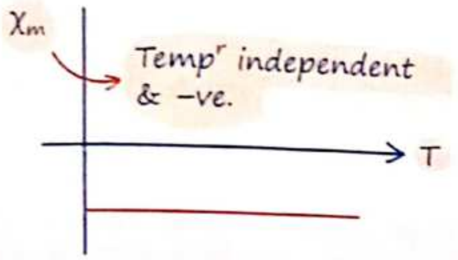
- Magnetic field lines are expelled by these substances.

Imp → Diamagnetism is universal property

MR***

Magnetise hota hai lenz law (law of inertia) se.

$\chi_m = -ve$
$B < B_0$
$\mu < \mu_0$
$\mu_r < 1$



Paramagnetic:-

Paramagnetic substance

$M_{atom} \neq 0$

$M_{material} = 0$ (in absence of external mag. field)

$M_{material} \neq 0$ (⊕ presence of B_{ext})

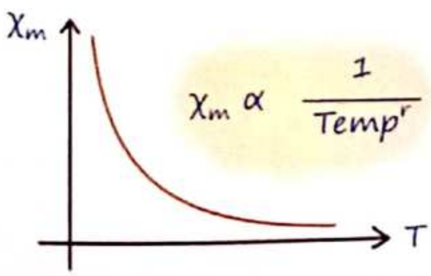
Tendency to move from Weak Magnetic field Region to strong magnetic field.

MR***

#Magnetise hota hai τ_{ext} se.

χ_m = +ve (Small +ve)

$B > B_0$
$\mu > \mu_0$
$\mu_r > 1$



Ferromagnetic :-

Ferromagnetic Material

strongly attracted to Magnetic Substance

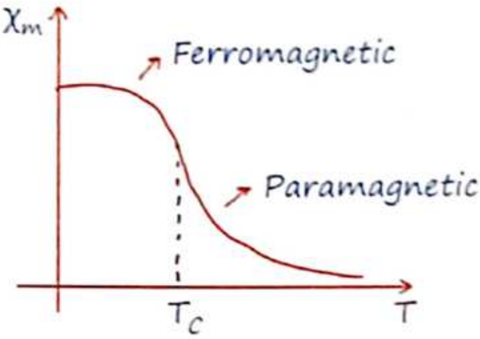
$\vec{M}_{domain} \neq 0$

$M_{material} = 0$ (∵ Random arrangement of domains)

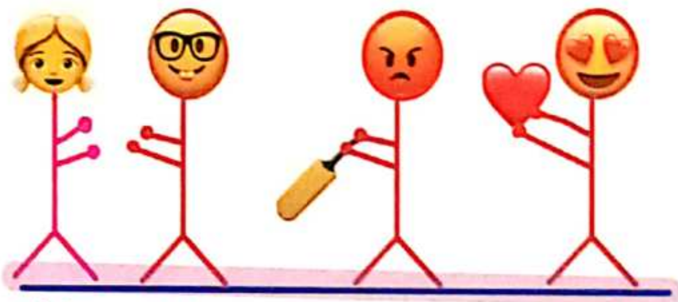
$M_{material} \neq 0$ (⊕ magnetic field external)

Domain Formation.

$\chi_m >>> 1$
$B >>> B_0$
$\mu >>> \mu_0$
$\mu_r >>> 1$



MR Speical***



Curie's Law:-

$$I = \frac{CB_0}{T}$$ Paramagnetic Sub^s.

$$\chi_m = \frac{C\mu_0}{T}$$

Curie Weiss Law :-

$$\chi_m = \frac{C}{T - T_C}$$

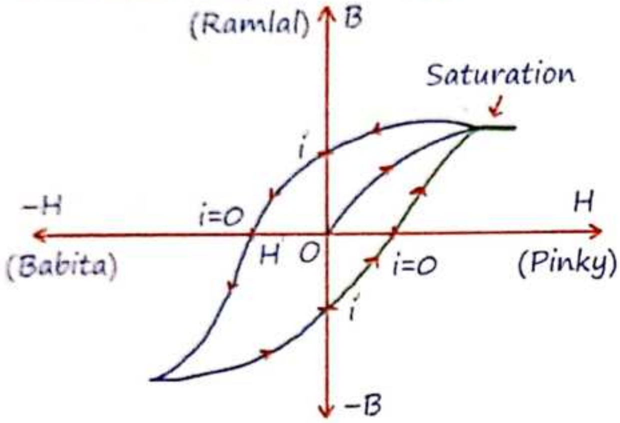
11 Hysteresis Loop or B/H Curve

MR Speical***

"Life of Ramlal"

Area under loop ∝ कितना कटा Ramlal का

Area under loop ∝ Energy loss



OI' = Retentivity (R)

Ramlal Pinky Ke Pyaar mein Kitne yaadein bachake rakh paya tha

OH' = Coercivity (C)

Babita Ko Jitna Pyaar dikhana pada pinky Ki yaadein Ko mitane Keliye.

- SOFT IRON :- Small Area.

  Low Retentivity & Coercivity.

- Permanent Mag. :- Large Area.

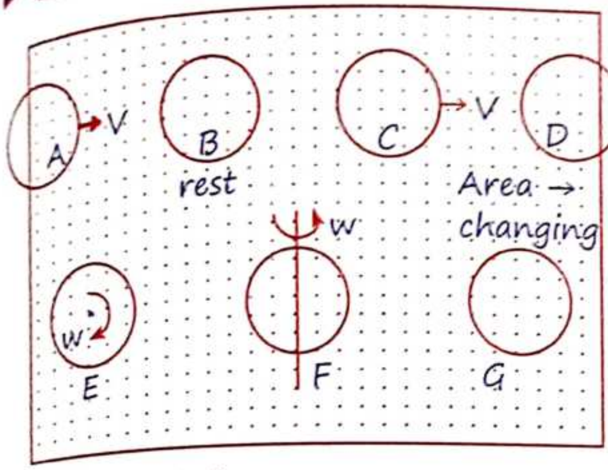
  High Retentivity & Coercivity.

MR*

Everyone including your society, family, friends sirf success ko he salute karte hai

27 Electromagnetic Induction

1 Faradays Experiment in Uniform Magnetic Field

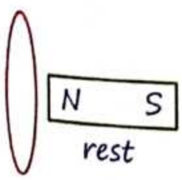


<u>In loop A, D, F, G :-</u>

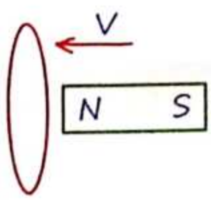
- Current induced in loop because of change in magnetic flux.
- This can be due to either the movement of the loop or a change in the area of the loop within the magnetic field.

<u>In loop B, C, E :-</u>

- No current induced, because flux is constant.
- These loops could be stationary, or their area within the magnetic field might not be changing.



No current



Current induced

2 Magnetic Flux

Counting of magnetic field lines passing through given cross-section area.

- gives the idea of magnetic energy.
- scalar S.I. unit → Tesla-m^2 = Weber C.G.S. unit → Gauss cm^2 = Maxwell

$$\phi = \vec{B}.\vec{A} = BA\cos\theta$$

θ = Angle between Mag. Field & Area

Unit :- 1 Wb = 10^8 Maxwell

- If $\vec{B}$ = Variable.

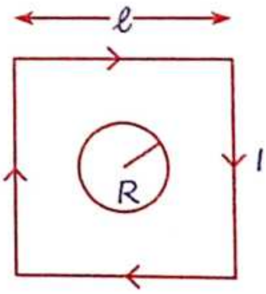
$$\phi = \int \vec{B}.\vec{dA}$$

- If magnetic field $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$ and $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$

  then (flux), $\phi = B_xA_x + B_yA_y + B_zA_z$

- Flux through circular loop ($\ell >> R$)

$$\phi = \frac{2\sqrt{2}\mu_0 I\pi R^2}{\pi\ell}$$



MR***

$\odot$ = outward = गधो की तरफ !

$\otimes$ = inward = गधो से दूर !

Note :-

ϕ

Variable | Constant

$\phi = \int_0^A B \, dA \cos\theta$ | $\phi = BA \cos\theta$

$I_{induced} \neq 0$ | $I_{induced} = 0$

Variable due to :-

$B \rightarrow$ time varying

$A \rightarrow$ Variable

θ = Changing

Induced current

Magnitude "Faraday's" Law | Direction "Lenz" Law

3 Lenz Law

*Joh "i" Ko Paida Krta hh "i" Ushika Oppose Krta hh.

MR* law :-

- Coil is in inertia (Pyaar) of Flux.

External Field | Induced Field B

$\odot B\uparrow \longrightarrow \otimes$ (Then I_{in} = CW)

$\odot B\downarrow \longrightarrow \odot$ (Then I_{in} = ACW)

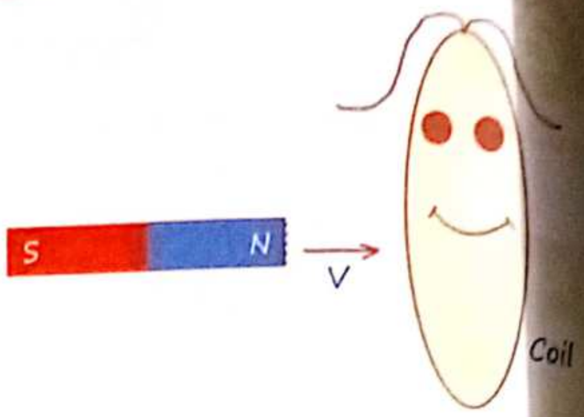
$\otimes B\uparrow \longrightarrow \odot$ (Then I_{in} = ACW)

$\otimes B\downarrow \longrightarrow \otimes$ (Then I_{in} = CW)

"Up $\odot$ Anti, down C"

- Coil Na Flux Ko pyaar Krta hai na Usko hate Krta hai woh bs Change in flux ko oppose Krta hai.

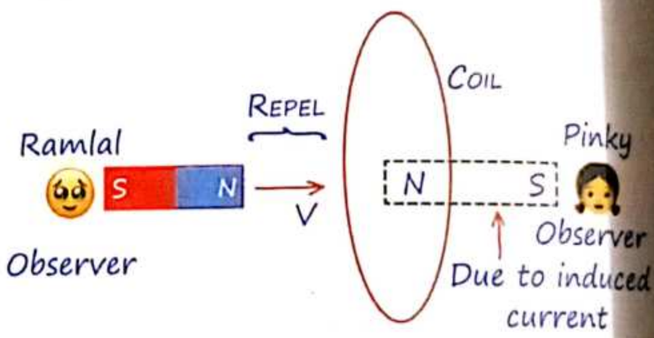
MR* law :-



Coil Kehti hai magnet से :-

Pass mt aana repel Karungi durr mt jana attract Karungi aaisehi padhai krte raho Exam Ke baad date Karungi.

Case :- I

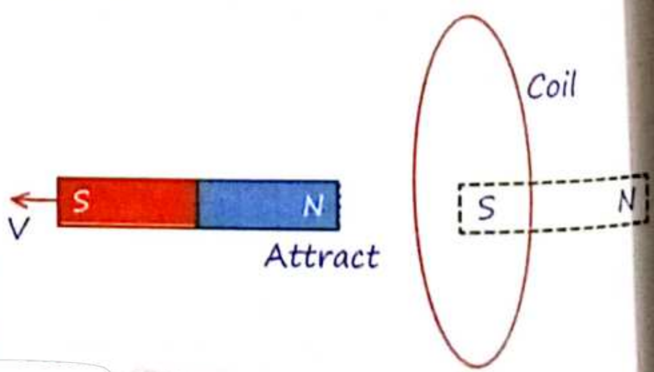


Current induced in loop

- w.r.t. Ramlal → ACW
- w.r.t. Pinky → CW

MR* → Aage se Anti then pichhe se clock

Case :- II

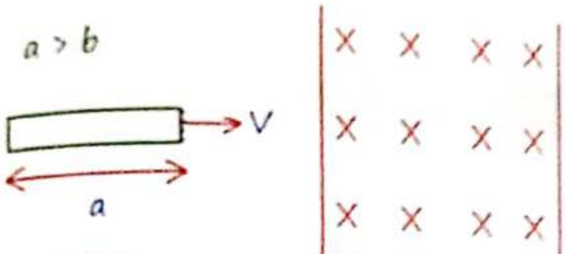


4 NOTE (AIIMS)

Q.1 Find time for which current will induced in ractangular loop?

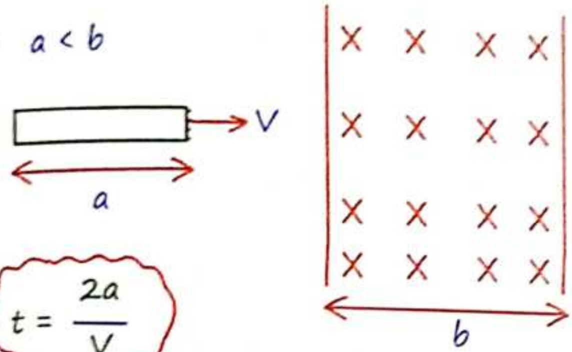
Sol.

I> $a > b$

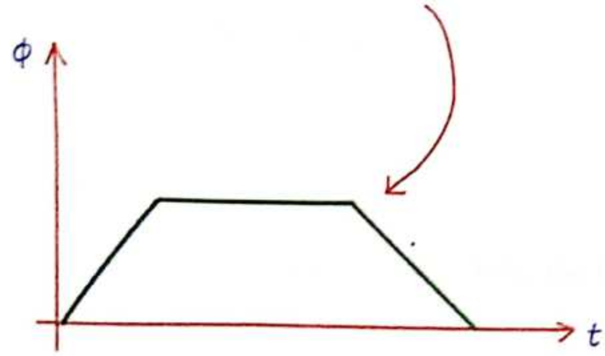


$$t = \frac{2b}{V}$$

II> $a < b$



$$t = \frac{2a}{V}$$



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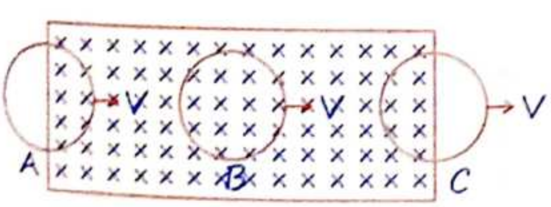
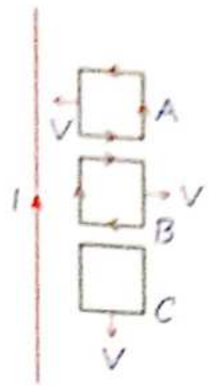


Current induced in loop

Ⓐ → ACW Ⓑ → O Ⓒ → CW

•



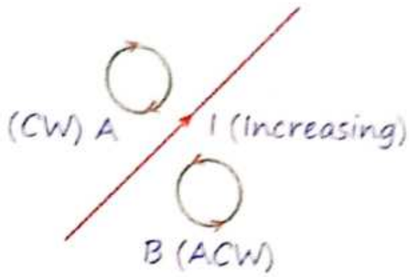
Current induced in loop

A → ACW

B → CW

C → Zero

•



5 Faraday's Law of EMI

Based on Energy Conservation.
[Loop]

ε_{inst}	ε_{avg}
$\varepsilon_{inst.} = \frac{-d\phi}{dt}$	$\varepsilon_{avg.} = -\frac{\Delta\phi}{\Delta t}$
$i_{inst.} = \frac{-d\phi}{R.dt}$	$i_{avg} = -\frac{\Delta\phi}{R\Delta t}$
$\lvert\varepsilon_{inst.}\rvert = \frac{d(BA\cos\theta)}{dt}$	$\Delta Q = -\frac{\Delta\phi}{R}$

Q.2 Radius of circular loop placed perpendicular to magnetic field increasing at rate r_o m/s then find induced emf in loop when radius is r.

Sol. emf

$$= -\frac{d(BA\cos 0^\circ)}{dt} = -B\frac{d\,\pi r^2}{dr} \times dr/dt$$

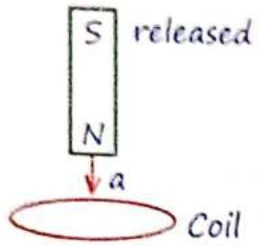
$$\text{emf} = -B\pi 2 r_o r \text{ Volt (Given, } \frac{dr}{dt} = r_o)$$

We know $\phi = BA\cos\theta$

$$|emf| = \frac{d\phi}{dt} = \frac{d(BA\cos\theta)}{dt}$$

In this formula, three variable B, Area and angle (θ), genrally 3 type ka question aayga B-time dependent, A-time dependent or θ-time dependent.

Falling a magnet through different coils :-

- 

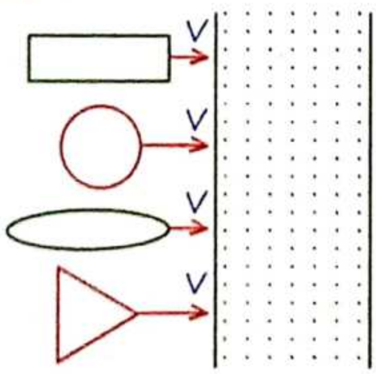
A non-conducting or a coil with a small cut:

- No current flows, hence no opposing magnetic field is created. The magnet falls with acceleration $a = g$.

Conducting coil:

- Induced current generates a magnetic field opposing the fall of the magnet, reducing the net acceleration i.e., $a < g$.

Q.3 In which loop induced emf will be uniform:



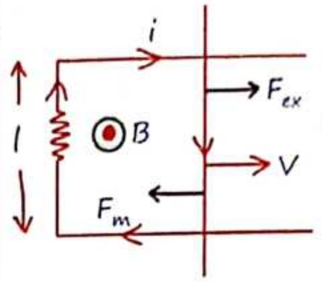
Sol. In rectangular loop because rate of change in flux is constant.

law :-

लोहा हो लकड़ी हो Plastic हो टूटा हुआ तार हो Kuch bhi ho sab mein EMF induced करवाएगा Tera Faraday.

6 Rail problems

Rod of length l moving with velocity V.



- Induced Emf: $\varepsilon = BVl$
- Current in the circuit $i = \frac{\varepsilon}{R} = \frac{BVl}{R}$

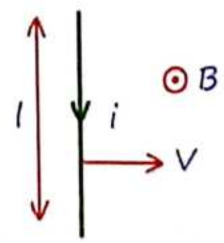
- External Force: $F_{ext} = F_m = Bil = \frac{B^2l^2V}{R}$

Note :- To maintain the constant velocity V, an external force F_{ext} must be applied to countereract the magnetic force.

- Power dissipated in the circuit

$$P = FV = \frac{B^2l^2V^2}{R}$$

Hall Bhaiyaa :- [Rod]



$\hookrightarrow B \perp V \perp l$

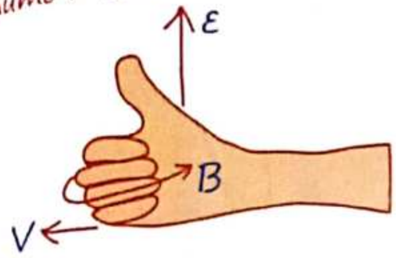
$\varepsilon = BVl_{eff.}$

Scalar Triple Product :-

$\varepsilon = (\vec{V} \times \vec{B}) \cdot \vec{l}$ $\quad \because (F_M = F_E)$

Direction of EMF :-

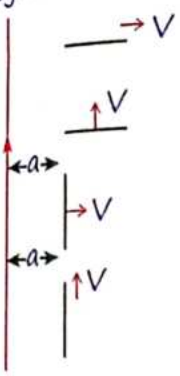
Plam :- Velocity
Curl Finger :- Magnetic Field
Thumb :- EMF.



MR* for Direction of higher potential
place four finger along velocity and slap magnetic field thumb will represent higher potential.

Induced emf in given Rod :-

The rod is divided into segments with specific lengths and velocities. The induced EMF in each segment depends on the length of the segment and the magnetic field.



$emf = 0$

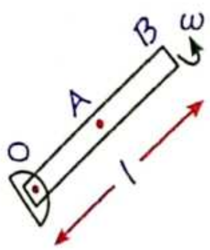
$emf = \dfrac{\mu_0 i V}{2\pi} \log\left(\dfrac{a+l}{a}\right)$

$emf = \dfrac{\mu_0 i lV}{2\pi a}$

$emf = 0$

Rotational Motion in a Magnetic Field :-

When a conductor rotates in a magnetic field, an emf is induced across its length due to the motion of the conductor through the magnetic field lines.



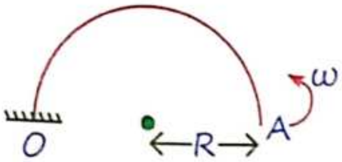
$(emf)_{OB} = \dfrac{1}{2} B\omega l^2$

$(emf)_{OA} = \dfrac{1}{2} \dfrac{B\omega l^2}{4}$

$(emf)_{AB} = \dfrac{3}{8} B\omega l^2$

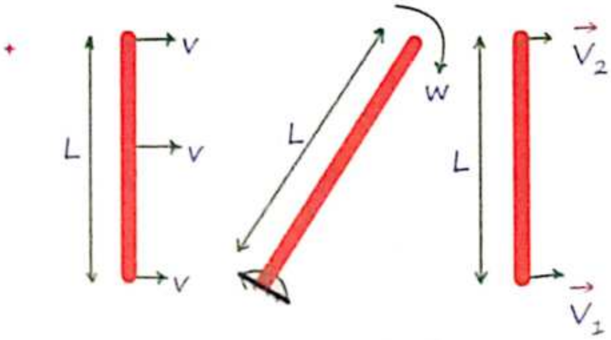
MR* :- *VVimp*

🔥Fire Concept 🔥



$|emf|_{OA} = \dfrac{1}{2} B\omega (2R)^2$

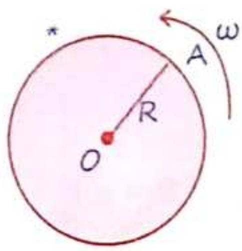
$l_{eff} = 2R$



Translational Motion (T)	Rotational Motion (R)	T + R Motion
$E_{motional} = BVl$	$E_R = \dfrac{1}{2} B\omega l^2$	$\varepsilon = BL\left[\dfrac{\vec{V}_1 + \vec{V}_2}{2}\right]$

dist^n between two points.

Note :-



$\varepsilon_{OA} = \dfrac{B\omega R^2}{2}$

Disc. ⟶ Collection of ∞ Rods.

7 Induced Electric Field

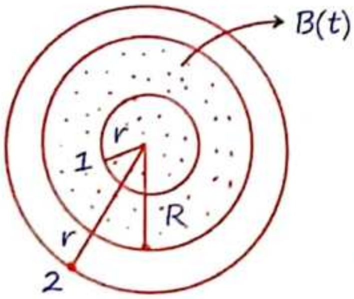
MR*

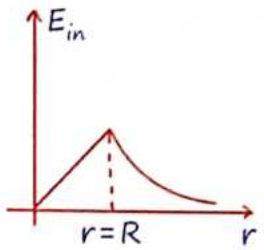
Electric Field

असली	नकली
• Due to Rest Charge.	• Due to time varying M. Field
• Electrostatic Field.	• Induced E. Field

असली	नकली
Conservative	Non-conservative
$\oint \vec{E}.\vec{dl} = 0$	$\oint \vec{E}.\vec{dl} = A.\dfrac{d\vec{B}}{dt}$
Does not Form Closed loop	Always form Closed loop. (Concentric Circle)

Value of induced E. Field :-





Inside the circular region (r < R)

$$\boxed{E_1 = \frac{r\, dB}{2\,.\,dt}}$$

Outside the circular region (r > R)

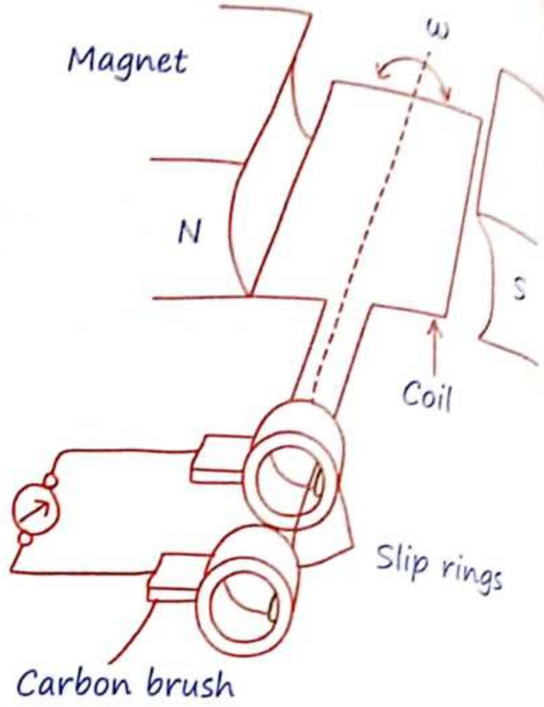
$$\boxed{E_2 = \frac{R^2}{2r}\frac{dB}{dt}}$$

8 Ac-generator

ϕ (flux) = NAB cos θ and $\theta = \omega t$

$$\boxed{\varepsilon = BAN\omega \sin\theta}$$

*Convert Mech. Energy into Electric Energy.



9 Self-inductance(L)

Aaj Kal Ke Rishte Kaha hai itne Ache isliye hm Akele hi hai Ache.

- $L = \mu_0 n^2 A l = \dfrac{\mu_0 N^2 A}{l}$
- $e = -L\dfrac{di}{dt}$
- $\phi = Li$

L (self-inductance) $\propto l$ ($n = \cos t^n$)

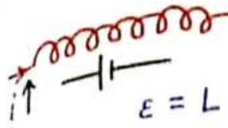
$\propto 1/l$ ($N = \cos t^n$)

Where n = no of turns per unit length

N = total no of turns

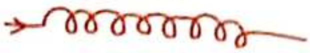
Unit :- Henry (H)

- ϕ = constant ;_ $L_1 i_1 = L_2 i_2$
- Direction buddhi से, Magnitude Formula से

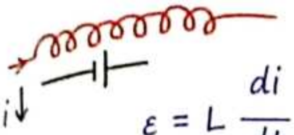


$$\varepsilon = L\frac{di}{dt}$$

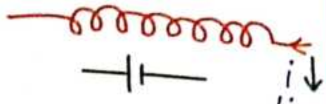
$I = cost^n$



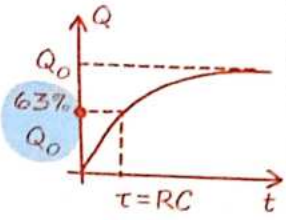
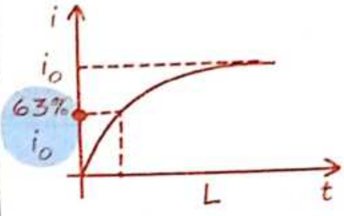
$$\varepsilon = 0$$



$$\varepsilon = L\frac{di}{dt}$$



$$\varepsilon = L\frac{di}{dt}$$

Capacitor	Inductor
$Q = Q_0[1 - e^{-t/RC}]$	$i = i_0[1 - e^{-Rt/L}]$
$\tau = RC$	$\tau = L/R$
• $t = \infty$ $Q = Q_0$	• $t = \infty$ $i = i_0$
	

Energy Store in Inductor :-

$$E = \frac{1}{2}Li^2 = \frac{B_0^2 Al}{2\mu_0}$$

• Energy Stored per unit Volume

$$= \frac{B_0^2}{2\mu_0}$$

Charging of Inductor :-

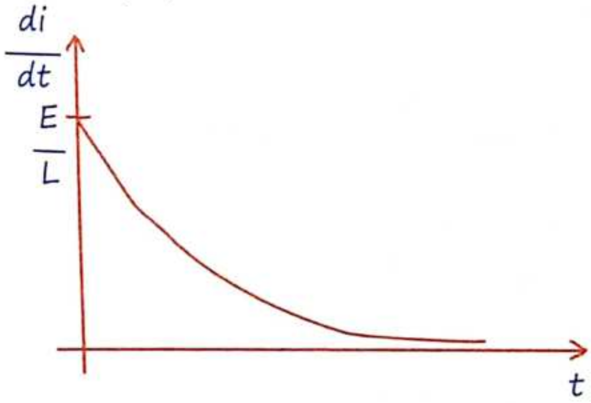
$t = 0$	$t = \infty$ [Steady State]
L = Open Wire C = Simple Wire	L = Simple Wire C = Open Wire
• $V_L = L\left(\frac{di}{dt}\right) = E$ • $V_R = 0$	• $V_L = 0$ • $V_R = E$ • $i = E/R$ $\boxed{E_{max} = \frac{1}{2}L\left(\frac{E}{R}\right)^2}$

Graph Between di/dt and Time :-

Rate of Change in Current :-

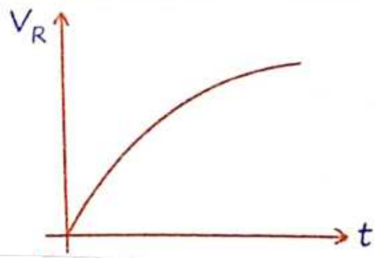
$$\frac{di}{dt} = \left(\frac{E}{L}\right)e^{-t/\tau}$$

At $t = 0$, $\frac{di}{dt} = \left(\frac{E}{L}\right)$ At $t = \infty$ $\frac{di}{dt} = 0$

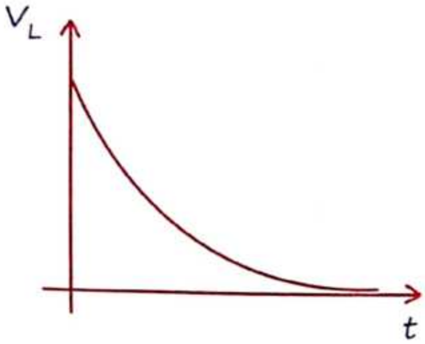


Graph Between V_R and Time :-

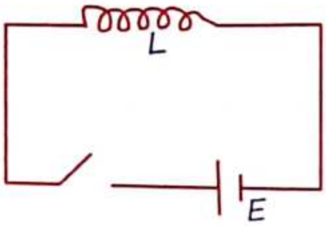
$$V_R = i_0(1 - e^{-t/\tau}).R$$



$$V_L = L\left(\frac{di}{dt}\right) = Ee^{-t/\tau}$$

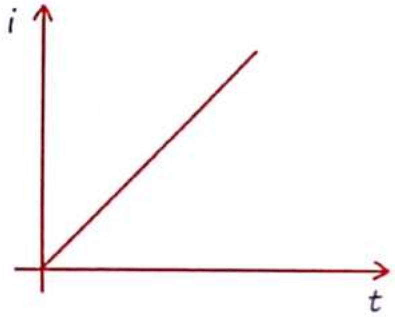


Current in Inductor as Function of Time :-

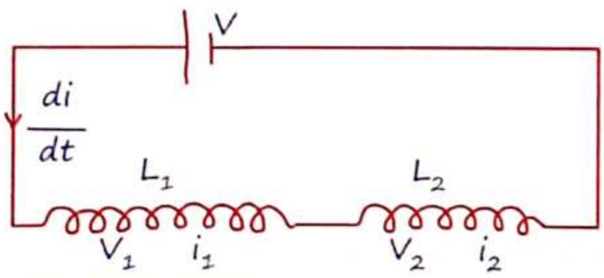


$$i = \frac{E}{L}t$$

Battery be like :- Raziya Gundo mein Fasgayi.



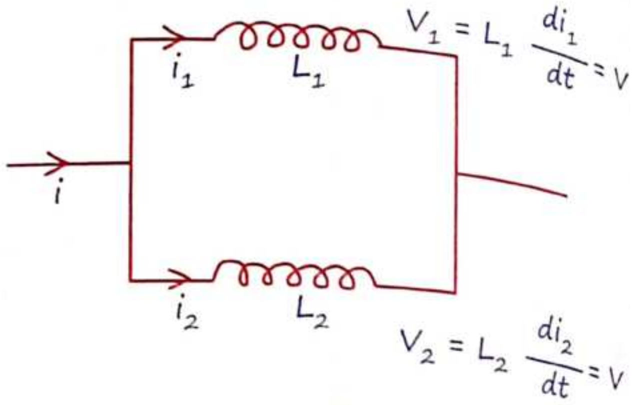
Series Combination of Inductor :-



$$\boxed{\frac{di}{dt} = \text{same}} \qquad \boxed{L_{eq} = L_1 + L_2}$$

Parallel Combination of Inductor :-

$$\frac{1}{Leq} = \frac{1}{L_1} + \frac{1}{L_2}$$



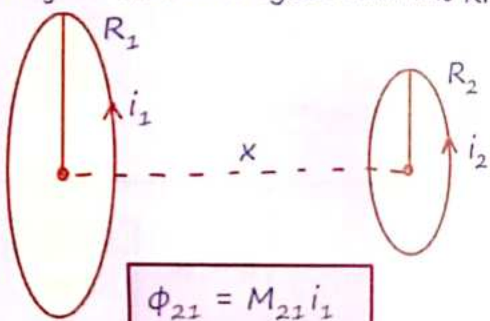
$$\boxed{\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}}$$

10 Mutual Inductance

Mutual inductance is the property of two coils (or inductors) where a change in current in one coil induces an emf in the other coil.

MR*

Jitne mein tumhe तंग Karunga utna tum bhi mujhe तंग Krna Jyda kam mt Krna...



$$\boxed{\phi_{21} = M_{21} i_1}$$

Mutual inductance of 2 due to 1 :-

$$\Rightarrow \boxed{M_{21} = \frac{\mu_0 N_1 N_2 R_1^2 R_2^2 \pi}{2x^3}}$$

Reciprocity Theorem :-

$$M_{12} = M_{21}$$

Magnetic flux	$\phi_{12} = M_{12} i_2$	$\phi_{21} = M_{21} i_1$
Induced e.m.f.	$\varepsilon_{12} = M_{12}\frac{di_2}{dt}$	$\varepsilon_{21} = M_{21}\frac{di_1}{dt}$

Note :- Coupling Factor

$0 \le K \le 1$

$$\boxed{M = K\sqrt{L_1 L_2}}$$

- L_1 & L_2 very close → $K = 1$.

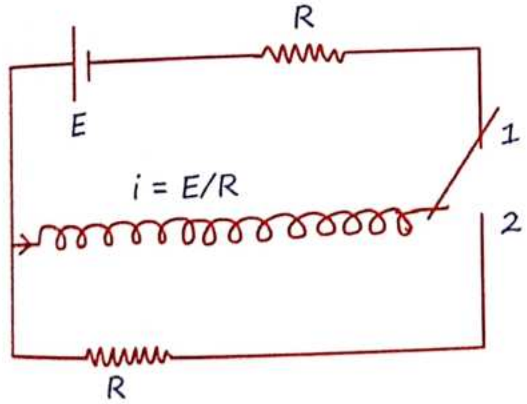
Series Combination of Inductor (Considering M) :-

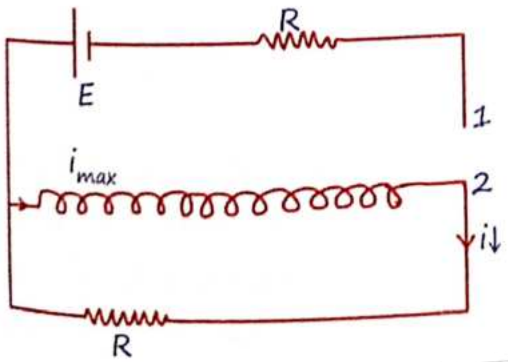
$L_{eq} = L_1 + L_2 \pm 2M$

$$\boxed{L_{eq} = L_1 + L_2 \pm 2M}$$

L_1 & L_2 are in same order	L_1 & L_2 are in Opposite order
$L_{eq} = L_1+L_2+2M$	$L_{eq} = L_1+L_2-2M$
$L_{eq}=L_1+L_2+2\sqrt{L_1L_2}$	$L_{eq}=L_1+L_2-2\sqrt{L_1L_2}$

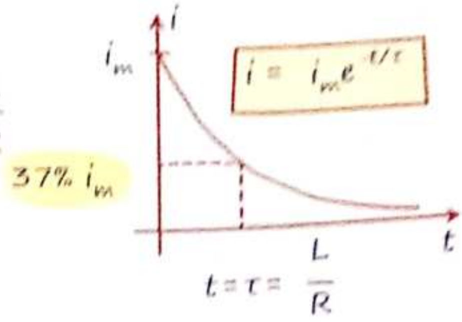
11 Discharging of Inductor



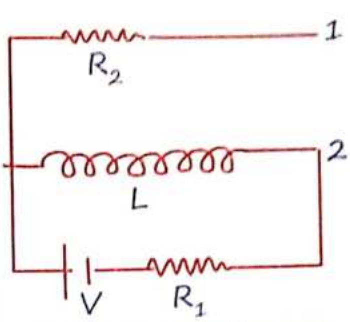


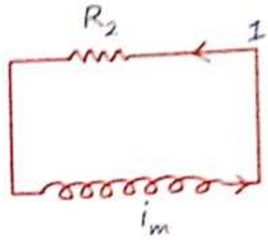
$\{i = i_m e^{-t/\tau}\}$ $\quad V_L = L\dfrac{di}{dt} = iR$

$\tau = \dfrac{L}{R}$



- Find total heat loss across R_2 when Key is shifted from (2) to (1) :-





$t = \infty$

$i_m = \dfrac{V}{R_1}$

$$\boxed{E_{stored} \text{ or } E_{loss} = \frac{LV^2}{2R_1^2}}$$

12 Eddy Current

Agar Kishi तार मे Current Flow करेगा तो उसे Kehte है Current और Plate मे Current Flow करेगा तो Eddy Current.

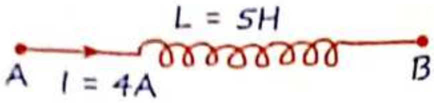
Application :-

1> Magnetic braking in train

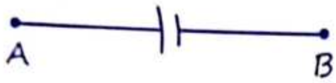
2> Electromagnetic damping.

3> Induction Furnance

4> Electric Power meters.



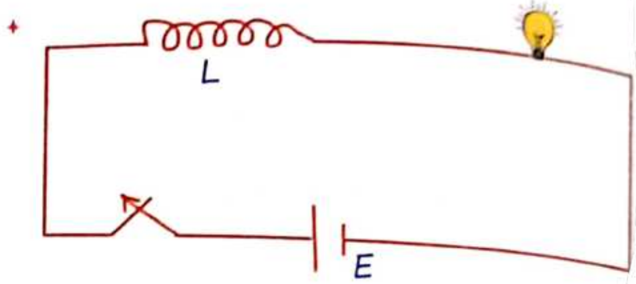
$\frac{di}{dt} = 2\ A/S \rightarrow i(\uparrow es)$



$$V_A - \frac{Ldi}{dt} = V_B$$

$$V_A - V_B = \frac{Ldi}{dt} = 5 \times 2 = 10\ V$$

Current is increasing hence B is at lowe potential.



Brightness of bull will suddenly increase when key is just opened.

MR*

"Khushi k liye kam karoge to khushi nahi milegi, lekin khush hokar kam karoge to khushi aur safalta jarur milegi."

28 Alternating Current

1 Alternating Current

- Alternating Current (A.C.) is an electric current that reverses its direction periodically. It is bidirectional, meaning it flows in both directions.

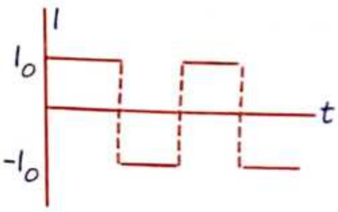
  Example: The household electricity supply is A.C., typically with a sinusoidal wave.

Direct Current (D.C.) :-

- Direct Current (D.C.) is an electric current that flows in one direction only. It is unidirectional and usually has a constant magnitude.

  Example: The current from a battery is D.C., providing a steady flow of electricity.

- A current of constant magnitude can be A/c current → Yes. It is square wave A/c.

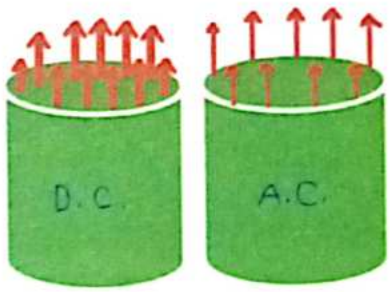


- A Variable current in magnitude only is D/c

  = A current is varying from +5A to +15A sinusoidally then this current is D/c or (AC+DC) mixture, not A/c because it is not a bidirectional.

- A/C → Bidirectional
- D/C → Unidirectional (Steady Current)

$i_{D/C} \propto r^{3/2}$
$i_{A/C} \propto$ S. Area.



2 Measuring of Current

Moving coil galvanometer	Hot wire ammeter
"T" on Current carrying coil.	Heat loss
Only D/C	Both D/C & A/C
θ (angle) $\propto i$	θ (angle) $\propto H \propto i^2$
Linear Scale	Non-linear Scale

Average Value :-

Discrete system

$$i_{avg} = \frac{i_1+i_2+...+i_n}{n}$$

Continuous system

$$\langle i \rangle = i_{avg} = \frac{\int i.dt}{\int dt}$$

RMS Current :-

$$i_{rms} = \sqrt{\langle i^2 \rangle} = \sqrt{\frac{\int i^2.dt}{\int dt}}$$

RMS Voltage :-

$$V_{rms} = \sqrt{\langle V^2 \rangle} = \sqrt{\frac{\int V^2 . dt}{\int dt}}$$

MR* Ratta

- Full Cycle (FC)

$\langle \sin\theta \rangle = 0$
$\langle \cos\theta \rangle = 0$

- Half Cycle (HC)

$\langle \sin\theta \rangle = \frac{2}{\pi}$

$\langle \cos\theta \rangle = \frac{2}{\pi}$

$$\langle \sin^2\theta \rangle = \langle \cos^2\theta \rangle = \left.\frac{1}{2}\right\} \text{Half/Full Cycle}$$

- $\langle 2\sin(\omega t). \cos(\omega t) \rangle_{F.cycle} = \langle \sin(2\omega t) \rangle_{F.cycle} = 0$
- Avg. value of $I_0 \sin(\omega t)$ in half cycle may be zero or $\frac{2i_0}{\pi}$ because it will depends half cycle ki limit kaha se liya hai.

Mathematical representation Alternating Current :-

$$i = i_0 \sin[\omega t + \phi]$$

where i_0 = peak current

ω = Angular frequency

ϕ = Phase angle

- Note :-

$\langle i \rangle_{FC} = 0$

$\langle i \rangle_{HC} = \frac{2i_0}{\pi}$

$$\sqrt{\langle i^2 \rangle} = i_{rms} = \frac{i_0}{\sqrt{2}}$$
- Virtual Current
- Effective Current

Reading of Ammeter.

$H = i_{rms}^2 Rt.$

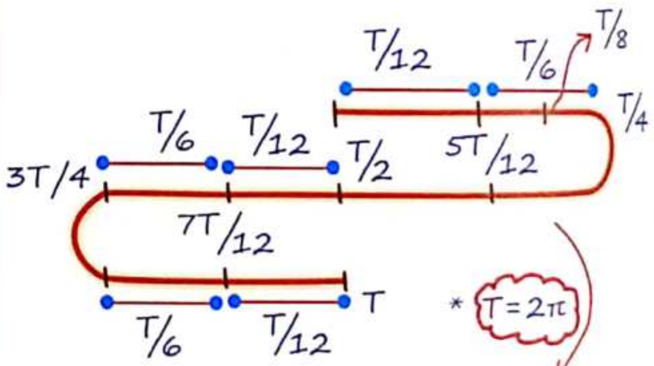
Voltage relation

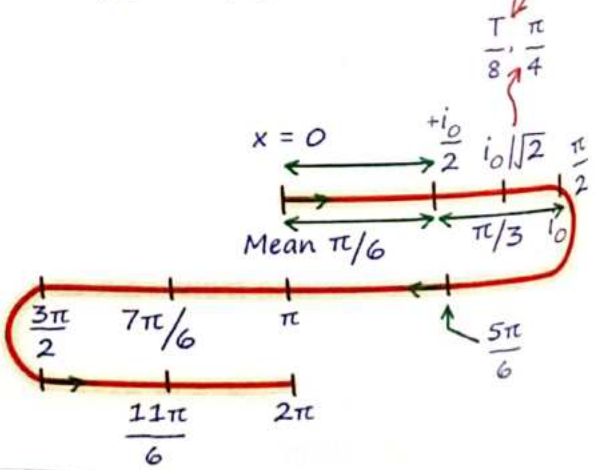
- $V = V_0 \sin(\omega t) = V_0 \sin(2\pi f t)$
- $V_{rms} = \frac{V_0}{\sqrt{2}}$

MR* Table

	Sinusoidal A/c	Square (A/c)	Trianglular	Full wave rectifier	Half wave rectifier
Avg. Full cycle	0	0	0	$\frac{2i_0}{\pi}$	$\frac{i_0}{\pi}$
Avg. Half cycle	$\frac{2i_0}{\pi}$	i_0	$\frac{i_0}{2}$	$\frac{2i_0}{\pi}$	$\frac{2i_0}{\pi}$, 0
R.M.S. value	$\frac{i_0}{\sqrt{2}}$	i_0	$\frac{i_0}{\sqrt{3}}$	$\frac{i_0}{\sqrt{2}}$	$\frac{i_0}{2}$

Fire Concept MR*





Alternating Current

MR* (i) $i = a + b\sin(\omega t)$

$a \uparrow$ DC, $b\sin(\omega t) \uparrow$ AC

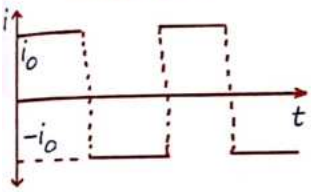
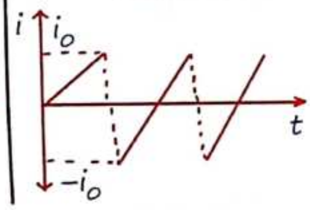
$$i_{rms} = \sqrt{a^2 + \frac{b^2}{2}} = \sqrt{\langle i^2 \rangle}$$

(ii) $i = a\sin\omega t + b\cos\omega t$

$a\sin\omega t \uparrow$ AC, $b\cos\omega t \uparrow$ AC

$$i_{rms} = \sqrt{\frac{a^2}{2} + \frac{b^2}{2}} = \sqrt{\langle i^2 \rangle}$$

Alternating Current :-

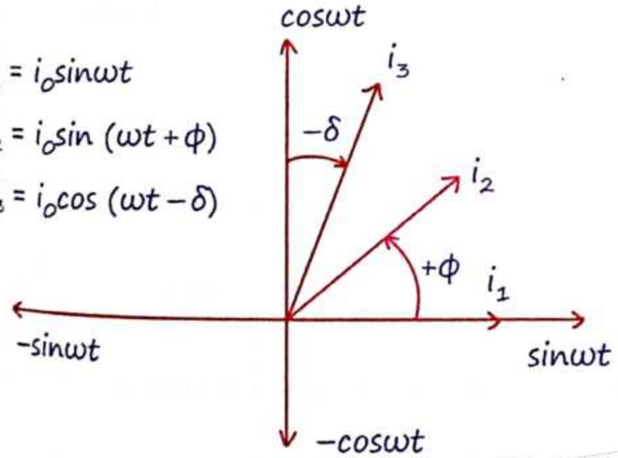
Square wave A/C	Triangular wave A/C
	
$\langle i \rangle_{FC} = 0$	$\langle i \rangle_{FC} = 0$
$\langle i \rangle_{HC} = i_0$	$\langle i \rangle_{HC} = i_0/2$
$\sqrt{\langle i^2 \rangle_{HC}} = i_0$	$\sqrt{\langle i^2 \rangle_{HC}} = \frac{i_0}{\sqrt{3}}$

Representation of A/c Current & Voltage by Phase Diagram :-

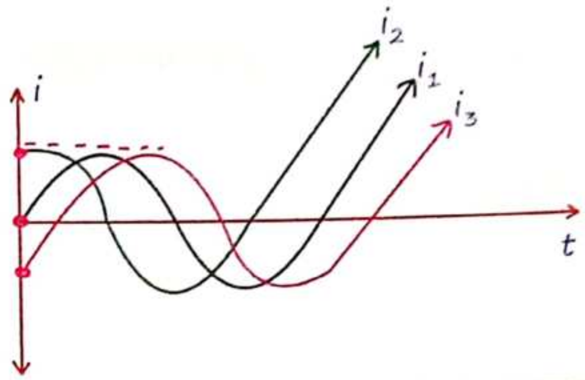
$i_1 = i_0\sin\omega t$

$i_2 = i_0\sin(\omega t + \phi)$

$i_3 = i_0\cos(\omega t - \delta)$



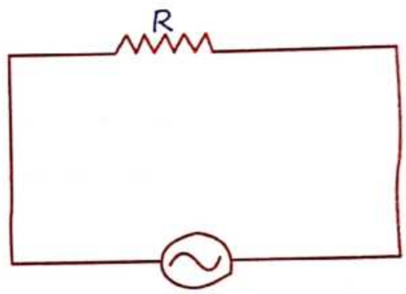
MR*



Joh jitna niche y-axis pe wo utna piche.

$i_3 < i_1 < i_2$

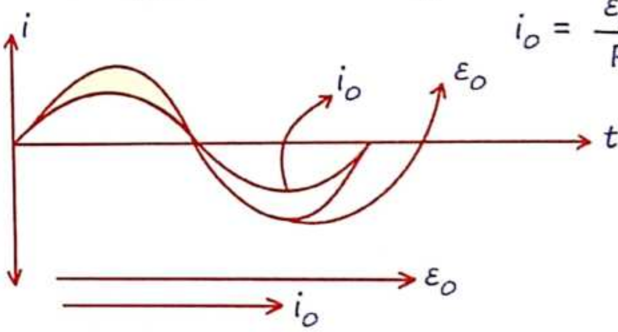
3 A/C Source Across Pure Resistance



Same phase

$\varepsilon = \varepsilon_0 \sin\omega t$ $\qquad$ $i = i_0 \sin\omega t$

$$i_0 = \frac{\varepsilon_0}{R}$$



+ Power factor, $\cos\phi = 1$

Power drop across R :-

$$\langle P \rangle = \varepsilon_{rms} i_{rms} = \frac{\varepsilon_{rms}^2}{R} = i_{rms}^2 R$$

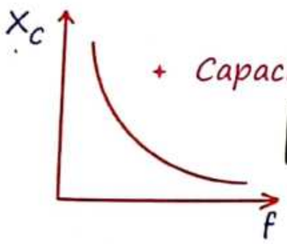
4 A/C Source Across Pure Capacitor

- $\varepsilon = \varepsilon_0 \sin\omega t$ — "i" leads "V"
- $\varepsilon = \varepsilon_0 \sin\omega t$

by $\pi/2$ i.e., $T/4$

- $i = \dfrac{\varepsilon_0}{1/C\omega}\cos\omega t = \dfrac{\varepsilon_0}{X_C}\cos\omega t$
- $i_0 = \dfrac{\varepsilon_0}{X_C}$
- $V_C = iX_C$

$$X_C = \frac{1}{C\omega} = \frac{1}{2\pi fC}\Bigg\}\ \text{Capacitive Reactance}$$



- Capacitor loves change.

$\boxed{\text{DC – Filter}}$

Capacitor Behaviour in D.C. and A.C. Circuit :-

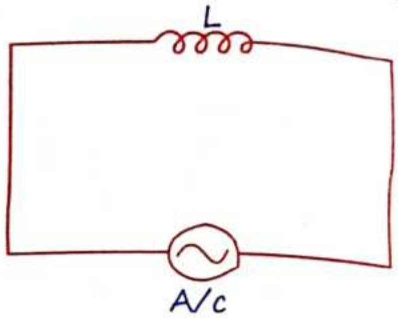
Parameter	D.C. (Direct Current)	Alternating current (A.C)
Frequency (f)	$f = 0$	$f = \infty$
Reactance (X_C)	$X_C = \infty$	$X_c = 0$
Behaviour	Acts as an open circle (no current flow)	Acts as a short circuit (current flow easily)

- Power drop across pure Capacitor :-

$$\cos\phi = 0$$

$\langle P \rangle = 0$ $\boxed{\text{Wattless Circuit}}$

5 A/C Source Across Pure Inductor



$V_L = \varepsilon_0 \sin\omega t$

"i" lags "V" by $\pi/2$ i.e., $T/4$

$$i = \frac{-\varepsilon_0}{\omega L}\cos\omega t = \frac{-\varepsilon_0}{X_L}\cos\omega t$$

- $i_0 = \dfrac{\varepsilon_0}{\omega L}$
- $V_L = iX_L$

$$X_L = \omega L \Big\}\ \text{Inductive Reactance}$$

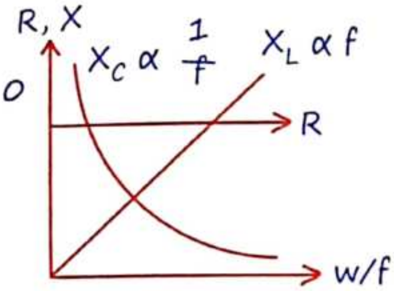
Parameter	D.C. (Direct Current)	Alternating current (A.C)
Frequency (f)	$f = 0$	$f = \infty$
Reactance (X_L)	$X_L = 2\pi fL = 0$	$X_L = \infty$
Behaviour	Acts as a short circuit (simple wire)	Acts as a open circle (no current flow)

For D/c	For A/c of high frequency
$f = 0$ $\Rightarrow$	$f = \infty \Rightarrow X_L = \infty$
$\Rightarrow$ Hence inductor be have as simple wire.	$\Rightarrow$ Hence inductor behave as open wire.

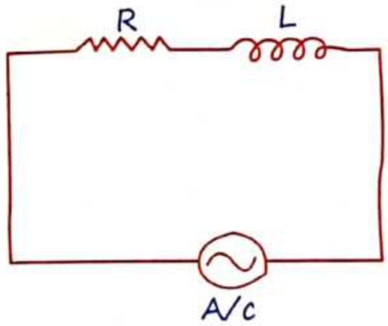
- Power loss across pure inductor

$\langle P \rangle = 0$

$\cos\phi = 0$

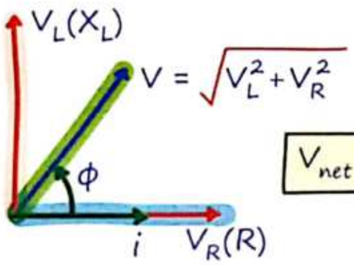


6 Series R-L Circuit



$V_{net} = V_B = \varepsilon_0 \sin\omega t$

$i = i_0 \sin(\omega t - \phi)$



V_{net} leads i by ϕ

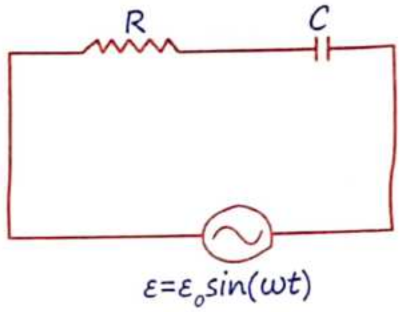
- Impedence $Z = \sqrt{R^2 + X_L^2}$
- $\tan\phi = \frac{V_L}{V_R} = \frac{X_L}{R}$
- $\cos\phi = \frac{V_R}{V_{net}} = \frac{R}{Z}$ } Power Factor

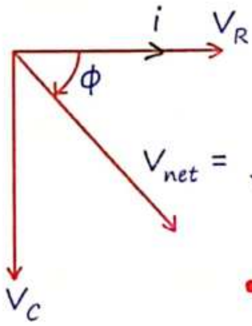
ϕ = Phase difference.

7 Series R-C Circuit

$i = i_0 \sin(wt + \phi)$

$i_0 = \frac{\varepsilon_0}{Z}$





V_{net} lags "i" by ϕ

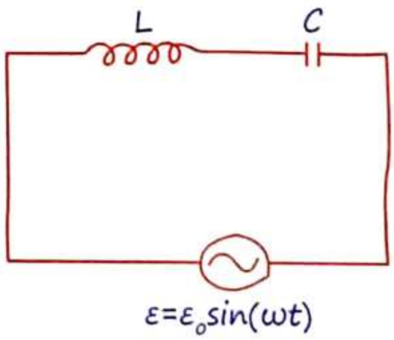
$V_{net} = \sqrt{V_R^2 + V_C^2}$

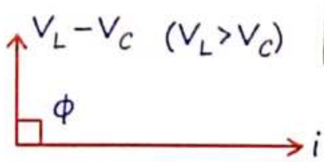
- $i_0 = \frac{\varepsilon_0}{Z}$
- $Z = \sqrt{R^2 + X_C^2}$

$\tan\phi = \frac{V_C}{V_R} = \frac{X_C}{R}$

$\cos\phi = \frac{V_R}{V_{net}} = \frac{R}{Z}$

8 Series L-C Circuit



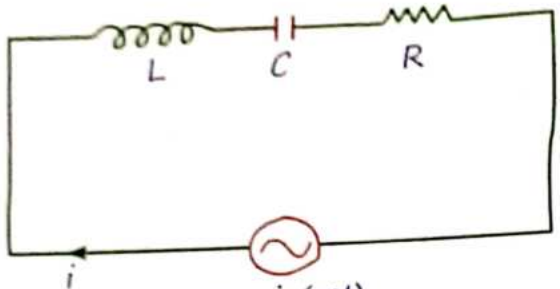


V_{net} leads i by ϕ

Net impedence :-

$Z = X_L - X_C \qquad i_0 = \varepsilon_0/Z$

9 Series LCR Circuit



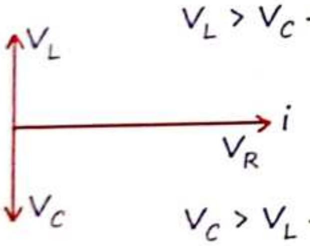
$\varepsilon_t = \varepsilon_0 \sin(\omega t)$

$i = i_L = i_C = i_R$ — Instantaneous current

$\varepsilon_t = V_L + V_C + V_R$ — Instantaneous voltage

$i_0 \neq i_{0L} \neq i_{0C} \neq i_{0R}$ — Peak Current

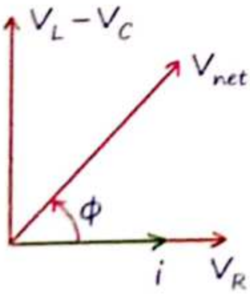
$\varepsilon_0 = V_{0L} + V_{0C} + V_{0R}$ — Peak voltage



$V_L > V_C \longrightarrow$ Inductive LCR

$V_C > V_L \longrightarrow$ Capacitive LCR

- If $V_L > V_C$ (Inductive LCR)



$V_{net} = \sqrt{(V_L - V_C)^2 + V_R^2}$

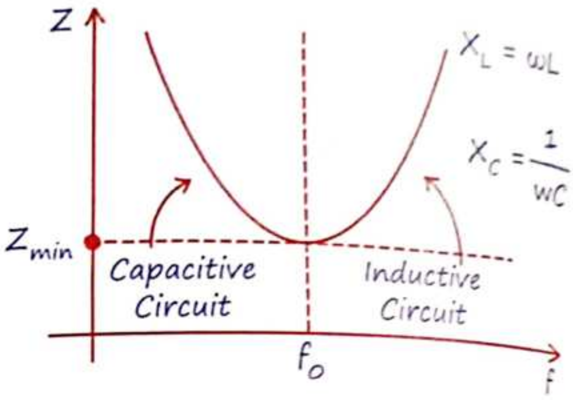
$\boxed{V_{net} \text{ leads } i \text{ by } \phi}$

- $\tan\phi = \dfrac{V_L - V_C}{V_R} = \dfrac{X_L - X_C}{R}$

- $\cos\phi = \dfrac{V_R}{V_{net}} = \dfrac{R}{Z}$

- $Z = \sqrt{(X_L - X_C)^2 + R^2}$

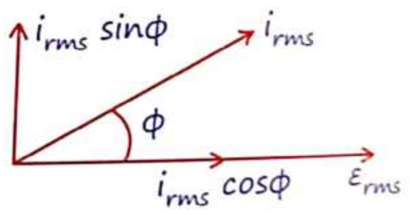
- $i_0 = \dfrac{\varepsilon_0}{Z}$



If Frequency increases from zero then impedence (Z) 1st decrease then increase

- $f = 0 \quad Z = \infty \qquad X_C = \infty \quad X_L = 0$
- $f = \infty \quad Z = \infty \qquad X_C = 0 \quad X_L = \infty$

Power drop :-



Vimp

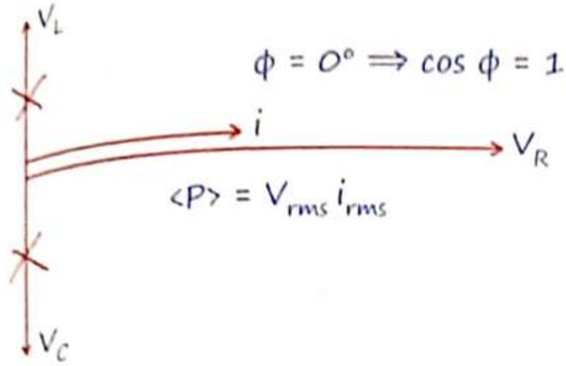
- $\langle P \rangle = \varepsilon_{rms}\, i_{rms} \cos\phi$

Resonance in series LCR circuit :-

$X_L = X_C, \quad V_L = V_C$

$\omega L = \dfrac{1}{\omega C} \Rightarrow \omega = \dfrac{1}{\sqrt{LC}} \Rightarrow f = \dfrac{1}{2\pi\sqrt{LC}}$

$Z_{min} = R \Rightarrow V_{net} = V_R$

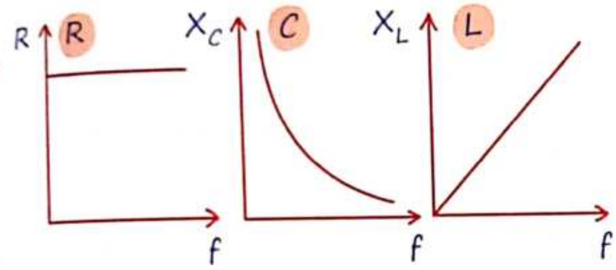


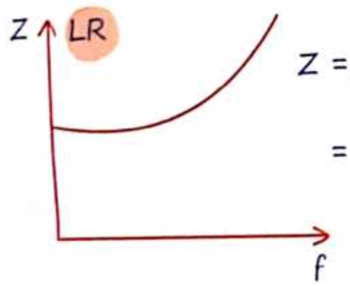
$\phi = 0° \Rightarrow \cos\phi = 1$

$\langle P \rangle = V_{rms}\, i_{rms}$

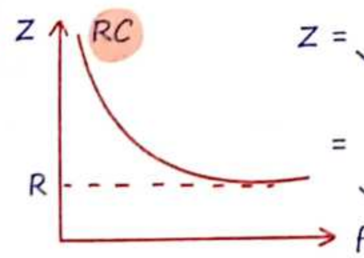
$i_{max} = \dfrac{\varepsilon}{R}$ $\qquad P_{max} = \dfrac{\varepsilon_0^2}{2R}$

Imp Graph :- MR*



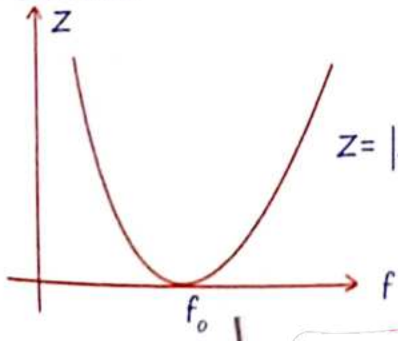


$$Z = \sqrt{X_L^2 + R^2}$$
$$= \sqrt{(2\pi f L)^2 + R^2}$$



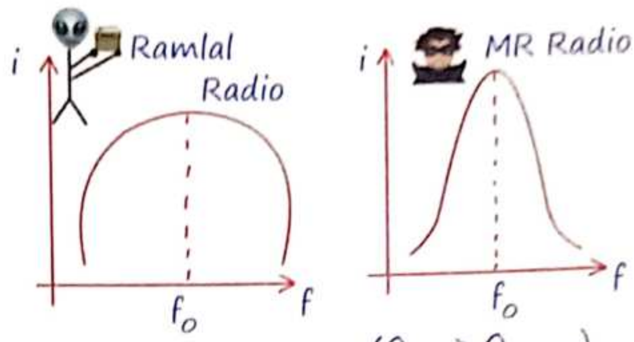
$$Z = \sqrt{X_C^2 + R^2}$$
$$= \sqrt{\frac{1}{(2\pi f C)^2} + R^2}$$

- L-C circuit



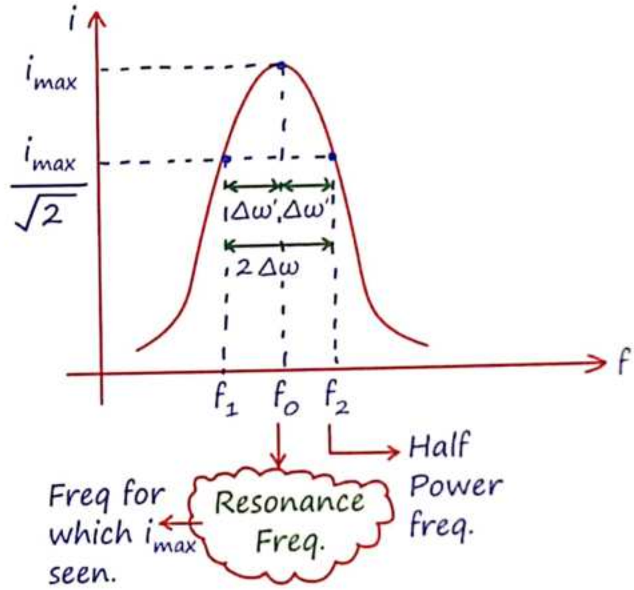
$$Z = |X_L - X_C|$$

Quality Factor :- (Q)



$(Q_{MR} > Q_{Ramlal})$

The sharper peak of MR Radio indicates a higher Q factor compared to Ramlal Radio. This means MR Radio has a better quality resonance.



$$Q = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0}{2\pi(f_2 - f_1)}$$

$$Q = \frac{\omega_0}{2\Delta\omega} = \frac{\omega_0}{2[2\pi(f_0 - f_1)}$$

$$2\Delta\omega = \frac{R}{L}$$

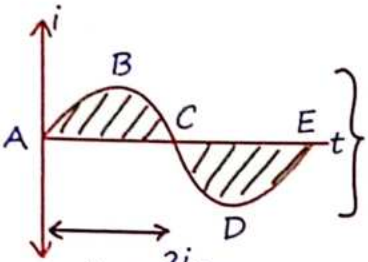
$$Q = \frac{\omega_0}{2\Delta\omega} = \frac{\omega_0 L}{R} = \frac{X_L}{R} = \frac{X_C}{R}$$

$$Q = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0}{R/L} = \frac{\omega_0 L}{R} = \frac{X_L}{R}$$

$$Q = \frac{X_L}{R} = \frac{X_C}{R}$$

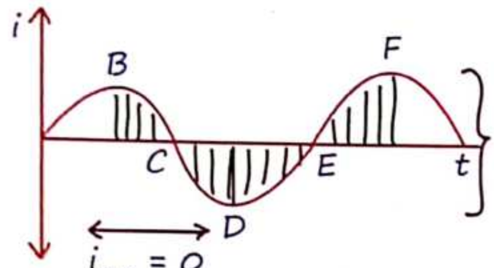
$$Q = \frac{1}{\sqrt{LC}}\frac{L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

MR*



A–B–C Half Cycle

$i_{avg} = \frac{2i_0}{\pi}$



B–C–D Half Cycle

$i_{avg} = 0$

Transformer :-

- Voltage Regulator
- Based on principle of Mutual Inductance.

*P = constⁿ $\qquad$ iV = constⁿ

*$i \propto \frac{i}{V} \propto \frac{i}{N}$ $\qquad$ *N = No. of turns.

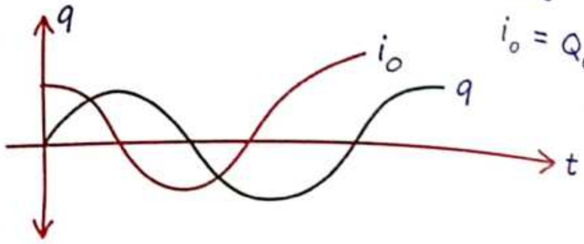
* $\boxed{\frac{i_0}{i_{in}} = \frac{V_{in}}{V_{out}} = \frac{N_P}{N_S}}$

Step-UP	Step-DOWN
$V_o > V_{in}$	$V_o < V_{in}$
$N_S > N_P$	$N_S < N_P$
$i_S < i_P$	$i_S > i_P$

L-C Oscillations :-

$q = Q_0 \sin\omega t$ $\qquad$ $i = i_0 \cos\omega t$

$i_0 = Q_0\omega$



$$\frac{d^2q}{dt^2} + \frac{q}{LC} = 0 \Rightarrow \omega = \sqrt{\frac{1}{LC}}$$

$$U_E = \frac{q^2}{2C} = \frac{Q_0^2 \sin^2\omega t}{2C}$$

$$U_B = \frac{1}{2}Li^2 = \frac{1}{2}Li_0^2\cos^2\omega t$$

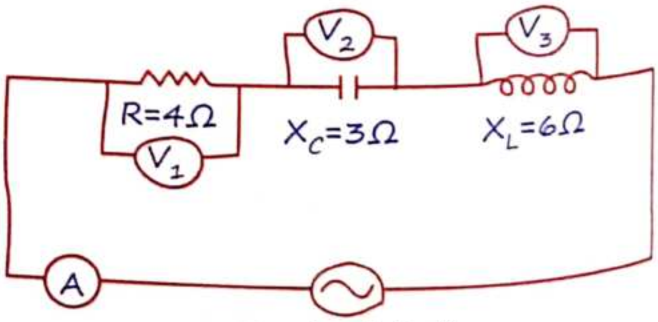
$$\boxed{TE = \frac{Q_0^2}{2C} = \frac{1}{2}Li_0^2 = const^n}$$

Choke Coil :-

- Control Current in A/C Circuit
- Divide Potential without Power loss.

(X_L, R) Choke Coil. → RL Circuit

Ideal	Practical	बेकार
$R = 0$	$R = Low$	$R = High$
$X_L \neq 0$	$X_L = High$	$X_L = Low$
$\phi = 90°$	$\phi < 90°$	$\phi = Low$
$\cos\phi = 0$	$\cos\phi = V.Low$	$\cos\phi = High$



$\varepsilon = 10\sin(\omega t)$

Physics

- $Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{4^2 + (6-3)^2}$
- $Z = 5\Omega$
- $i_0 = \frac{\varepsilon_0}{Z} = \frac{10}{5} = 2$ Amp
- i_{RMS} (Reading of Ammetre) $= \frac{i_0}{\sqrt{2}} = \sqrt{2}$
- Power factor $\cos\phi = \frac{R}{Z} = \frac{4}{5}$
  ($\phi = 37°$)
- $X_L > X_C \rightarrow$ Inductive circuit
  Voltage will leads by current by 37°
- $i = i_0 \sin(\omega t - \phi)$
  $i = 2 \sin(\omega t - 37°)$

Reading of $V_1 \Rightarrow \quad V_1 = i_{Rms} R = \sqrt{2}(4)$

Reading of $V_2 \Rightarrow \quad V_2 = i_{Rms} X_C = 3\sqrt{2}$

Reading of $V_3 \Rightarrow \quad V_3 = i_{Rms} X_L = 6\sqrt{2}$

RMS voltage across 'R' and 'C'

$$= \sqrt{(4\sqrt{2})^2 + (3\sqrt{2})^2} = 5\sqrt{2}$$

RMS voltage across 'C' and 'L'

$$= 6\sqrt{2} - 3\sqrt{2} = 3\sqrt{2}$$

RMS voltage across 'R' and 'L'

$$= \sqrt{(6\sqrt{2})^2 + (4\sqrt{2})^2} = \sqrt{104}$$

$\langle P \rangle = i_{rms} V_{rms} \cos\phi$

$$= \sqrt{2} \times \frac{10}{\sqrt{2}} \times \frac{4}{5} = 8W$$

The Ultimate MR Star* Table

Circuit	Phase diff. Between i&V	Power factor $\cos\phi$ = R/Z	Impedance (Z)	Who leads!	Power loss
Pure resistive	0	1	R	Same Phase	$P = i_{rms}V_{rms}$
Pure Capacitive	$\pi/2$	Zero	$X_C = \frac{1}{\omega C}$	Current	Zero
Pure inductive	$\pi/2$	Zero	$Z_L = \omega L$	Voltage	Zero
RL	$0 < \phi < \frac{\pi}{2}$	b/n 0 to 1	$Z = \sqrt{R^2 + X_L^2}$	Voltage	$P = i_{rms}V_{rms}\frac{R}{Z}$
RC	$0 < \phi < \frac{\pi}{2}$	b/n 0 to 1	$Z = \sqrt{R^2 + X_C^2}$	Current	$P = i_{rms}V_{rms}\cos\phi$
LC	$\pi/2$	Zero	$Z = X_L - X_C$	Depends	Zero
Series LCR	$0 \le \phi < \frac{\pi}{2}$	1 or b/n 0 to 1	$Z = \sqrt{R^2 + (X_L - X_C)^2}$	Depends	$P = i_{rms}^2 R = \frac{V_{rms}^2}{R}$

ELECTRIC OSCILLATION

EQUIVALENT OF L.C. OSCILLATION :-

Mechanical system	Electrical system
Mass	Inductance
Displacement	Charge
Velocity	Current
Potential Energy	Electric Energy
Kinetic Energy	Magnetic Energy
$\frac{1}{K}$(K=Spring const.)	Capacitance

Q.1 In L.C. Oscillation Maximum Current is Io at t=o then find current in circuit when magnetic energy is half of electrical energy.

Sol. Total Energy Conserved

$$\frac{1}{2} L I_0^2 = Ue + Um$$

$$\frac{1}{2} L I_0^2 = 2Um + Um$$

$$\frac{1}{2} L I_0^2 = 3\frac{1}{2} LI^2$$

$$I = \frac{I_0}{\sqrt{3}}$$

Q.2 The number of turns in the primary and secondary coils of a step-down transformer are 200 and 50 respectively. If the power in the input is 100 watt at 1A then the output power and current will respectively be

Sol. Power in an transformer (ideal) does not change, the total electric power remains same.

Power = 100 Watt

for current, the proportionality is

$$\frac{N_1}{N_2} = \frac{V_1}{V_2} = \frac{I_2}{I_1}$$

$$\frac{200}{50} = \frac{I_2}{1}$$

$$I_2 = \frac{200}{50} = 4Amp.$$

MR*

"Aaj kuch karlo asa ki bhavishya me khud ko kosne ki nobat na aaye."

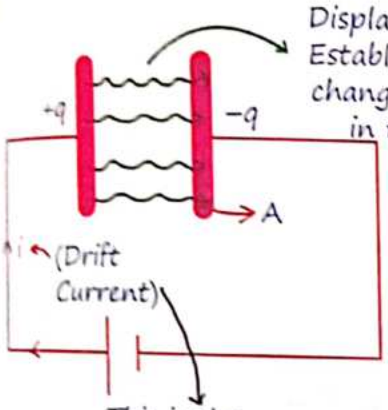
29 Electromagnetic Wave

1 Behaviour of charges in Different conditions

Charge	E.F.	MF	EM Wave
Rest	✓	✗	✗
V = Const^n	✓	✓	✗
Accelerated	✓	✓	✓

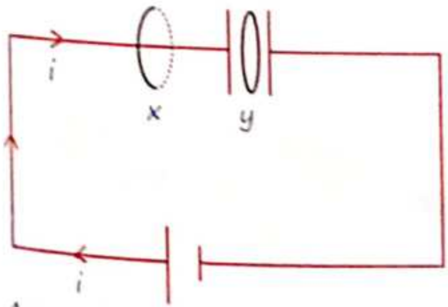
2 Charging of Capacitor

Displacement Current Established due to the changing electric field in the capacitor.



This is the conventional current flowing through the circuit, causing the capacitor to charge.

$$E = \frac{\sigma}{\varepsilon_0} = \frac{q}{A\varepsilon_0}$$



According to amperes law

for x: $\int \vec{B}_x \cdot d\vec{\ell} = \mu_0 i$

for y: $\int \vec{B}_y d\vec{\ell} = \mu_0 i \to 0$

hence $B_y = 0$

But maxwell found experimentally,

$B_y \neq 0$

and gives reason of this is Displacement current b/w the plate of capacitor.

Displacement Current :-

$Q = CV$ $\boxed{i_{Drift} = i_{Displacement}}$

$$i = \frac{dQ}{dt} = C \cdot \frac{dV}{dt} = \frac{\varepsilon_0 A}{\ell} \frac{dV}{dt}$$

ℓ = Distance between the plates

$$i = \varepsilon_0 A \frac{dE}{dt} = \varepsilon_0 \frac{d\phi}{dt}$$

- Formed only due to changing E. Field.
- Not exist under steady current (ϕ = constant)
- Flows between c/s Area of Capacitor plate.

3 Maxwell Equation (4 equations)

Gauss law of Electrostat.

$$\phi = \oint \vec{E} \cdot \vec{ds} = \frac{q_{in}}{\varepsilon_0}$$

- Electric fields are generated by electric charges.

Gauss law of Magnetism.

$$\oint \vec{B} \cdot \vec{ds} = 0$$

- Magnetic fields have no beginning or end (no monopoles)

Electromagnetic Wave

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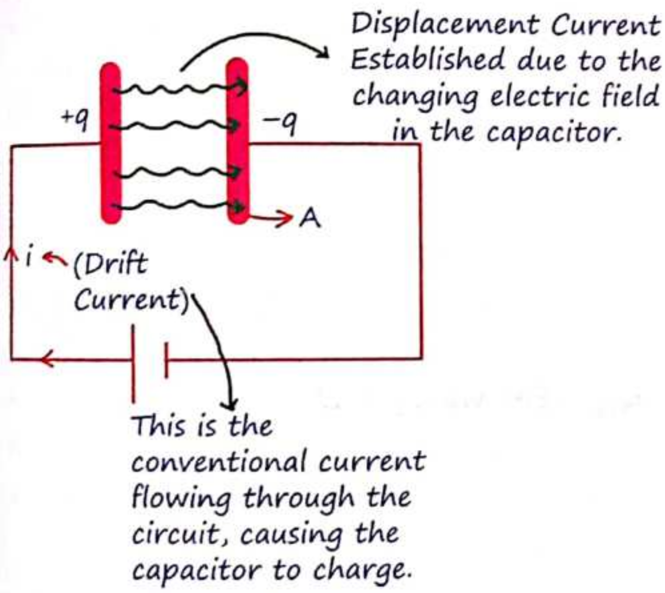

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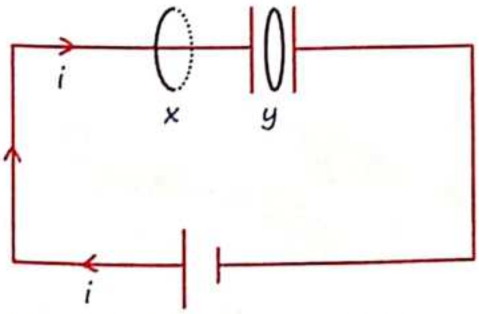

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Induced E. Field :- Faraday's Law of induction

$$-\oint \vec{E}_{in} \cdot \vec{d\ell} = EMF = -\frac{d\phi}{dt}$$

$$\oint \vec{E}_{in} \cdot \vec{d\ell} = \frac{d\phi}{dt}$$

- A changing magnetic field creates an electric field.

Ampere's Maxwell Law :-

$$\oint \vec{B} \cdot \vec{d\ell} = \mu_o [i_{Drift} + i_{Disp}]$$

$$= \mu_o \left[i_{Drift} + \varepsilon_o \frac{d\phi}{dt}\right]$$

- Electric currents and changing electric fields create magnetic fields.

4 Source of EMW

Accelerating Charge :-

When a charged particle accelerates, it emits electromagnetic waves. This is because the changing velocity of the charge causes disturbances in the electric and magnetic fields, which propagate as waves.

Example:

- An electron in an antenna oscillates back and forth, producing radio waves.

Transition of Electrons from n^{th} orbit :-

When an electron transitions from a higher energy orbit to a lower one in an atom, it releases energy in the form of electromagnetic radiation.

Example:

- The emission of light from hydrogen atoms when electrons move between energy levels.

Retardation of Electrons in High Atomic Weight Targets (X-rays) :-

When high-speed electrons are suddenly decelerated upon hitting a target of high atomic weight, X-rays are produced. This is known as Bremsstrahlung radiation.

Example:

- X-ray machines in medical imaging use this principle to produce X-rays by bombarding a metal target with electrons.

De-excitation of Nucleus in Radioactivity (γ-rays) :-

When the nucleus of a radioactive atom de-excites, it releases energy in the form of gamma rays. These are high-energy electromagnetic waves.

Example:

- Gamma rays emitted from radioactive materials used in cancer treatment (radiotherapy).

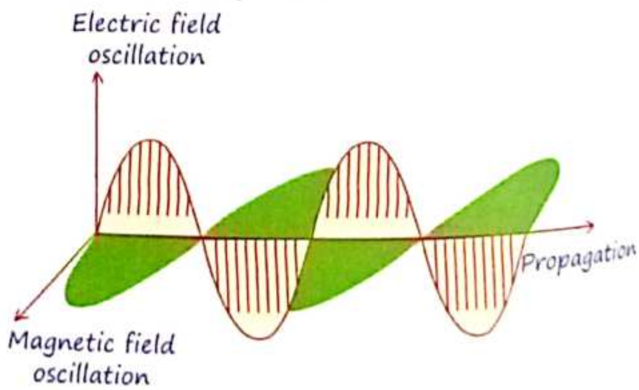
EM Wave in LC Oscillation :-

- LC Oscillation: In an LC circuit (inductor-capacitor circuit), the energy oscillates between the electric field of the capacitor and the magnetic field of the inductor. This oscillation produces electromagnetic waves.

Example:

Radio transmitters use LC circuits to generate specific frequencies of radio waves.

- No Medium required, Non-mechanical wave.
- Q_{net} [EM wave] = 0: This is because the wave consists of oscillating electric and magnetic fields, not actual moving charges.
- Transverse in nature
- E&B oscillate perpendicular to each other but in same phase.



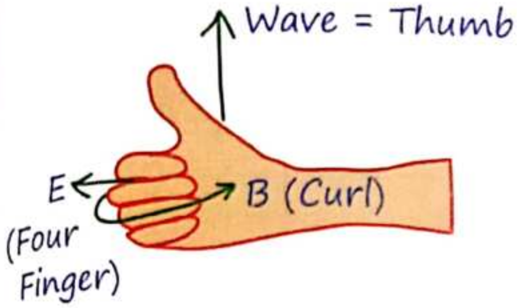
$E_y = E_0 \sin\omega t$
$B_z = B_0 \sin\omega t$

$E_0 = Q_0/A\varepsilon_0$

$\boxed{E_0 = B_0 C}$

E.F. & Mag. F are in Same phase

Direction of Wave :- $\vec{E} \times \vec{B}$



MR* for direction

Direction of wave = $\hat{E} \times \hat{B}$

Place Your four finger (with Palm) of right hand along electric field and Slap magnetic field. thumb will Indicate direction of wave.

5 Nature of EMW

Speed of EM-wave in Vaccum:

$$C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8 \text{ m/s}$$

Speed of EM-wave in a medium:

$$V = \frac{1}{\sqrt{\mu_m \varepsilon_m}} = \frac{C}{\sqrt{\mu_r \varepsilon_r}}$$

$$\boxed{V = \frac{C}{\mu}}$$

μ = Mag. Permeability

μ → Refractive index

ε = Electric Permittivity.

- Angular Frequency :-

$$\omega = \frac{2\pi}{T} = 2\pi f$$

- Angular wave no :-

$$K = \frac{2\pi}{\lambda} = \frac{2\pi\nu}{C}$$

- Speed of EMW :-

$$\boxed{C = \lambda\nu = \frac{\lambda}{T} = \frac{\omega}{K}}$$

If Medium is Changed !

λ = Changes

f = Same.

$C_1 = \lambda_1 \nu$

6 Energy Density

- EM Wave (U) :-

$$(U) = \varepsilon_0 E^2 = \frac{B^2}{\mu_0}$$

$$U_{avg} = \langle U \rangle = \frac{\varepsilon_0 E_0^2}{2} = \frac{B_0^2}{2\mu_0}$$

- Electrostatic (U_E) :-

$$U_E = \frac{1}{2}\varepsilon_0 E^2$$

$$\langle U_E \rangle_{avg} = \frac{1}{4}\varepsilon_0 E_0^2$$

- Magnetic (U_B) :-

$$U_B = B^2/2\mu_0$$

$$\langle U_V \rangle_{avg} = B_0^2/4\mu_0$$

7 EM-Waves

The electromagnetic (EM) spectrum encompasses all types of electromagnetic radiation, from gamma rays to radio waves. Gaddi SUV in My Range

Energy

Gaddi SUV In My Range

- Gamma Ray (γ-ray) (1 MeV)
- X-ray (1 KeV)
- Ultraviolet (UV) (10-100 eV)
- Visible (1-2 eV)
- Infrared (IR) (1 meV)
- Microwave (1 μeV)
- Radiowave (10^{-8} eV)

E $\downarrow$ (Decreases)

f $\downarrow$ (Decreases)

λ $\uparrow$ (increases)

low energy

Intensity of EM Wave :-

The intensity of an electromagnetic wave represents the power per unit area carried by the wave.

$$\boxed{\langle I \rangle = \frac{1}{2}\varepsilon_0 E_0^2 C = \frac{1}{2}\frac{B_0 E_0}{\mu_0}}$$

8 Force & Radiation Pressure

Surface

Completely reflecting	Completely absorbing
Momentum :-	
$\lvert\Delta\vec{P}\rvert = 2P = \frac{2E}{C}$	$\lvert\Delta\vec{P}\rvert = P = \frac{E}{C}$
Force :-	
$F = \frac{\Delta P}{t} = \frac{2E}{Ct}$	$F = \frac{\Delta P}{t} = \frac{E}{Ct}$
Pressure :-	
$P = \frac{F}{A} = \frac{2E}{CAt}$	$P = \frac{F}{A} = \frac{E}{CAt}$
$P = \frac{2I}{C}$	$P = \frac{I}{C}$
I = Intensity	$\mu_r = \frac{\mu_m}{\mu_0}$
E = Energy	$\varepsilon_r = \frac{\varepsilon_m}{\varepsilon_0}$ = Dielec. Const^n

9 Poynting Vector

Direction :- II^{el} to wave & $\perp$ to electric field and the magnetic field.

Magnitude :- Intensity of EM Wave.

$$\boxed{\vec{S} = \frac{\vec{E_0} \times \vec{B_0}}{2\mu_0}}$$

Type of Radiation	Frequency Range (Hz)	Wavelength Range
gamma-rays	$10^{20} - 10^{24}$	$< 10^{-12}$ m
X-rays	$10^{17} - 10^{20}$	0.01 nm – 10 nm
ultraviolet	$10^{15} - 10^{17}$	400 nm – 1 pm
visible	$4 - 7.5 \times 10^{14}$	750 nm – 400 nm
near-infrared	$1 \times 10^{14} - 4 \times 10^{14}$	2.5 μm – 750 nm
infrared	$10^{13} - 10^{14}$	25 μm – 2.5 μm
microwaves	$3 \times 10^{11} - 10^{13}$	1 mm – 25 μm
radio waves	$< 3 \times 10^{11}$	> 1 mm

	Radiowaves	Microwaves	IR	Visible	UV	X-Ray	Gamma-Rays
Production	Accl^n of e^- in Antenna	Klystron, Magneton & Gunn Diode	Heated Bodies	e^- Excitation	Transfer of e^- from E.S. to G.S.	X-ray Tube Innner Shell e^-	Radioactive Nucleus Nuclear Explosion
Detection	Receiver's Antenna.	Point Contact Diode	Bolometer, IR Photographic Plate, Photodiode	Human eye, Photographic Films, Photocells	Diodes & Photographic Films.	Gieger Counter Ionisat^n Chamber Photographic Plates	Gieger Counter Ionisat^n Chamber Photographic Plates
Uses	FM, AM, TV, Cellular Network	Radar Navigation, Measuring Speed of balls, Microwave Oven (3GHz, Reso. of H_2O).	Remotes, Hi Fi System	To See Objects	UV Fitter, Lasik Laser, Sanitization	Diagnostic tool, Radio-therapy.	Radio therapy High level Sanitization.

MR*

"बदल जाओ वक्त के साथ,
या फिर वक्त बदलना सीखो।
मजबूरियों को मत कोसो,
हर हाल में चलना सीखो।।"

1 Light

- It is a form of electromagnetic energy which gives the Sensation of vision.
- It is an essential part of the electromagnetic spectrum that is visible to the human eye.
- Light itself is not visible; rather we see objects because they reflect or emit light.
- Visible light has a wavelength range of approximately 380 nm to 700 nm, which corresponds to the portion of the electromagnetic spectrum that human eyes can detect."
- [V I B G Y O R] : visible light Ka range

  The visible light spectrum includes colors in the following order: Violet (V), Indigo (I), Blue (B), Green (G), Yellow (Y), Orange (O), and Red (R)."
- Speed of light does not depends on Speed of Source and speed of observer.
- Frequency and sense of colour does not depends on medium.
- Intensity, wavelength, speed of light depends on medium.

For example, light travels slower in water than in air."

Speed of Light in Vaccum

Speed of light in medium,

$$C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}, V = \frac{1}{\sqrt{\mu_m \varepsilon_m}}$$

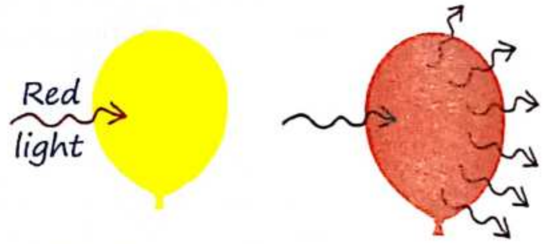
$$V = \frac{C}{\sqrt{\mu_r \varepsilon_r}} = \frac{C}{\mu}$$

$$\therefore \mu_1 V_1 = \mu_2 V_2$$

$$V = \lambda f, \ \lambda \propto \frac{1}{\mu} \propto V$$

MR*

Koi object jis light Ko emit Karega woh waisa dhikega.



Heat up

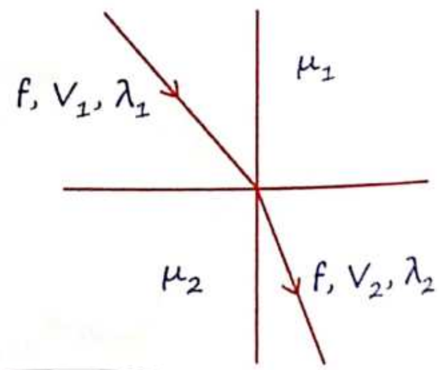
$\therefore$ Yellow balloon will burst.

2 Plane Mirror

Effect of Reflection & Refraction :-

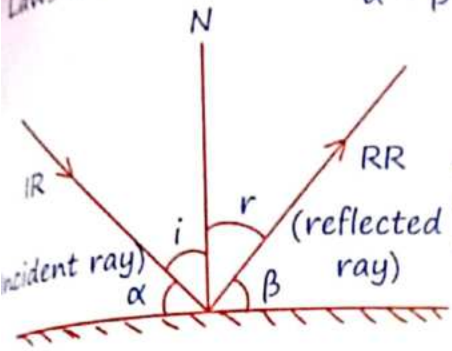
Frequency of light

f = Medium independent, only depend on Source



flection :-

Laws of reflections :- $\angle i = \angle r$
$\alpha = \beta$



Angle b/n IR & RR is always $2i$.

viation from Plane Mirror :-

a) By a Mirror :-

$\delta = 180 - 2i$

Normal Incidence	Grazing Incidence
$\delta = 180°$	$\delta \approx 0$

b) By two inclined Mirror :-

$\delta = 360 - 2\theta$

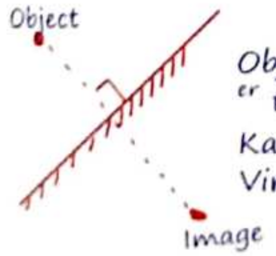
θ = Angle b/n Mirror.

Note :-

- If incident ray rotated "θ" then reflected ray rotates "θ" in Opposite Sense.
- Plane Mirror is rotated by "θ", reflected ray turns by "2θ" in same sense.
- If plane minor is rotated by θ about axis perpendicular to plane then No effect on reflected ray.

Image Formation by Plane Mirror :-

MR•



Object से ek line ⊥ er to Mirror draw Karo उसी line पर Virtual image hogi.

Some Properties of Image formation by Plane mirror :-

- Focal length of Plane mirror is Infinite
- On covering part of mirror, No change in size of Image but brightness will decrease because less light is reflected."
- Real Object → Virtual Image: A plane mirror always forms a virtual image of a real object. The image appears to be behind the mirror and cannot be projected onto a screen.
- Virtual object → Real Image: A plane mirror can form a real image if the object is virtual. This occurs in certain optical setups, such as with converging lenses or other mirrors that create a virtual object.
- Object and Image always of same distance from mirror.
- Lateraly inverted image: The left and right sides of the image are reversed.
- Plane mirror can form Inverted Image of real object

Height of Mirror :-

(A) To see full height of object :-

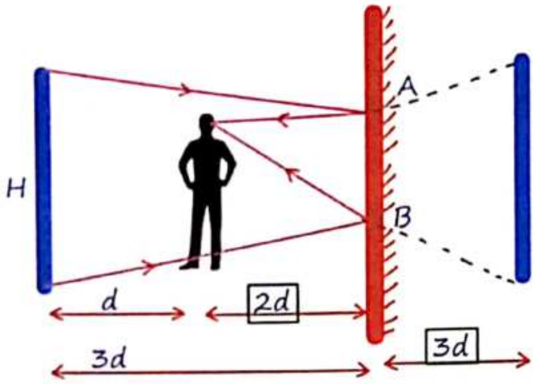
$H_{mirror} = H_{object}/2.$

(B) To see full wall behind object ?

(Observer at Centre).

$$H_{mirror} = \frac{H_{wall}}{3}$$

(C) To see full wall behind object if man is not centre ?



$$\frac{AB}{2d} = \frac{H}{5d} \qquad H_{mirror} = \frac{2H}{5}$$

Clock System :-

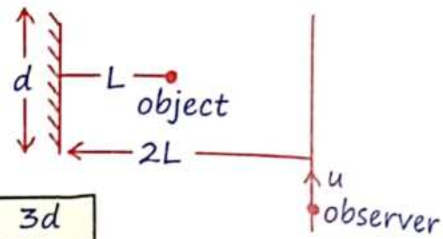
- The reflection inverts the left and right sides, effectively reversing the direction of the clock's hands." therefore the time shown in the image of a clock reflected in a plane mirror can be given as follow.
    - If time in clock → $(H_{hr} : M_{mint} : S_{sec})$ then mirrored time will be → $(11-H)_{hr} : (59-M)_{mint} : (60-S)_{sec}$
    - When only HR and MINT is given $= (11-H)_{hr} : (60-M)_{mint}$
    - If only HR is given mirror time $= (12-H)_{Hour}$

3 Velocity of Image

Object moving $\parallel^{el}$ to plane mirror :-

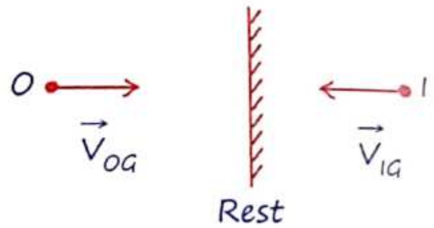
- Velocity of image w.r.t object, $\vec{V}_{IO} = 0$
- Velocity of object w.r.t. ground $\vec{V}_{OG}$ = Velocity of image w.r.t. ground $\vec{V}_{IG}$

- Observer is moving with constant speed u then time for which observer Can See image of object



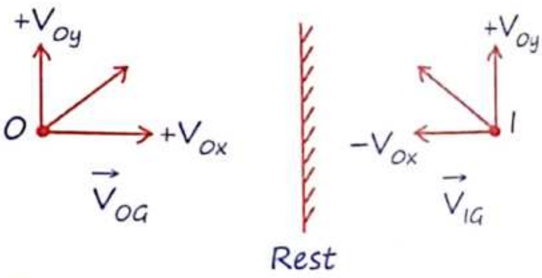
$$\boxed{t = \frac{3d}{u}}$$

Object is moving $\perp^{er}$ to plane mirror :-



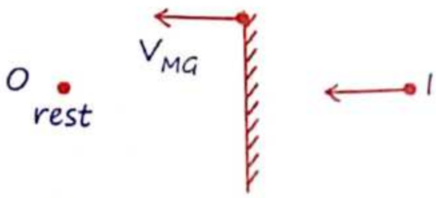
$$\vec{V}_{IG} = -\vec{V}_{OG}$$
$$\vec{V}_{IO} = -2\vec{V}_{OG}$$

Object is moving at an angle to mirror :-



$$^{*}V_{IO} = -2V_{Ox}$$

Object at rest & Mirror is moving with V_{MG}:-



$$\vec{V}_{IO} = \vec{V}_{IG} = 2\vec{V}_{MG}$$

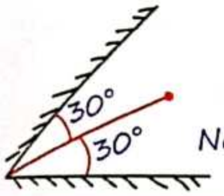
Image Formation by Two Plane Mirror :-

$$m = \frac{360}{\theta}$$ → θ = Angle b/n Mirror

- Even → No. of images formed. $n = m-1$
- Odd
  - Bisector ✓ → If the object is placed on the angle bisector, the number of images formed. $n = m-1$
  - Bisector ✗ → If the object is not placed on the angle bisector, the number of images formed. $n = m$

MR*

- $m = \frac{360°}{\theta}$ fraction then ush fraction Se choti integer lenge approximate Nahi lena,
- Ex $M = 7.8 \rightarrow m = 7$ Image



$$m = \frac{360°}{60°} = 6 \text{ (even)}$$

No of images $= m - 1 = 5$

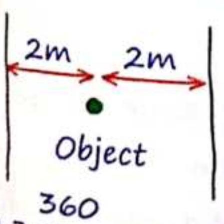
M_1	M_2
30°	30°
	+60°
90°	90°
	+60°
150°	150°

Pahila image Kaha banega

Angle b/n two mirror add Karo.

Ye coincide Karega.

Infinite Image When Two Plane Mirror is Parallel :-



$$m = \frac{360}{0} \Rightarrow n = \infty$$

Add dist^n b/n two mirror.

M_1	M_2
2m	2m
	+4m
6m	6m
10m	10m
14m	14m
⋮	⋮

4 Spherical Mirror

Property	Concave Mirror	Convex Mirror
Nature of mirror	Converging	Diverging
Focal length (f)	Negative ($f = -ve$)	Positive ($f = +ve$)
Object distance (U)	Negative ($U = -ve$) (Real object)	Negative ($U = -ve$) (Real object)
Image distance (V)	Negative ($V = -ve$) (Real Image)	Positive ($V = +ve$) (Virtual Image)

- Along Incident ray distance taken as +ve.

Mirror Equation :-

- $f = \frac{R}{2}$
- $\frac{1}{V} + \frac{1}{U} = \frac{1}{f}$

Transverse magnification

$$m_T = \frac{H_I}{H_O} = -\frac{V}{U} = \frac{f}{f-U} = \frac{f-V}{f}$$

$$m_L = \frac{-dV}{dU} = m_T^2 = \frac{V^2}{U^2}$$

* only valid for small object

Longitudinal Magnification

$$m = -\frac{V}{U}$$

$m = +ve$

- > 1 = Erect & Magnified
- < 1 = Erect & Diminished.

$m = -ve$

- > 1 = Inverted Magnified
- < 1 = Inverted & Diminished.

Joh Chahiye uska sign convention nai lete !

Image Formation by Concave Mirror :-

Object	Image
+ ∞	+ f[R, I, −ve]
+ b/n ∞ & C	+ b/n C & f [R, I, −ve]
+ A + C	+ At C[R, I, −1]
+ b/n C & f	+ b/n ∞ & C [R, I, −ve↑]
+ At f	+ At ∞. [R, I, −ve]
+ b/n f & Pole	+ Behind Mirrror [V, E, +ve]

Image Formation by Convex Mirror :-

Object	Image
+ at ∞	+ At focus. [V, E, m = +ve]
+ b/n ∞ & Pole	+ b/n Pole & Focus.

MR*

Tum mirror Ko kahi bhi rakho uska "f" change nai hoga lens Ka ho sakta hai.

Newton's Formula :-

$f = \sqrt{xy}$

x = Object dist^n from focus.

y = image dist^n from focus.

Velocity of Image in Case of Concave Mirror :-

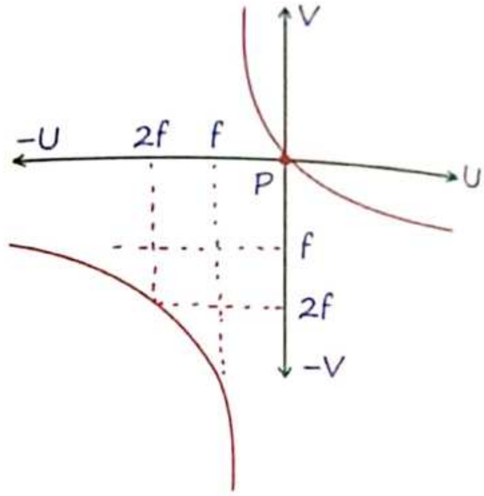
1> Object is moving $\perp^{er}$ to Principle axis :-

- $V_I = m_T V_O$.

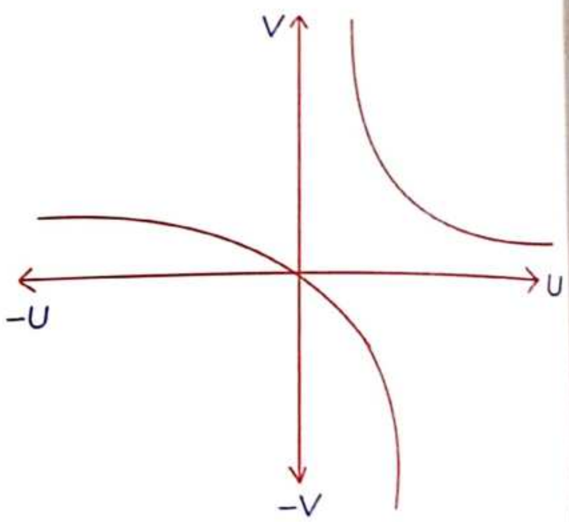
2> Object is moving $\parallel^{el}$ to Principle axis :-

- $V_I = m_L V_O = m_T^2 V_O$.

Graph for Concave Mirror :-



Graph for Convex Mirror :-



5 Refraction

+ When light travels from a rarer medium (R) to a denser medium (D), the angle of deviation (δ)

$\delta = i - r$

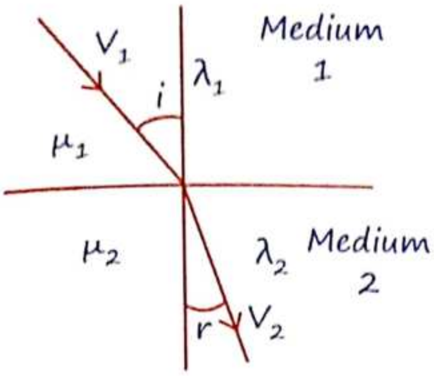
Where i is the angle of incidence, and r is the angle of refraction.

+ When light travels from a denser medium (D) to a rarer medium (R), the angle of deviation (δ)

$\delta = r - i$

Snell's Law :-

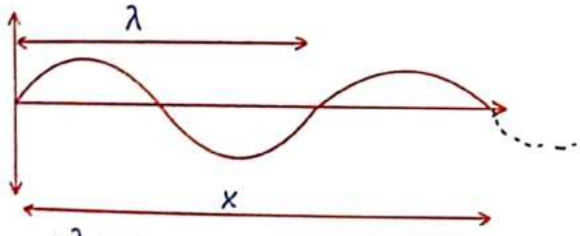
$$\mu_{21} = {}^1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{\sin i}{\sin r} = \frac{V_1}{V_2} = \frac{\lambda_1}{\lambda_2}$$



MR*

Snell's law can be directly applied b/n 1st & last medium irrespective of intermediate medium.

- "N" Number of waves in a "X" Distance if Wavelength is λ.



$n\lambda = x$

$$\boxed{\text{no. of waves, } n = \frac{x}{\lambda}}$$

Optical Path (d) :-

*d_{vacuum} & d_{medium} for same time.

$$\boxed{d = \mu x} \qquad x = d_{medium}, \quad d = d_{vacuum}$$

$$\boxed{t = \frac{\mu x}{c}}$$

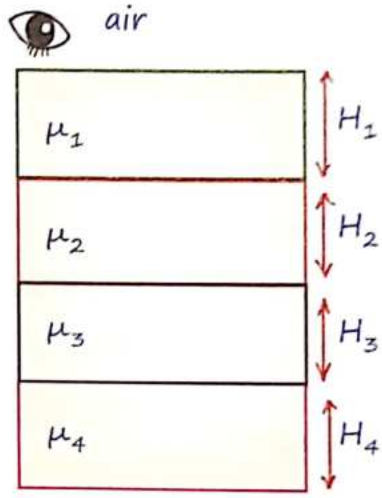
6 Real & Apparent Depth

MR** Special

$$\boxed{\frac{d_{app}}{\mu_{obr}} = \frac{d_{real}}{\mu_{obj}}}$$

This means that apparent depth seen by an observer is influenced by the refractive indices of the media through which the light travels.

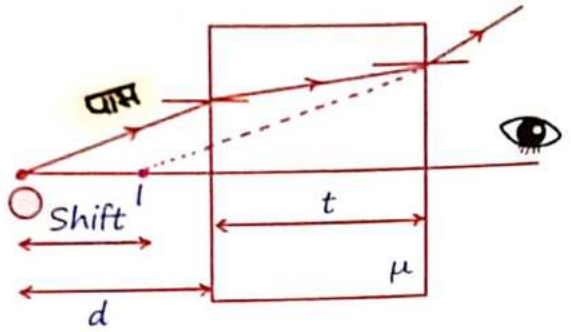
Apparent depth in multiple layers :-



$$\frac{H_{app}}{1} = \frac{H_1}{\mu_1} + \frac{H_2}{\mu_2} + \frac{H_3}{\mu_3} + \frac{H_4}{\mu_4}$$

Diverging ray :-

- When a diverging ray passes through a medium with thickness t and refractive index μ, there is a lateral shift in the path of the ray.
- The real path is indicated by the solid line, and the apparent path is shown by the dotted line.



- $d_{real} = d + t$, $d_{app} = d + \frac{t}{\mu}$
- Shift $= d_{real} - d_{app}$

$$= (d+t) - \left(d + \frac{t}{\mu}\right)$$

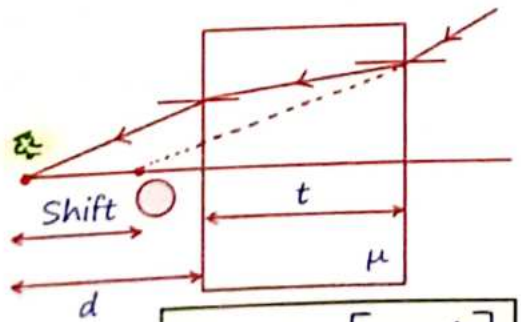
$$\boxed{\text{Shift} = t\left[1 - \frac{1}{\mu}\right]}$$

MR*

Jab bhi glass pr diverging ray aata hai toh glass ke taraf Image Shift kr jaata hai. Aur Converging ray Ke liye durr.

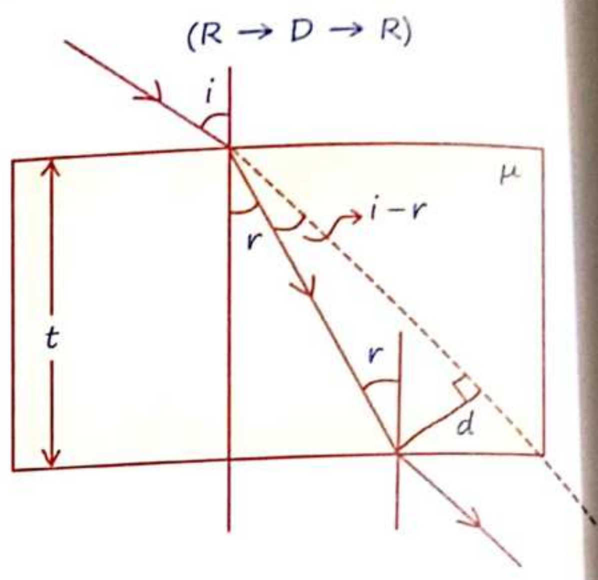
Converging ray :-

When a converging ray passes through a medium with thickness t and refractive index μ, there is a lateral shift similar to the diverging ray.



$$\boxed{\text{Shift} = t\left[1 - \frac{1}{\mu}\right]}$$

7 Lateral Shift



$$d = \frac{t}{\cos r}\sin(i-r)$$

- If $i \to$ small.

For small angles of incidence i, the sine and cosine functions can be approximated:

$\sin i \approx i$

$\cos r \approx i$

$$d = ti\left[1 - \frac{1}{\mu}\right]$$

8 Total Internal Reflection [$i > ic$]

- Total internal reflection occurs when a light ray traveling from a denser medium (D) to a rarer medium (R) strikes the boundary at an angle greater then critical angle ($i > i_c$).
- The light ray is completely reflected back into the denser medium, with no refraction occurring in the rarer medium.

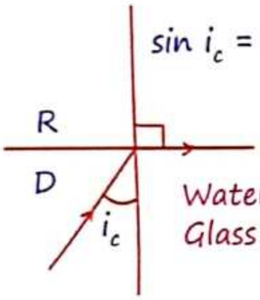
- Condition for total internal reflection: light must travel from D → R Medium.
- Angle of incidence (i) must be greater than the critical angle (i_c)

Critical Angle :-

MR*

$$\sin i_c = \frac{\mu_{Kam}}{\mu_{Jyada}} = \frac{\mu_{rarer}}{\mu_{Denser}}$$

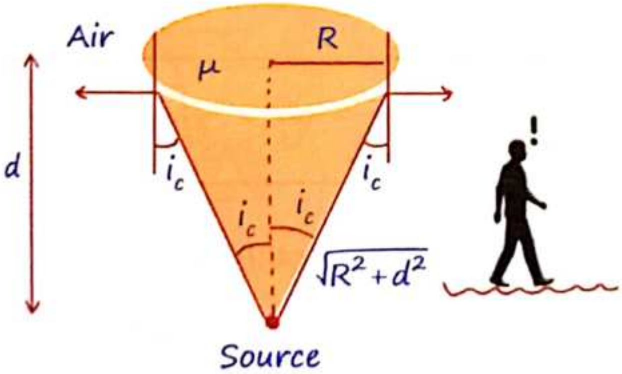
$$\sin i_c = \frac{\mu_2}{\mu_1} = \frac{V_1}{V_2} = \frac{\lambda_1}{\lambda_2}$$



Water – Air :- $i_c = 49°$
Glass – Air :- $i_c = 42°$

Radius of Visibility :-

The radius of visibility refers to the maximum distance at which an observer can see an object submerged in a medium (such as water) from above the surface.



$$R = \frac{d}{\sqrt{\mu^2-1}}$$

- Water – Air

$$h = \frac{\sqrt{7}}{3} r$$

Velocity of Image in Refraction :-

1> Object is moving parallel to boundary :-

- $V_{IG} = V_{OG}$
- $V_{IO} = 0$

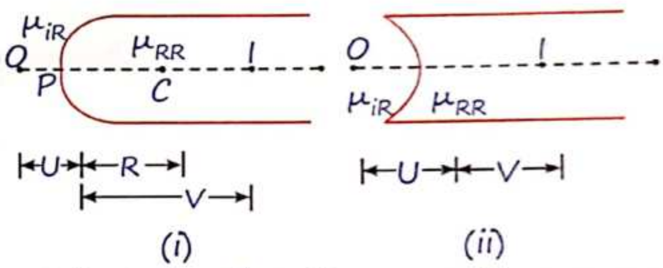
2> Object is moving $\perp^{er}$ to boundary :-

MR*

$$\boxed{\frac{V_{app}}{\mu_{obr}} = \frac{V_{real}}{\mu_{obj}}}$$

9 Refraction at Spherical Surfaces

MR* One stop solution



Spherical surface (i) convex (ii) concave

Refraction Formula:-

The refraction formula at a spherical surface separating two media with refractive indices μ_{IR} (incident ray medium) and μ_{RR} (refracted ray medium) is given by:

$$\boxed{\frac{\mu_{RR}}{V} - \frac{\mu_{IR}}{U} = \frac{\mu_{RR} - \mu_{IR}}{R}}$$

- Normal will pass through Center of Curvature.
- Nature of Surface will be decided by position of object.
- Concave :- R = –ve

  Convex :- R = +ve

Magnification :-

$$m_T = \frac{H_I}{H_O} = \frac{\mu_{IR} \cdot V}{\mu_{RR} \cdot U}$$

$$m_L = \frac{dV}{dU} = \frac{\mu_{IR} \cdot V^2}{\mu_{RR} \cdot U^2}$$

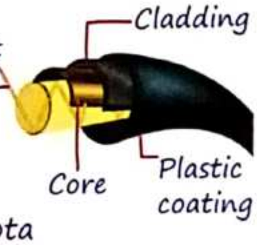
10 Optical Fiber

[Based on TIR]

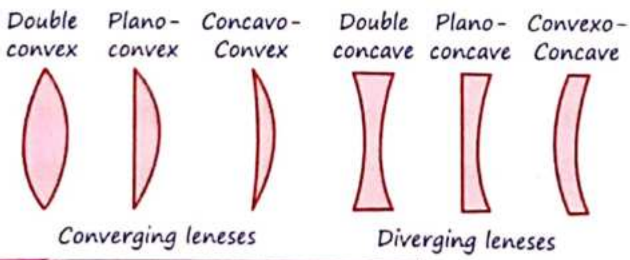
The angle at which ray must be incident so as information gets transmitted :-

$$\boxed{\sin i \le \sqrt{\underbrace{\mu_B^2}_{\text{Bada}} - \underbrace{\mu_C^2}_{\text{Chota}}}}$$

$(\mu_{core} > \mu_{cladding})$



11 Lens



12 Lens-Maker Equation

$$\boxed{\frac{1}{V} - \frac{1}{U} = \left(\frac{\mu_L}{\mu_M} - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]}$$

Where:

μ_L = refractive index of the lens material.

μ_M = refractive index of the surrounding medium.

Lens Equation :-

$$\frac{1}{V} - \frac{1}{U} = \frac{1}{f}$$

$$\frac{1}{f} = \left(\frac{\mu_L}{\mu_M} - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

Focal Length of lens :-

- Focal length depends on medium.

$$f = \frac{R}{2(\mu - 1)}$$

Biconvex (+)

Biconcave(−)

μ = refractive Index of lens w.r.t medium

$$f = \frac{R}{(\mu - 1)}$$

Planoconvex (+)

Planoconcave (−)

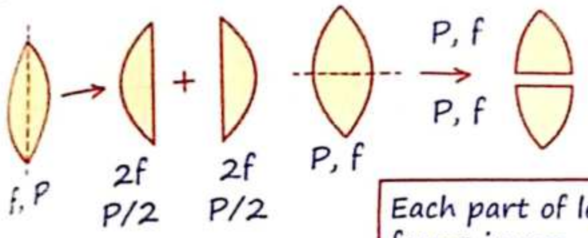
$$\boxed{m_T = \frac{H_I}{H_O} = \frac{V}{U} = \frac{f}{f+U} = \frac{f-V}{f}}$$

$m_T = -ve$ Real Image

$m_T = +ve$ Virtual Image

Small Object $\boxed{m_L = \frac{dV}{dU} = m_T^2}$

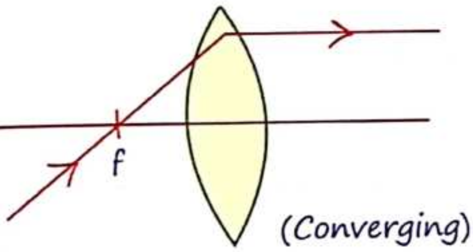
Cutting of Lens :-

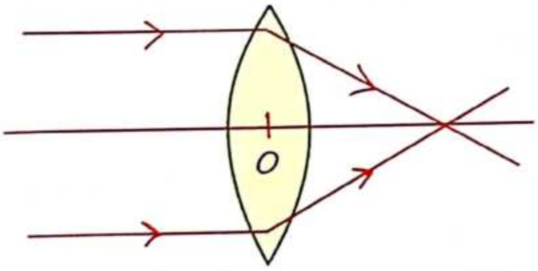


Each part of lens forms image.

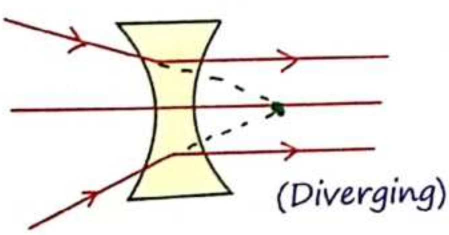
13 Ray-Diagram

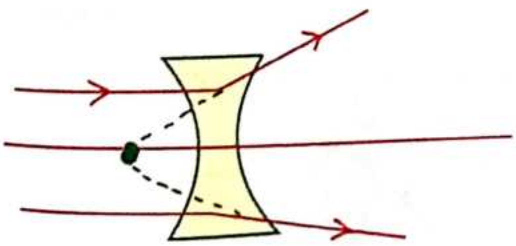
(a) Biconvex lens :- $f = +ve$





(b) Biconcave lens :- $f = -ve$





Nature of Lens Considering R.I. of Surrounding & Lens

Condition	Behaviour
$\mu_L = \mu_M$	Glass Plate: There is no refraction, and the lens does not converge or diverge light.
$\mu_L > \mu_M$	Same Nature: A convex lens converges light, and a concave lens diverges light.
$\mu_L < \mu_M$	Opposite Nature: A convex lens diverges light, and a concave lens converges light.

14 Image Formation by Equiconvex Lens

Object	Image
• ∞	• f[R, I, –ve]
• b/n ∞ & 2f	• b/n 2f & f [R, I, –ve]
• at 2f	• at 2f [R, I, –1]
• b/n 2f & f	• b/n ∞ & 2f [R, I, –ve↑]
• at f	• at ∞ [R, I, –ve↑↑]
• b/n f & Pole	• on same side of lens [V, E, +ve]

Convex lens ⇅ Concave Mirror (Converging)

Can a Convex lens be have as diverging lens.

→ Yes, when $(\mu_m > \mu_L)$

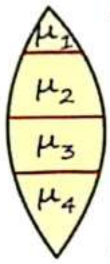
15 Image Formation by Concave Lens

Object	image
• at ∞	• at focus. [V, E, m = +ve]
• b/n ∞ & Pole	• b/n Pole & Focus.

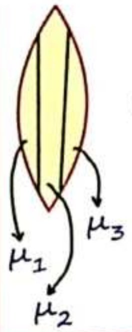
Concave Lens $\rightleftharpoons$ Convex Mirror

(Diverging)

No. of Image Form :-



No. of images = No. of Materials



No. of images = No. of Object

16 Combination of Lens

Power :-

Lens

$P_L = \dfrac{1}{f_L}$

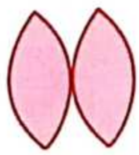
Mirror

$P_M = \dfrac{1}{f_M}$

(a) When lens are in contact :-

$P = P_1 + P_2$

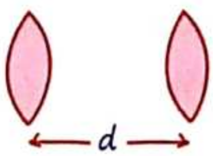
$\dfrac{1}{f} = \dfrac{1}{f_1} + \dfrac{1}{f_2}$



(b) When lens are separated by a distance "d":-

$P = P_1 + P_2 - dP_1P_2$

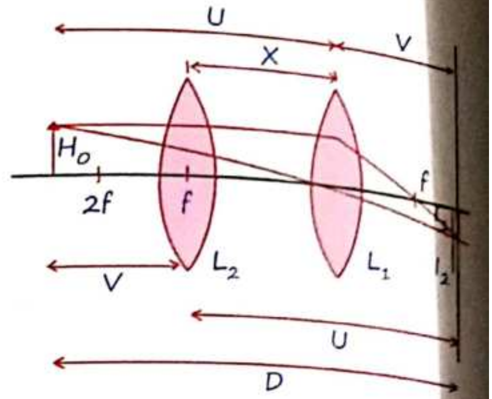
$\dfrac{1}{f} = \dfrac{1}{f_1} + \dfrac{1}{f_2} - \dfrac{d}{f_1f_2}$



Take P with sign !

17 Displacement Method

Image Ka dist^n Object Ko, Object Ka dist^n Image Ko



$$f = \frac{D^2 - x^2}{4D}$$

D = dist^n b/n Object & Screen.

x = Dist^n b/n two position of lens.

$H_o = \sqrt{I_1 I_2}$ $\qquad m_1 = \dfrac{V}{U} = \dfrac{I_1}{H_o}$

$|f| = \dfrac{x}{m_1 - m_2}$ $\qquad m_2 = \dfrac{U}{V} = \dfrac{I_2}{H_o}$

Combination of a Lens & a Mirror (Silvering of Lens) :-

MR*

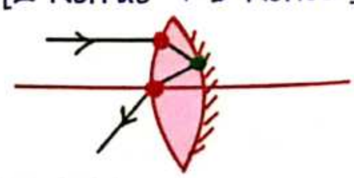
Koi sign mt rakho bs direct lens ka opposite nature mirror Ko dedo.

$$\boxed{\frac{1}{|f_{net}|} = \frac{2}{|f_L|} + \frac{1}{|f_M|}}$$

Convex lens Silvered $\longrightarrow$ Concave Mirror $f = -ve$

Concave lens Silvered $\longrightarrow$ Convex Mirror $f = +ve$

[2 Refrac^n + 1 Reflec^n]



(a) Convex Lens :-

$$f_{eq} = \frac{R}{2(2\mu-1)} \Bigg\} \text{Concave Mirror}$$

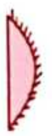
(b) Plano-convex lens :-

$$f_{eq} = \frac{R}{2(\mu-1)} \Bigg\} \text{Concave Mirror}$$

(c) Plano-convex lens :-

$$f_{eq} = \frac{R}{2(\mu-1)} \Bigg\} \text{Concave Mirror}$$

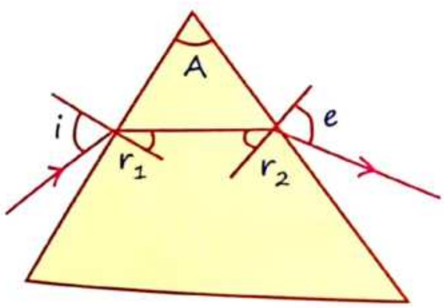
$$f_{eq} = \frac{R}{2(\mu-1)} \Bigg\} \text{Convex Mirror}$$



$$f_{eq} = \frac{R}{2\mu} \Bigg\} \text{Concave Mirror}$$

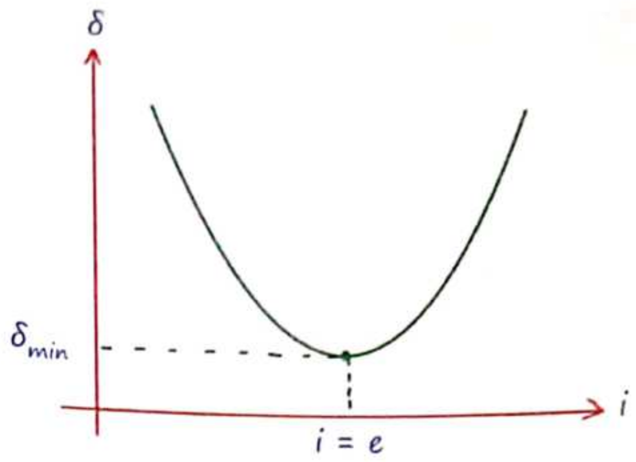
$$f_{eq} = \frac{R}{2\mu} \Bigg\} \text{Convex Mirror}$$

18 Prism



Angle of prism, $A = r_1 + r_2$ Total deviation $\delta_{Total} = i + e - A$

$$\mu = \frac{\sin i}{\sin r_1} \qquad \frac{1}{\mu} = \frac{\sin r_2}{\sin e}$$



For Minimum deviation :-

$$r_1 = r_2 = \frac{A}{2} \qquad i = e$$

$$\delta_{min} = 2i - A, \quad \mu = \frac{\sin\left[\frac{\delta_m + A}{2}\right]}{\sin\left[\frac{A}{2}\right]}$$

For thin prism (A = small)

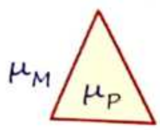
$$\delta_{min} = A(\mu - 1)$$

A = Refracting Angle.

+ Half Angle Formula :-

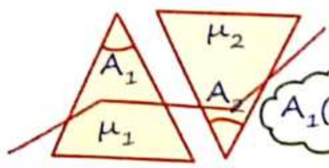
$$\sin\theta = 2\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2}$$

Prism is placed in medium :-



$$\delta = A\left[\frac{\mu_P}{\mu_M} - 1\right]$$

Condition for no deviation :-



$$\delta_1 + \delta_2 = 0$$

$$A_1(\mu_1 - 1) = -A_2(\mu_2 - 1)$$

19 Dispersion of light

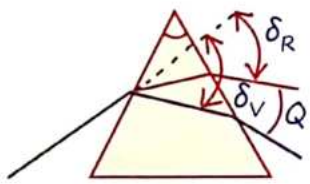
Dispersion of light occurs when white light passes through a prism, separating into its constituent colors (VIBGYOR - Violet, Indigo, Blue, Green, Yellow, Orange, Red).

$$\mu \propto \frac{1}{\lambda^2} \xrightarrow[\delta\downarrow]{\text{VIBGYOR}} \lambda\uparrow \ \mu\downarrow$$

20 Angular dispersion (Q)

$$Q = \delta_V - \delta_R$$

$$Q = A\,[\mu_V - \mu_R]$$



Mean deviation :-

$$\delta_{mean} = \frac{\delta_V + \delta_R}{2}$$

$$\delta_{mean} = A\left[\frac{\mu_R + \mu_V}{2} - 1\right]$$

$$\delta_{mean} = A\,[\mu_{mean} - 1]$$

$$\delta_{mean} = A\,[\mu_{yellow} - 1]$$

Dispersive Power :- (ω)

$$\omega = \frac{Q}{\delta_{mean}} = \frac{\delta_V - \delta_R}{\delta_{mean}}$$

$$\omega = \frac{\mu_V - \mu_R}{\mu_y - 1} = \frac{\delta_V - \delta_R}{\frac{\delta_V + \delta_R}{2}}$$

Dispersion without Deviation :-

$$\delta_y = -\delta_y'$$

$$A_1(\mu_1 - 1) = -A_2(\mu_2 - 1)$$

(emergent ray $\parallel^{el}$ incident ray)

Deviation without Dispersion :-

$$Q = -Q'$$

$$\delta_V - \delta_R = -[\delta_V' - \delta_R']$$

(White light bahar)

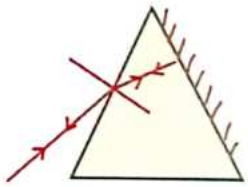
Total dispersion :- In case of dispersion without deviation

$$\theta = \theta_1 + \theta_2 = A[\mu - 1](\omega_1 - \omega_2)$$

Silvering of Prism :-

$A = r_1 \qquad r_2 = 0$

$$\mu = \frac{\sin i}{\sin A}$$



$\mu \propto 1/\lambda$

21 Optical Instruments

Visual Angle :-

Angle from horizontal with which a observer sees an object.

$$\boxed{Q_o = \frac{H_o}{D}}$$

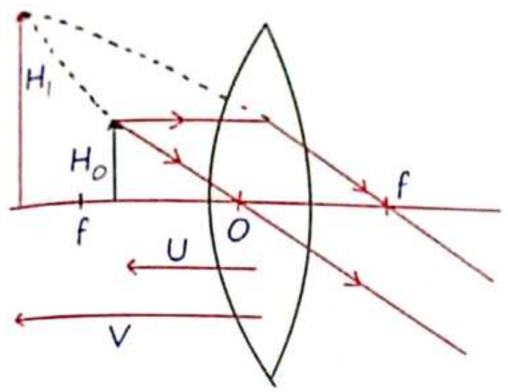
Eye की औकात !

D = 25 cm

Distance of Distinct Vision.

Myopia	Hypermetropia
Concave lens. Near Sightedness	Convex lens. Far-Sightedness

Simple Microscope



$$M = \frac{D}{U} = D\left[\frac{1}{f} + \frac{1}{V}\right]$$

I

- Strained eye position
- Near point
- $V = D$
  image distance — Distance of distinct vision

$$M_{max} = 1 + \frac{D}{f}$$

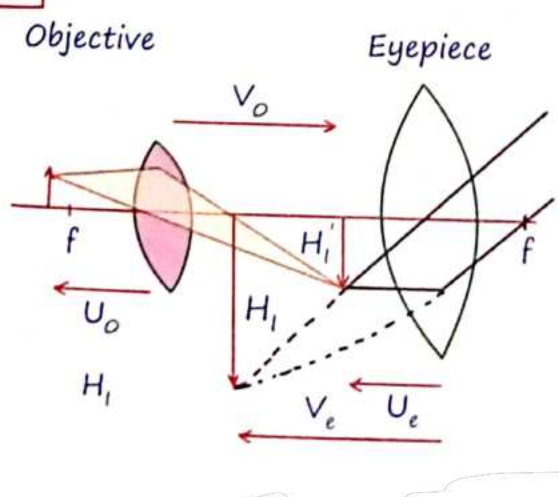
II

- Relaxed eye position
- Far point
- $v = \infty$.

$$M_{min} = \frac{D}{f}$$

Compound Microscope :-

$\boxed{f_o < f_e}$



$$M = M_O M_e = \frac{V_O}{U_O}\frac{D}{U_e}$$

$$M = \frac{DV_O}{U_O}\left[\frac{1}{f_e} + \frac{1}{V_e}\right]$$

- Note :- $M = \frac{Q_1}{Q_O}$

I

- Strained eye position
- Near point
- $V = D$

$$M = \frac{V_O}{U_O}\left(1 + \frac{D}{f_e}\right)$$

II

- Relaxed eye position
- Far point
- $v = \infty$.
- v_e = infinity
  $u_e = f_e$

$$M = \frac{V_O D}{U_O f_e}$$

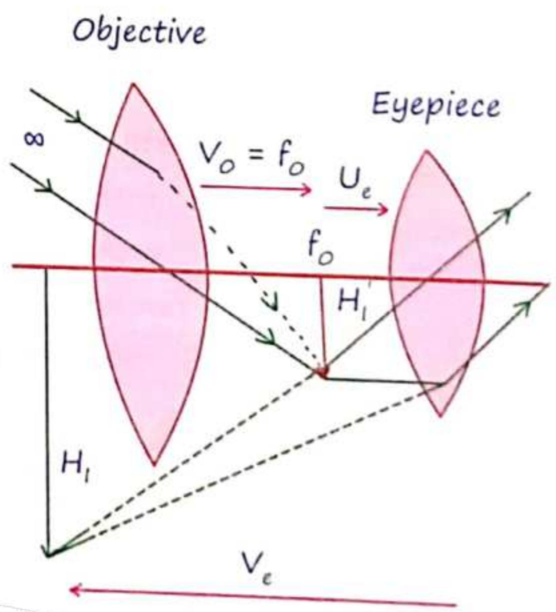
$$M = \frac{LD}{f_O f_e}$$

Length of Compound Microscope :-

$L = V_O + U_e \cong V_O$

In case of far point $L = V_O + f_e$

Astronomical Telescope



$$\boxed{M = \frac{f_o}{U_e} = f_o\left[\frac{1}{f_e} + \frac{1}{V_e}\right]}$$

I

- Strained eye position
- Near point
- $V = D$

$$M = f_o\left[\frac{1}{f_e} + \frac{1}{D}\right]$$

II

- Relaxed eye position
- Far point
- $v = \infty$.

$$M = \frac{f_o}{f_e}$$

- **Length of Telescope :-**

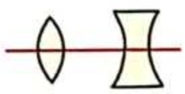
$L = f_o + U_e$

$L = f_o + f_e$ → For image at ∞.

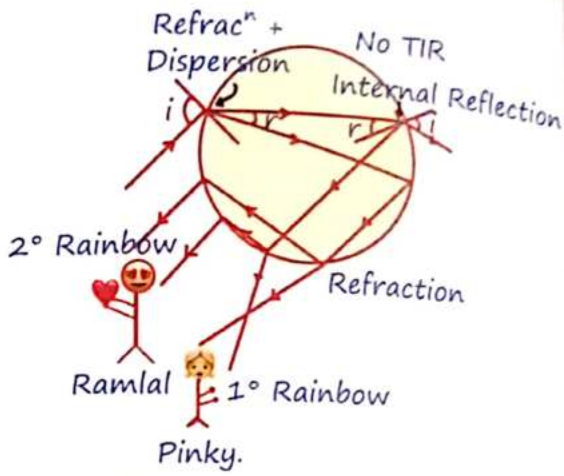
- **Galileo Telescope :-**

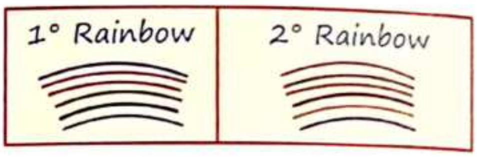
$L = f_o - f_e$

Rest Same.



22 Rainbow formation





1° Rainbow	2° Rainbow
+ Red on upper side	+ Red on lower side
+ violet on lower	+ violet on upper

- **Scattering of Light :-**

E.g.:- Blue colour of Sky, Red Sky & sun on the time of sunrise or sunset.

Scatering $\propto \frac{1}{\lambda^4}$

MR*

"Beta duniya gol hai, tuhare kiye gaye mehnat tum tak zarur lautega.... mehnat karna mat chhodo...jitna tumhare hath main hai h utna karo, baki bhagwan par chhod do."

31 Wave Optics

(Mech + Longitudinal)

Huygen:- Wave

Einstein:- Particle

Maxwell:- EM wave

} Light

(Non Mech + Transverse)

$$\boxed{\frac{2\pi}{\phi} = \frac{\lambda}{\Delta x} = \frac{T}{\Delta t}}$$

Equation of Wave for Light/ Sound / EM Wave / ac: -

$y = A\sin(\omega t + kx + \phi)$

Wave Sabhi Medium particle Ko SHM Deta hai!

1 Newton corpuscular theory

Explain:-

- Rectilinear propagation: Light travels in straight lines.
- Reflection: Bouncing of light from a surface.
- Refraction of Light: Bending of light when it passes from one medium to another.

Can't explain:-

- Interference: When two waves meet and combine.
- Diffraction: Bending of light around obstacles.
- Polarization: Orientation of light waves in particular directions.

2 Huygens wave theory

Explain:-

- Rectilinear propagation, Interference, Reflection, Refraction, Diffraction.

Can't explain:-

- Polarisation, PEE, Compton effect.

Types of waves:-

1. **Medium:-**

   Mechanical ✓ EMW ✗

2. **Propagation:-**

   Progressive ∞ Stationary Finite

3. **Vibration:-**

   Transverse $\perp$ Longitudinal $\parallel^{el}$

3 Wave front

A wavefront is a surface that connects all points vibrating in sync at the same phase of a wave. It helps visualize how waves propagate through different mediums.

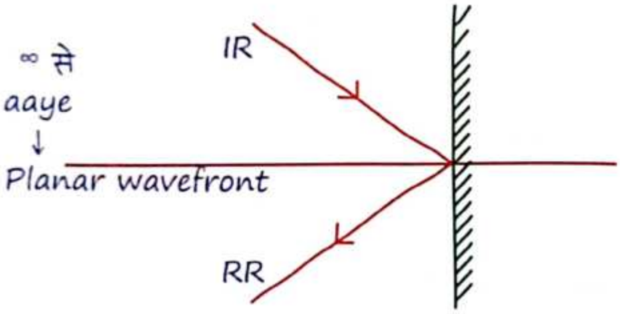
Wavefront Type	Source of Light	Amplitude (A)	Intensity (I)
Spherical Wavefront	Point source	$A \propto \frac{1}{r}$	$I \propto (A)^2 \propto \frac{1}{r^2}$
Cylindrical Wavefront	Linear source	$A \propto \frac{1}{\sqrt{r}}$	$I \propto (A)^2 \propto \frac{1}{r}$
Plane Wavefront	Very distant source	$A \propto r^0$ (constant)	$I \propto (A)^2 \propto r^0$ (constant)

Intensity of wave:-

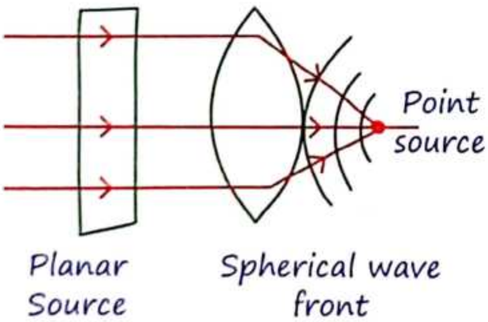
$$I = \frac{1}{2}\rho V A^2 \omega^2$$

4 Behaviour of plane wavefront on reflection & refraction

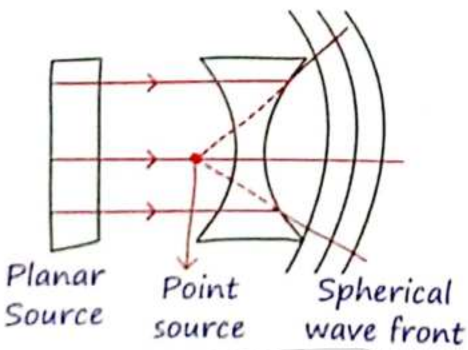
Plane Mirror:-



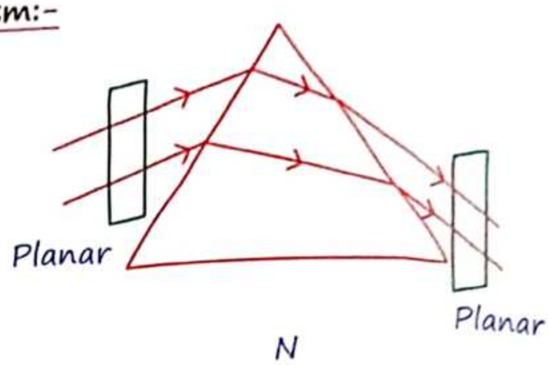
Convex Lens:-

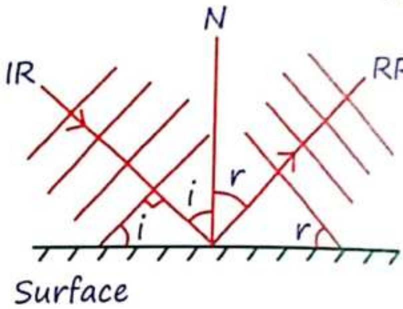


Concave lens:-



Prism:-





5 Sources

Phase difference

Time के साथ constant	Time के साथ Variable
Coherent	Incoherent
f_{req} = same	

Incoherent sources:-

$$\Delta\phi = [\omega_1 - \omega_2]t + [K_1x_1 - K_2x_2] + [\phi_1 - \phi_2]$$

Time Dependent Hai!

Coherent sources:-

$$\Delta\phi = K(x_1 - x_2] + \phi_1 - \phi_2$$

$$\boxed{\Delta\phi = \frac{2\pi}{\lambda}.\Delta x.} \quad \because \quad \phi_1 = \phi_2$$

Single wavelength — Monochromatic source

6 Interaction of light

- **$\lambda << l_{object}$ = RAY :-**

When the wavelength (λ) of light is much smaller than the size of the object it interacts with light can be treated as rays.

- **$\lambda \approx l_{object}$ = WAVE**

When the wavelength (λ) is comparable to the size of the object, light exhibits wave-like behavior.

- **e-/p+/α-particle = PARTICLE**

Light also behaves as particles (photons), particularly in interactions at the atomic or subatomic level, such as in the photoelectric effect or Compton scattering.

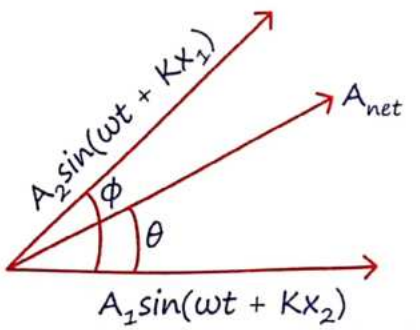
(Photon)

7 Interference

Interference is a phenomenon where two or more coherent light waves superimpose to form a resultant wave of greater, lower, or the same amplitude.

Requirements:

+ For interference to occur, two coherent sources (sources with constant phase difference) are required.



Resultant amplitude

$$A_{net} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}$$

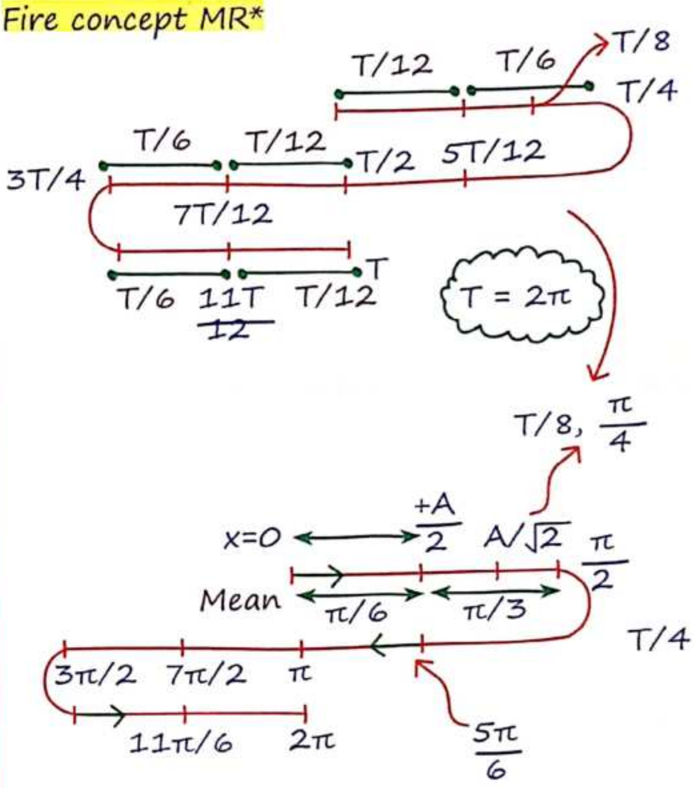
Resultant intensity

$$I_{net} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

Phase Relation and angle (θ)

$$\tan\theta = \frac{A_2\sin\phi}{A_1 + A_2\cos\phi}$$

Fire concept MR*



8 Intensity

$$I = \frac{1}{2}\rho V A^2 \omega^2$$

Where:

ρ = density of the medium

V = velocity of the wave

A = amplitude of the wave

ω = angular frequency of the wave

$I \propto d \propto A^2$ $\quad$ d = slit width

Average Intensity

$$I_{av} = \frac{I_{min} + I_{max}}{2}$$

Ratio of Maximum to minimum Intensity

$$\frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_2}+\sqrt{I_1}\right)^2}{\left(\sqrt{I_2}-\sqrt{I_1}\right)^2} = \frac{\left(A_2 + A_1\right)^2}{\left(A_2 - A_1\right)^2}$$

$$\frac{I_{max}}{I_{min}} = \frac{\left[\sqrt{\frac{I_2}{I_1}}+1\right]^2}{\left[\sqrt{\frac{I_2}{I_1}}-1\right]^2} = \frac{\left[\frac{A_2}{A_1}+1\right]^2}{\left[\frac{A_2}{A_1}-1\right]^2}$$

9 Constructive Interference

$\phi = 0°$ $\cos\phi = 1$

$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$

$I_{max} = \left(\sqrt{I_1}+\sqrt{I_2}\right)^2$

$A_{max} = A_1 + A_2$

$I_1 = I_2 = I_0$

$I_{max} = 4I_0$

$\phi = 2n(\pi) \rightarrow$ even

$\Delta x = n(\lambda) \rightarrow$ integral

$n = 0, 1, 2, 3, ...$

MR*

$\lambda = 2\pi$

- **YDSE:-**

$$Y = n\left(\frac{\lambda D}{d}\right) \text{ Posit}^n \text{ of } n^{th} \text{ Bright.}$$

- **NOTE:-**

1> "n" source of same intensity I_0.
Find I_{Result^n}:-
$I_{net} = nI_0$ (scalar Add^n)

2> "n" source of same intensity I_0 then find I_{max}:- (Const. Inter.)
$*I_{max} = n^2 I_0 = (n\sqrt{I_0})^2$

Fringe visibility:-

$$\text{Visibility Ratio} = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$$

$$= \frac{\left(\sqrt{\frac{I_1}{I_2}}+1\right)^2 - \left(\sqrt{\frac{I_1}{I_2}}-1\right)^2}{\left(\sqrt{\frac{I_1}{I_2}}+1\right)^2 + \left(\sqrt{\frac{I_1}{I_2}}-1\right)^2}$$

10 Destructive interference

$\phi = 180°$ $\cos\phi = -1$

$I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$

$I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$

$A_{min} = A_1 - A_2$

$I_1 = I_2 = I_0$

$I_{max} = 0, I_{min} = 0$

$\phi = (2n+1)\pi \rightarrow$ odd

$\Delta x = (2n+1)\frac{\lambda}{2} \rightarrow$ odd

$n = 0, 1, 2, 3, ...$

- **YDSE:-**

$$Y = (2n-1)\frac{\lambda D}{2d} \text{ Posit}^n \text{ of } n^{th} \text{ Dark.}$$

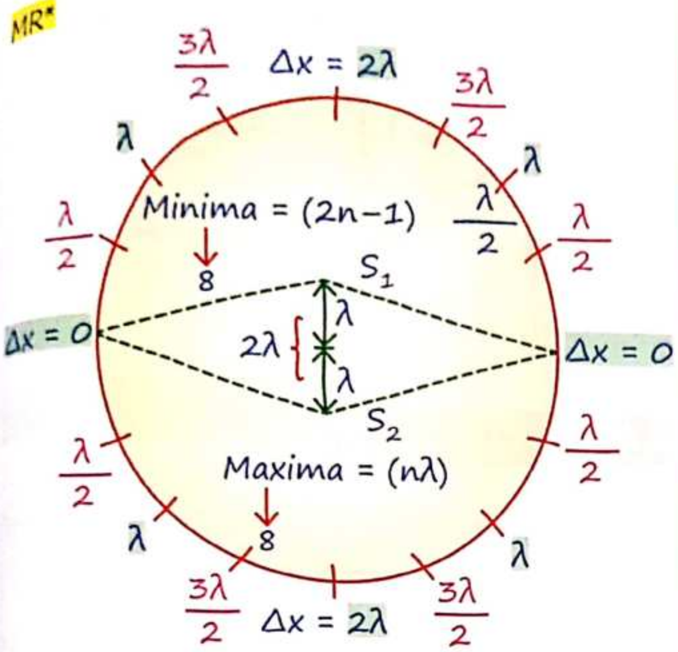
IIT Neet 2016 $\frac{I_1}{I_2} = \alpha$

$$\text{Visibility Ratio} = \frac{2\sqrt{\alpha}}{1+\alpha}$$

✲ Two wave of "I_0" intensity I_{res} If phase diff is ϕ:-

$$I = 4I_0\cos^2\frac{\phi}{2}$$

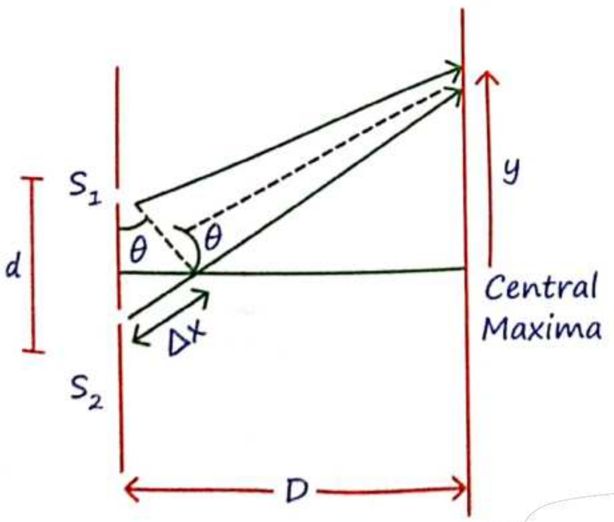
MR*



11 Young Double slit Experiment

Basic Setup :-

- S1 and S2 are two slits separated by a distance d.
- A screen is placed at a distance D from the slits.
- Light waves passing through the slits interfere and create a pattern of bright and dark fringes on the screen.



Path Difference (Δx) :-

$d \simeq \lambda$ $\qquad \Delta x = d\sin\theta$

$\qquad\qquad\qquad = d\tan\theta$

$$\Delta x = \frac{Yd}{D}$$

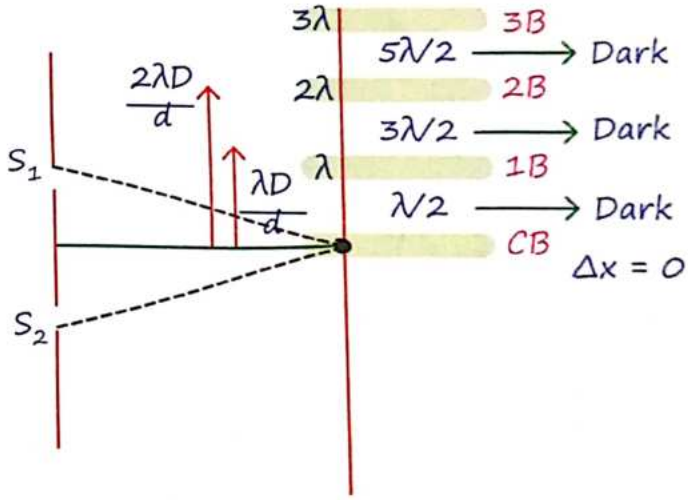
Positions of Bright and Dark Fringes :-

- The position of the n-th bright fringe (nB) on the screen is given by:

$$y_n = \frac{n\lambda D}{d}$$

- The position of the n-th dark fringe is given by:

$$y_n = \frac{(n+\frac{1}{2})\lambda D}{d}$$



Fringe Width (β):-

$\beta = Yn^{th}$ Bright $- Y(n-1)^{th}$ Bright

$$\boxed{\beta = \frac{\lambda D}{d}}$$

Angular Fringe Width:-

$$\theta = \frac{\beta}{D} \qquad \theta = \frac{\lambda}{d}$$

MR*

☼ Angular kuch bhi puche aap "D" se divide kardena!

☼ $\mu_{air} > \mu_{vacuum}$

$\beta \propto \frac{1}{\mu}$ $\therefore$ In vacuum,

interference with large "β" seen!

YDSE in air	YDSE in liq.
1> $\beta = \frac{\lambda D}{d}$	1> $\beta' = \frac{\beta}{\mu}$
2> $\theta = \frac{\lambda}{d}$	2> $\theta' = \frac{\theta}{\mu}$
3> Max:- $Y_n = \frac{n\lambda D}{d}$	3> $Y_n' = \frac{n\lambda D}{\mu d}$
Min:-	$Y_n' = \frac{(2n-1)\lambda D}{2\mu d}$
$Y_n = (2n-1)\frac{\lambda D}{2d}$	f = same $\lambda' = \lambda/\mu$

Q.1 Two light of $\lambda_1 > \lambda_2$ is used in YDSE, then central Maxima & 1st Maxima will 2.

Sol. Central Maxima $\Rightarrow$ At same position.

☼ 1st Maxima:- Not at same place.

$\downarrow$

Jiska $\lambda\uparrow$ $Y\uparrow$ $\because Y = \frac{n\lambda D}{d}$

$\therefore \lambda_1$ = dur rahega!

White light:-

☼ sabse pahle maxima = violet ka ayega.

☼ sabse pahle dikhega = red.

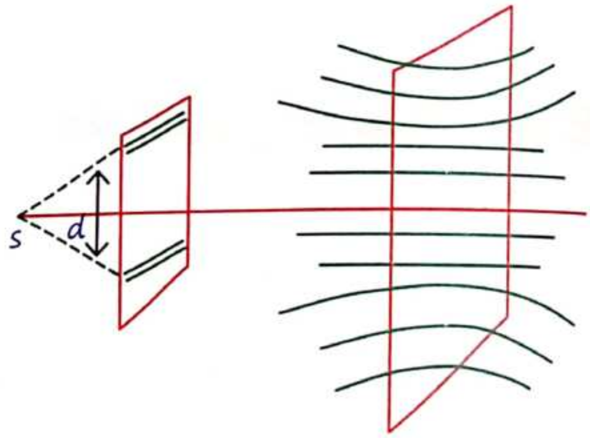
☼ Used to find central maxima.

Fringe

Nearest	Central	Farthest
$\downarrow$	$\downarrow$	$\downarrow$
Blue	चमकीला White	Red.

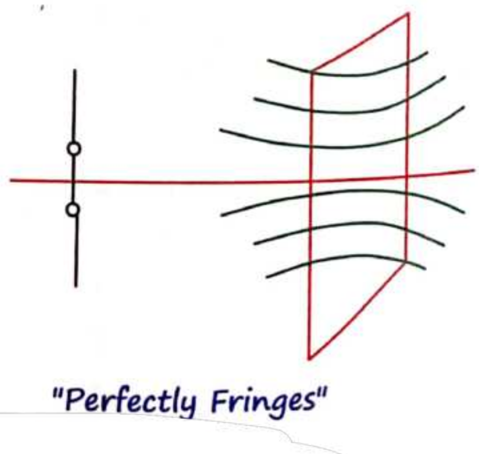
12 Shapes of fringes

Two slit used:-



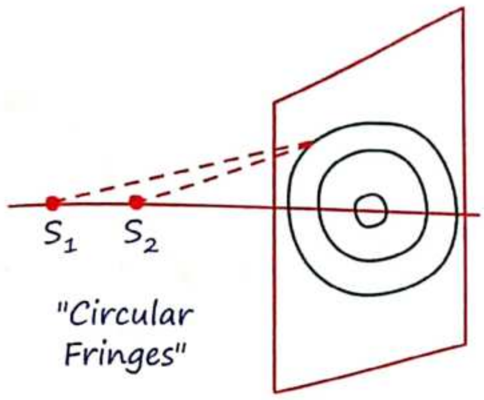
"Hyperbolic" Fringes

Two pin holed used:-



"Perfectly Fringes"

When two hole is along the line joining of source and screen:-



"Circular Fringes"

13 Optical path

air:- $\Delta\phi = \frac{2\pi}{\lambda} t$

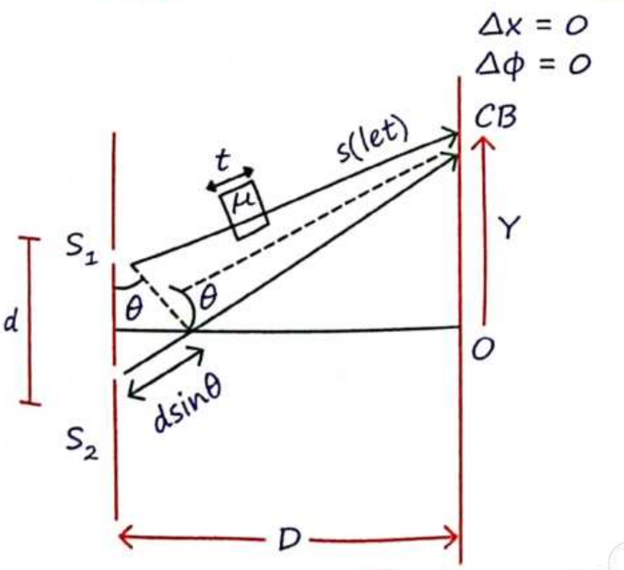
Med^m:- $\lambda' = \frac{\lambda}{\mu}$ $\Delta\phi' = \frac{2\pi}{\lambda}(\mu t)$

YDSE when a slab of "μ" R.I. inserted in path of S_1:-

$d\sin\theta = t(\mu-1)$

$$\frac{dY}{D} = t(\mu-1)$$

$$\beta = \frac{\lambda D}{d}$$



MR*

Jiske path mein slab jayega woh "μt" jyada chalega agar $\Delta x = 0$ krna hai toh usko niche lao yaneki kam karo toh S_2 ke path ko badao dono barabar hojayenge aur $\Delta x = 0$!

Position of CB:-

$$Y = \frac{Dt(\mu-1)}{d}$$

Kuch nai Normal YDSE ka

$\lambda = (\mu-1)t$ rakhdo!

Number of bright & dark fringes:-

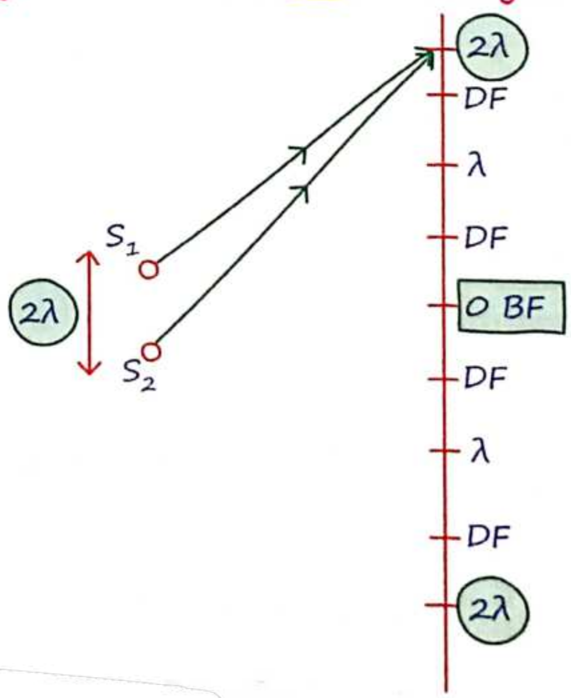
Total No. of:-

(a) Bright Fringe $= \left[\frac{2d}{\lambda} + 1\right]$

(b) Dark Fringe $= 2\left[\frac{d}{\lambda} + \frac{1}{2}\right]$

$\frac{d}{\lambda}$ = minimum integer $\begin{cases} 2.5=2 & 2.9=2 \\ 3.6=3 \end{cases}$

Majduri se duri MR ★ hai jaruri



$$\text{No. of BF} = \frac{2(2\lambda)}{\lambda} + 1 = 5$$

$$\text{No. of DF} = 2\left[\frac{d}{\lambda} + \frac{1}{2}\right] \leftarrow \text{Integer}$$

$$= 2\,(2+0.5) = 2\,[2] = 4$$

14 Polarisation

(Only for Transverse wave)

$E = cB$

$\longrightarrow$ negligible compared to E.F

$\therefore$ Restricting of $\vec{E}$ = Polaris^n.

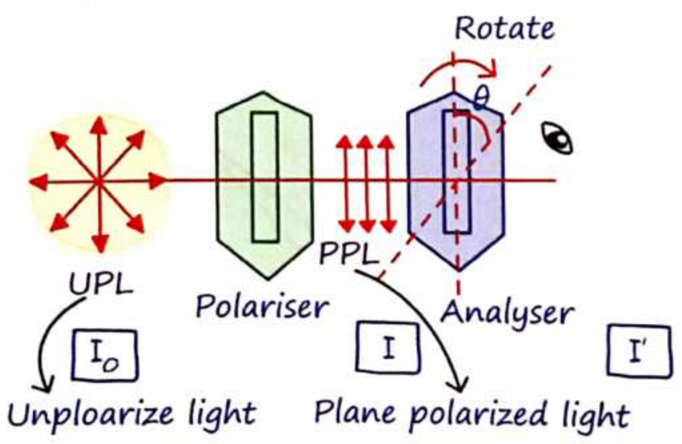
Law of malus:-

I_o = Intensity of unpolarised light.

θ = Angle b/n Analyser & Polarizer.

I = Intensity of light transmitted through the polariser.

$I = I_o \cos^2\theta$.



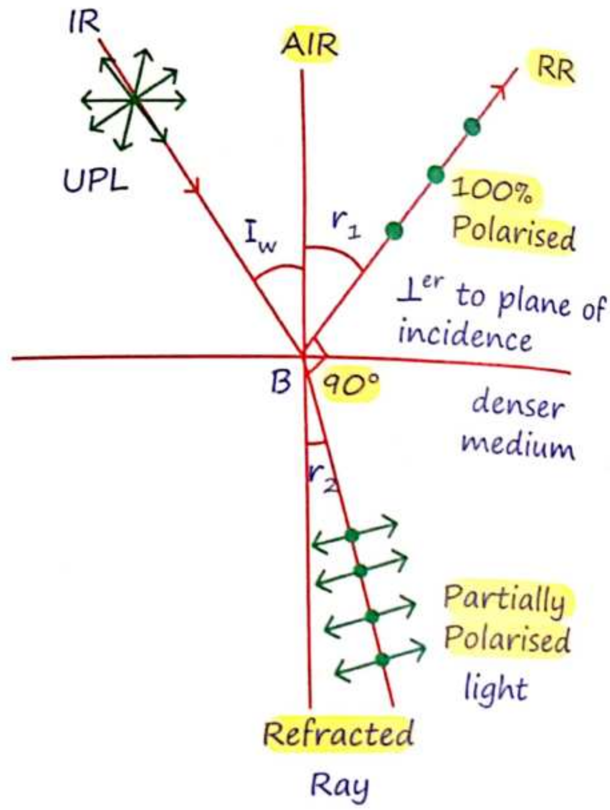
$$\boxed{I = \frac{I_o}{2}}$$

PPL
After polaroid

$$\boxed{I' = I\cos^2\theta = \frac{I_o}{2}\cos^2\theta}$$

After Analyser

15 Brewster's Law (Polarisation by reflection)



$$i_w = \tan^{-1}(\mu)$$

$$\tan i_w = \frac{\mu_2}{\mu_1}$$

16 Diffraction

Bending of wave around an object.

जोरदार :- Diffrac^n $\quad s \approx \lambda_{wave}$

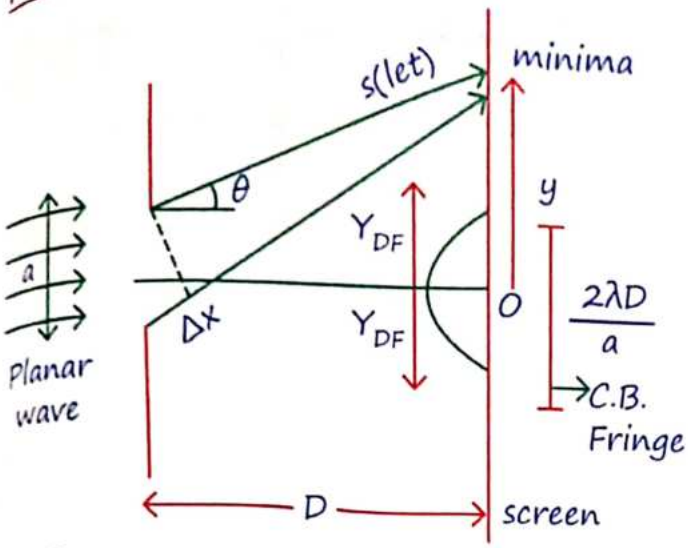
Fresnel

- Source & screen at Finite dist^n
- Spherical wave front

Fraunhoffer

- Source & screen at ∞ dist^n
- Planar wave front

Fraunhoffer Diffraction:-



उल्टा है YDSE का!

Neet

✲ Width of C.B. Fringe $= \frac{2\lambda D}{a}$

✲ Angular width of C.B.F. $= \frac{2\lambda}{a}$

Minima (D.I)	(C.I) Maxima
$a\sin\theta = n\lambda$	$a\sin\theta = (2n-1)\frac{\lambda}{2}$
$n = 1,2,3,4...$	$n = 2,3,4...$

$$\tan\theta = \frac{Y}{D}$$

✲ $a\sin\theta \approx a\tan\theta$

17 Validity of ray-optics

$$Z = \frac{a^2}{\lambda}$$

Iske pahile ray

Iske baad wave

Dist^n of source & screen (Z)

a = width of C.B. Fringe.

Important points:-

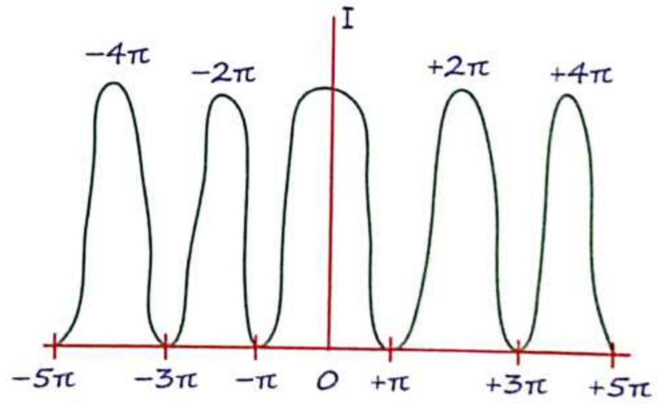
- 1 inch = 2.54 cm
- For max R.L. we use max λ.
- $\lambda_w = \frac{\lambda_{air}}{\mu}$
- * $d\theta_w = \frac{d\theta_a}{\mu}$ *
- If two coherent source of equal intensity are taken then fringe visibility is 100%. because $I_{min} = 0$.

18 Doppler's effect of light

$$r' = r\left[\frac{1 \pm \frac{V_r}{C}}{\sqrt{1-\left(\frac{V_r}{C}\right)^2}}\right] \Rightarrow \lambda' = \frac{\lambda}{1+\frac{V_r}{C}}$$

$$\frac{\Delta r}{r} = \frac{\Delta\lambda}{\lambda} = \pm\frac{V_r}{C}$$

19 Interference



Interference due to thin filim.

- For reflected light:

Maxima – $2\mu t\cos r = (2n+1)\frac{\lambda}{2}$

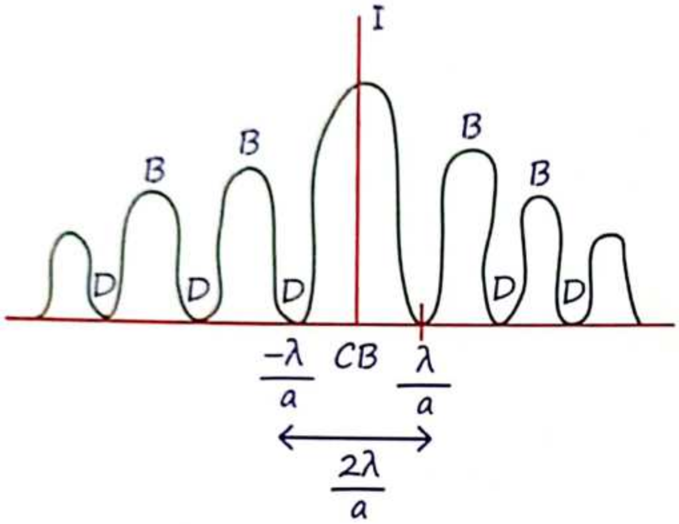
Mininima – $2\mu t\cos r = n\lambda$

- For transmitted light:

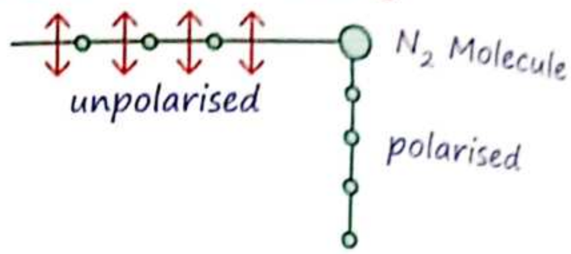
  Maxima – $2\mu t \cos r = n\lambda$

  Minima – $2\mu t \cos r = (2n-1)\dfrac{\lambda}{2}$

20 Diffraction



Polarisation by scattering



32

Dual Nature

1 Dual Nature of light:

Newton Baba → Light is a Particle (corpuscule)
→ Explain reflection / refraction

Huygen Chacha → Light is a mechanical wave
→ Ether medium Propsed by Huygen
→ Explain Diffraction / refraction.

Maxwell Chacha → Light is a Non-mechanical transverse wave.
→ No medium required.

de Broglie Bhaiya → Nature loves Symmetry
→ Light have dual Nature.

Davission & Geomer → e^- is a Wave exp^m verification

G.P. thomson → verification of electron as wave.

2 Photon

$$E = h\nu = \frac{hc}{\lambda} = \frac{2\times10^{-25}J}{\lambda} = \frac{12400\ eV}{\lambda(A^\circ)}$$

$$v_{photon} = \frac{1}{\sqrt{\mu_0\varepsilon_0}},$$ independent of frequency

$$(v_{photon})_{Med.} = \frac{c}{\sqrt{\mu_r\varepsilon_r}}$$

Energy of light beam:

$$E = nhf = \frac{nhc}{\lambda} \qquad n = \text{no. of Photons}$$

$1eV = 1.6 \times 10^{-19}$ joule

Properties of Photon:

- charge and rest mass =0
- Momentum of Photon $P = \frac{h}{\lambda}$
- Moving mass of photon $m = \frac{h}{\lambda C} = \frac{E}{c^2}$
- Photon does not deviate in electromagnetic field but deviate in gravitational field.

Power of light:

Total energy falling per second is called power of light beam.

$$P = \frac{E}{t} = \frac{nh\nu}{t}$$

No. of photons emitted per unit time $= \frac{n}{t}$

$$= \frac{P\lambda}{hc}$$

Intensity of light:

Total energy falling per unit area per-sec is called intensity.

$$I = \frac{nE}{At} = \frac{nh\nu}{At} = \frac{nhc}{\lambda At} = \frac{\text{Power}}{\text{Area}}$$

If source is same, frequency is same.

$I \propto$ no. of photons.

Q.1 If Intensity and frequency both becomes double then no. of Photon will be??

Ans. n = remains same

$I \propto nf$

Q.2 For a given source. Intensity becomes double then no. of Photon will?

Ans. becomes double $\boxed{I \propto n}$

Fractional change in frequency of photon when it travels "h" distance on earth surface:

$$\frac{\Delta\nu}{\nu} = \frac{gh}{C^2} \qquad h\nu_i + \frac{h\nu_i}{c^2}gh = h\nu_2 + 0$$

Radiation pressure:

- $\boxed{P = \frac{I}{C}[2-\rho]}$

  *Complete Absorption : (ρ = 1)

  $P = \frac{I}{C}$

  $F = PA = \frac{IA}{C} = \frac{Power}{C}$

- $\boxed{P = \frac{I}{C}[\sigma+1]}$

  *Complete Reflective : $\sigma = 1$

  $P = \frac{2I}{C}$

  $\boxed{F = \frac{2IA}{C} = \frac{2Power}{C}}$

  Where,
  I = Intensity of light Beam
  c = Speed of light
  P = Pressure
  ρ = Absorption coefficient
  σ = Reflection coefficient

3 Photoelectric effects:

- Conversion of light energy to electrical energy.
- Explained by Einstein
- Instantaneous process; time lag b/w falling of photon and emiting of electron is 10^{-9} sec.
- Efficiency = $\frac{\text{no of electron}}{\text{no of photon}} = 10^{-3}$ to 10^{-4}.
- One to one intraction one photon can eject only one electron.
- Work Function, ϕ = Material Properties

  $= h\nu_o = \frac{hc}{\lambda_o}$

  where ν_o and λ_o are thereshold frequency and wavelength.

- For Photoelectric effect :

  $\nu_{light} \geq \nu_o$ [Jyada hogi]
  $\lambda_{light} \leq \lambda_o$ [kam hogi]
  $(KE)_{max} = eV_o$
  V_o = Stopping Potential

- $E_{photon} = \phi + (KE)_{max}$

  $h\nu = h\nu_o + eV_o$

  $\frac{hc}{\lambda} = \frac{hc}{\lambda_o} + \frac{1}{2}MeV^2_{max}$

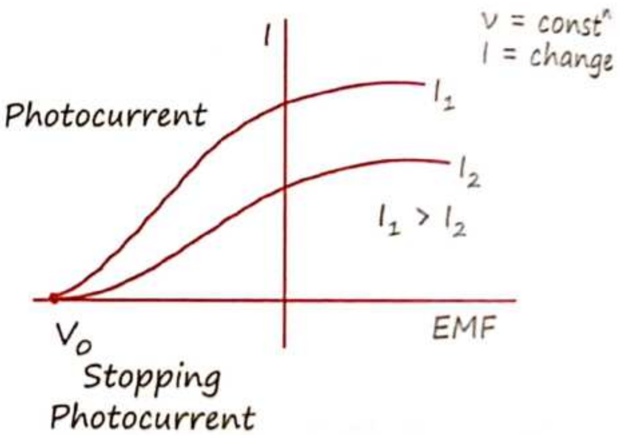
MR*

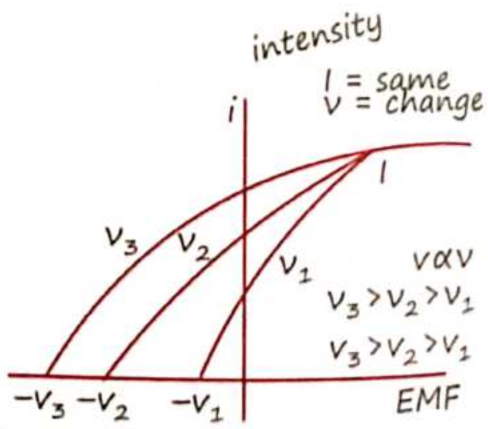
Intensity ∝ no. of Photon ∝ no of electron ∝ Photocurrent

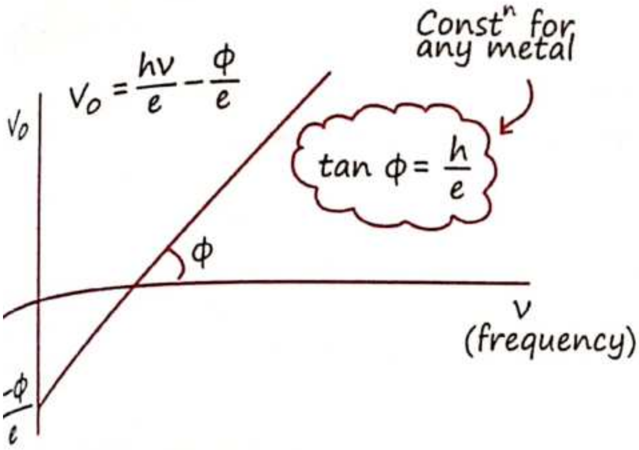
Frequency ∝ energy of Photon ∝ energy of e^- (K.E) ∝ stopping potential

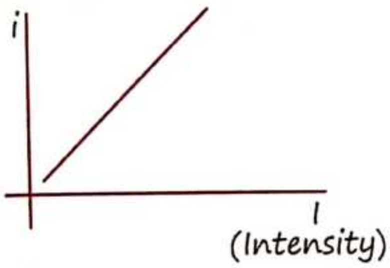
- Photocurrent depends on Intensity not on frequency.
- Stopping Potential depends on frequency not on Intensity

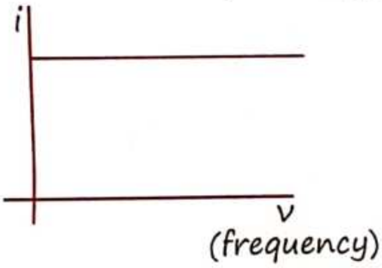
Graphs:

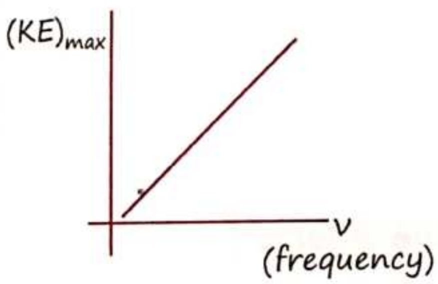


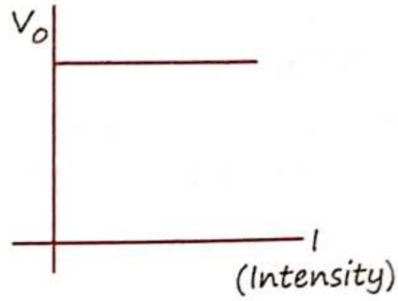


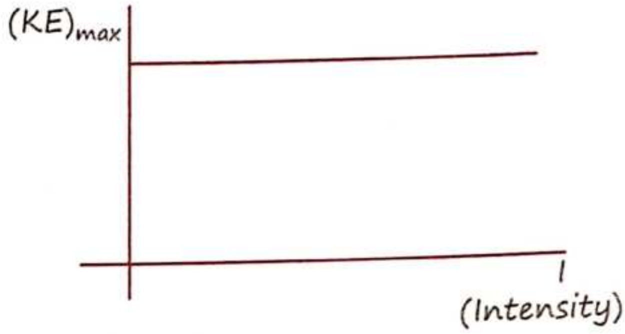












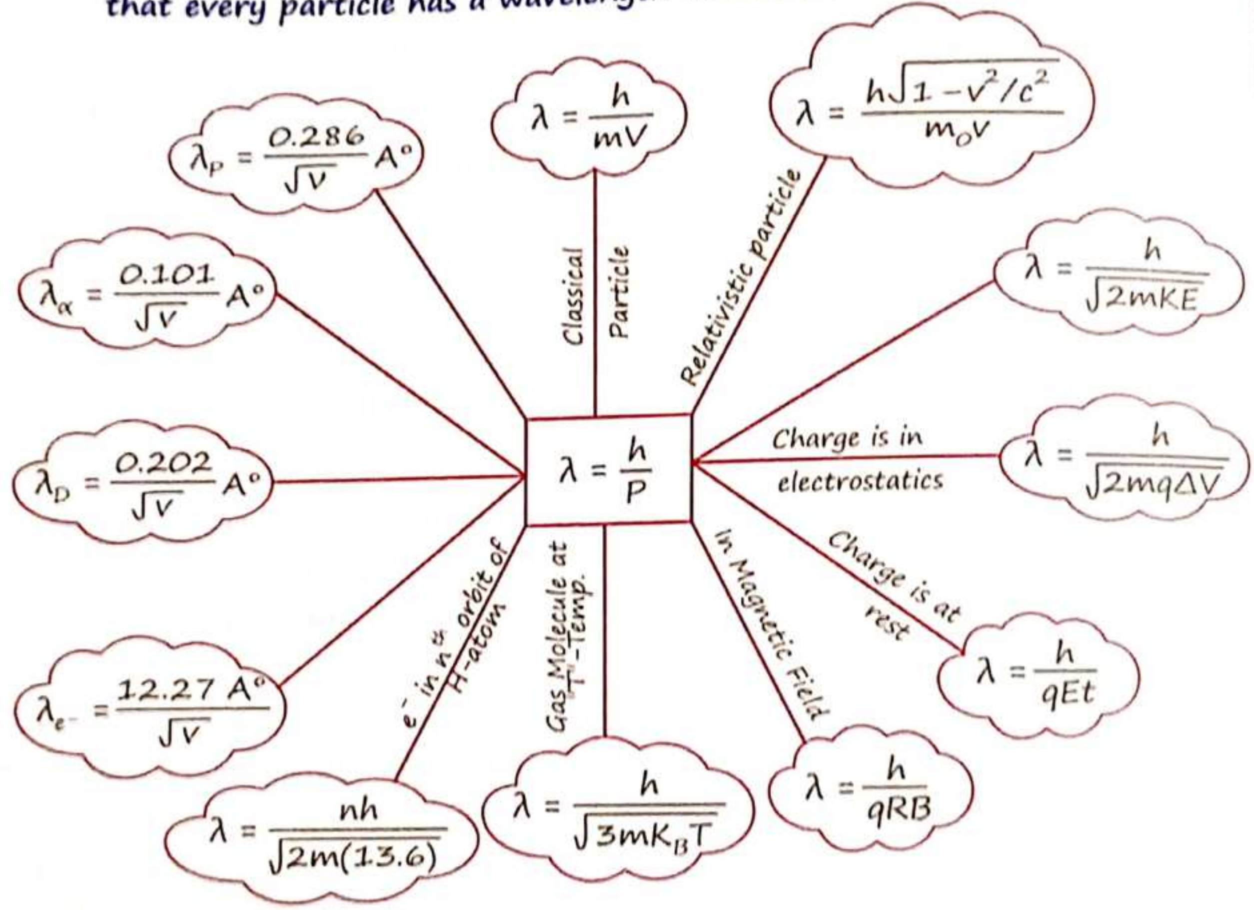
+ If distance b/w Source and plate becomes double then stopping potential remains same but photocurrent becomes one fourth.
+ If frequency becomes double then K.E of electrons becomes more than double.

Q.3 Saturation Current and stopping Potential are I_o and V_o when frequency of light is 1.5 times of threshold frequency. Now if frequency becomes half then saturation current and stopping potential will?

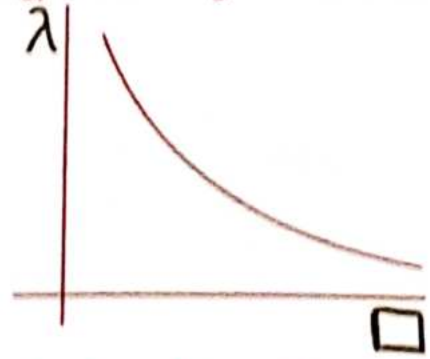
Sol. Both becomes zero, Photoelectric effect hoga hi Nahi

4 Matter waves:

Matter waves, or de Broglie waves, describe the wave-like behavior of particles. Illustrating that every particle has a wavelength associated with its motion.



V. imp. λ v/s $\square$ = Always Reactangular hyperbola : $K_B = 1.38 \times 10^{-28}$



Wavelength of revolving e^- for Bohr's orbit:
$n\lambda = 2\pi r$

5 Compton Effect:

$E = 2ev$ Visible Region

Photoelectric Effect:

$E = 10eV$ X-ray Region

$\Delta\lambda = \lambda_2 - \lambda_1 = \frac{h}{m_e C}(1 - \cos\phi)$, ϕ = scattering angle

$\frac{h}{m_e C} = 2.44$ pm

Pair Production: $E = 1.02$ MeV

6 X-ray:

(Neutral) $E = KeV$

X-RAY

HARD
$\lambda = 1\ A^\circ\ (E\uparrow)$

SOFT
$\lambda = 4\ A^\circ\ (E\downarrow)$

CONTINUOUS
(Retard^n of e^-)

CHARACTERISTICS (Collision b/n accelerated e^- and bonded e^-)

7 Inverse Phenomena of Photoectric effect

X-ray target atom → High atomic number, High melting Point, High Conductivity

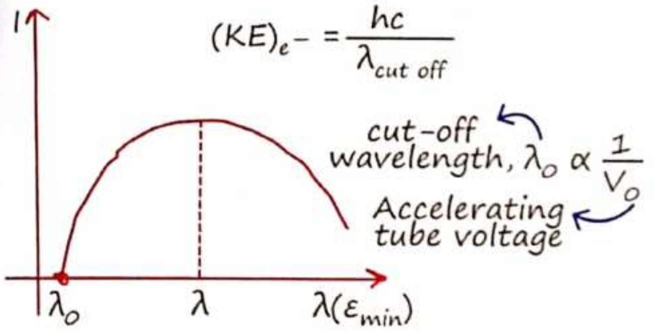
Intensity of X-ray $\propto$ Tube Current	Frequency of X-ray $\propto$ Tube Voltage

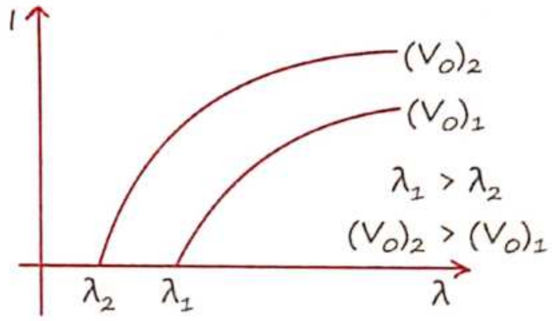
X-RAY

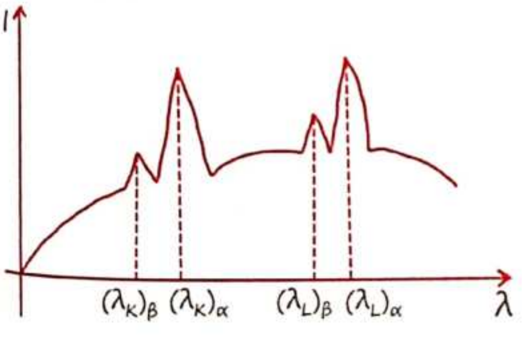
CONTINUOUS
(Retard^n of e^-)

CHARACTERISTICS
(Collision b/n accelerated e^- and bonded e^-)

$(KE)_{e^-} = \frac{hc}{\lambda_{cut\ off}}$







For K-Series:
$f = K^2(z-1)^2$

Q.4 Wavelength is λ for K_α line for atomic no $z = 43$ then find wavelength for K_α line for atomic no $z = 29$.

Sol. $\frac{\lambda}{\lambda^1} = \left(\frac{29-1}{43-1}\right)^2,\ \frac{\lambda}{\lambda^1} = \left(\frac{28}{42}\right)^2 = \left(\frac{2}{3}\right)^2$

$\frac{\lambda}{\lambda^1} = \frac{4}{9}$ $\quad \lambda^1 = \frac{9}{4}\lambda$

8 Heisenberg uncertainty principle:

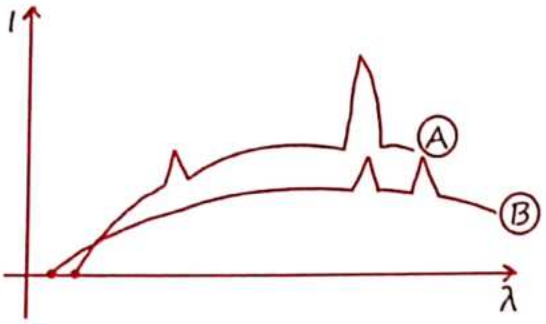
$\Delta x.\Delta p \geq \frac{h}{4\pi}$ $\hbar = \frac{h}{2\pi}$

$\Delta L.\Delta Q \geq \frac{\hbar}{2}$ $\Delta t.\Delta E \geq \frac{\hbar}{2}$

Where,

Δx = uncertainity in position
Δp = uncertainity in momentum
ΔL = uncertainity in angular momentum

+ Graph is given for two different atom A and of atomic no. Z_A and Z_B at cut-off voltage V_A and V_B



$V_A < V_B$ cut-off voltage, $Z_A > Z_B$ atomic no.

Q.5. Given below are two statements:

Statements-I: Two photons having equal linear momenta have equal wavelengths.

Statements-II: If the wavelength of photon is decreased, then the momentum and energy of a photon will also decrease.

In the light of the above statements, choose the correct answer from the options given below.

(a) Both Statement-I and Statement-II are fasle

(b) Statement-I is false but Statement-II is true

(c) Both Statement-I and Statement-II are true

(d) Statement-I is true but Statement-II is false.

Ans. (d)

Q.6. A 2mW laser operates at a wavelength of 1500 nm. The number of photons that will be emitted per second is

[Given Planck's constant $h = 6.6 \times 10^{-34}$ Js, speed of light $c = 3.0 \times 10^8$ m/s]

(a) 2×10^6 (b) 1.5×10^{16}

(c) 5×10^{15} (d) 1×10^{16}

Ans. (d) $P = 2mW = 2 \times 10^{-3}\ w = \frac{nhc}{\lambda t}$

$$\frac{n}{t} = \frac{2\times10^{-3}\times\lambda}{hc}$$

$$= \frac{2\times10^{-3}\times500\times10^{-9}}{2\times10^{-25}}$$

$$= 5 \times 10^{15}$$

Q.7. Given below are two statements:

Statements-I: Two photons having equal linear momenta have equal wavelengths.

Statements-II: If the wavelength of photon is decreased, then the momentum and energy of a photon will also decrease.

In the light of the above statements, choose the correct answer from the options given below.

(a) Both Statement-I and Statement-II are false

(b) Statement-I is false but Statement-II is true

(c) Both statement-I and Statement-II are true

(d) Statement-I is true but statement-II is false.

Ans. (d) $P\uparrow = \frac{h}{\lambda\downarrow}$ & $E\uparrow = \frac{hc}{\lambda\downarrow}$, $\lambda = \frac{h}{P}$

Q.8. A 2 mW laser operates at a wavelength of 500 nm. The number of photons that will be emitted per second is

[Given Planck's constant $h = 6.6 \times 10^{-34}$ Js, speed of light $c = 3.0 \times 10^{8}$ m/s]

(a) 2×10^{6} (b) 1.5×10^{16}

(c) 5×10^{15} (d) 1×10^{16}

Ans. (c) $P = 2mW = 2 \times 10^{-3}$ w $= \frac{nhc}{\lambda t}$

$$\frac{n}{t} = \frac{2 \times 10^{-3}\lambda}{hc}$$

$$= \frac{\cancel{2} \times 10^{-3} \times 500 \times 10^{-9}}{\cancel{2} \times 10^{-25}}$$

$$5 \times 10^{15}$$

Q.9. From the photoelectric effect experiment, following observations are made. Identify which of these are correct

A. The stopping potential depends only on the work function of the metal.

B. The saturation current increases as the intensity of incident light increases.

C. The maximum kinetic energy of a photo electron depends on the intensity of the incident light.

D. Photoelectric effect can be explained using wave theory of light.

Choose the correct answer from the options given below:

(a) B, C only (b) A, C, D only

(c) B only (d) A, B, D only

Ans. (c) $E_p = \phi + eV_0$

Q.10. Given below are two statements:

Statements-I: Stopping potential in photoelectric effect does not depend on the power of the light source.

Statements-II: For a given metal, the maximum kinetic energy of the photoelectron depends on the wavelength of the incident light.

(a) Statement-I is incorrect but Statement-II is correct

(a) Both Statement-I and Statement-II are incorrect

(c) Statement-I is correct but Statement-II is incorrect

(d) Both Statement-I and Statement-II are correct

Sol. (d)

$P = \frac{n\,hf}{t}$

$I = \frac{n\,hf}{At}$

Stopping potential,

$V_s = \dfrac{\frac{hc}{\lambda} - \phi}{e}$

$= \dfrac{(K.E)_{max} - \phi}{e}$

Q.11. In photo electric effect

A. The photocurrent is proportional to the intensity of the incident radiation

B. Maximum Kinetic energy with which photoelectrons are emitted depends on the intensity of incident light.

C. Max. K.E with which photoelectrons are emitted depends on the frequency of incident light.

D. The emission of photoelectrons require a minimum threshold intensity of incident

E. Max. K.E of the photoelectrons is independent of the frequency of the incident light Choose the correct answer from the options given below:

(a) A and C only (b) A and E only

(c) B and C only (d) A and B only

Ans. (a)

Q.12. The threshold frequency of metal is f_o When the light of frequency $2f_o$ is incident on the metal plate, the maximum velocity of photoelectron is v_1 When the frequency of incident radiation is increased to $5f_o$ The maximum velocity of photoelectrons emitted is v_2 The ratio of v_1 to v_2 is:

(a) $\frac{v_1}{v_2} = \frac{1}{2}$ (b) $\frac{v_1}{v_2} = \frac{1}{8}$

(c) $\frac{v_1}{v_2} = \frac{1}{16}$ (d) $\frac{v_1}{v_2} = \frac{1}{4}$

Ans. $2hf_o = hf_o + \frac{1}{2} mV_1^2$

$$5hf_o = hf_o + \frac{1}{2} mV_2^2$$

$$\frac{hf_o}{4hf_o} = \left(\frac{V_1}{V_2}\right)^2$$

$$\sqrt{\frac{1}{4}} = \frac{V_1}{V_2}$$

Q.13. The stopping potential in the context of photoelectric effect depends on the following property of incident electromagnetic radiation

(a) Frequency

(b) Amplitude

(c) Intensity

(d) Phase

Ans. (a)

Q.14. Two identical photocathodes receive the light of frequencies f_1 and f_2 respectively. If the velocities of the photo-electrons coming out are v_1 and v_2 respectively, then

(a) $v_1^2 - v_2^2 = \frac{2h}{m}[f_1 - f_2]$

(b) $v_1^2 + v_2^2 = \frac{2h}{m}[f_1 + f_2]$

(c) $v_1 + v_2 = \left[\frac{2h}{m}(f_1 + f_2)\right]^{1/2}$

(d) $v_1 - v_2 = \left[\frac{2h}{m}(f_1 - f_2)\right]^{1/2}$

Ans. (d)

Q.15. When radiation of wavelength λ is incident on a metallic surface, the stopping potential of ejected photoelectrons is 4.8 V. If the same surface is illuminated by radiation of double the previous wavelength, then the stopping potential becomes 1.6 V. The threshold wavelength of the metal is

(a) 6λ (b) 8λ

(c) 2λ (d) 4λ

Ans. (c) $\dfrac{\dfrac{hc}{\lambda}-\dfrac{hc}{\lambda_o}}{\dfrac{hc}{2\lambda}-\dfrac{hc}{\lambda_o}} = \dfrac{48ev}{16ev} = 3$

$$\cancel{hc}\left(\frac{1}{\lambda}-\frac{1}{\lambda_o}\right) = \cancel{hc}\left(\frac{3}{2\lambda}-\frac{3}{\lambda_o}\right)$$

$$\frac{3}{2\lambda}-\frac{1}{\lambda} = \frac{3}{\lambda_o}-\frac{1}{\lambda_o}$$

$$\frac{1}{\lambda} = \frac{2}{\lambda_o}\ \lambda_o = 2\lambda$$

Q.16. Electron beam used in an electron microscope, accelerated by a voltage of 20 kV has a de-Broglie wavelength of λ_o. If the voltage is increased to 40 kV then the de-Broglie wavelength associated with the electron beam would be:

(a) $3\lambda_o$ (b) $9\lambda_o$

(c) $\lambda_o/2$ (d) $\lambda_o/\sqrt{2}$

Ans. (d) $\lambda \propto \dfrac{1}{\sqrt{2}}$

Q.17. An α-particle, a proton and an electron have the same kinetic energy. Which one of the following is correct in case of their De-Broglie wavelength

(a) $\lambda_\alpha > \lambda_p > \lambda_e$ (b) $\lambda_\alpha < \lambda_p < \lambda_e$

(c) $\lambda_\alpha = \lambda_p = \lambda_e$ (d) $\lambda_\alpha > \lambda_p < \lambda_e$

Ans. (b) $\lambda = \dfrac{h}{\sqrt{2m\,K.E}}$

Q.18. An electron moving with speed v and a photon moving with speed c have same de-Broglie wavelength. The ratio of kinetic energy of electron to that of photon is

(a) $\dfrac{v}{2c}$ (b) $\dfrac{2c}{v}$

(c) $\dfrac{v}{3c}$ (d) $\dfrac{3c}{v}$

Ans. (a) $e^- \rightarrow v$

$$\lambda_e = \frac{h}{m_e v}$$

$$E_p = \frac{hc}{\lambda_p}$$

$$\lambda_p = \frac{\cancel{h}c}{\lambda_p} = \frac{\cancel{h}}{m_e V}$$

$$E_p = meVc$$

$$\frac{K.E_e}{E_p} = \frac{\frac{1}{2}m_e V^2}{m_e V\,C} = \frac{V}{2c}$$

Q.19. A particle is traveling 4 times as fast as an electron. Assuming the ratio of de-Broglie wavelength of a particle to that of electron is 2 : 1, the mass of the particle is:

(a) 1/16 times the mass of e^-

(b) 8 times the mass of e^-

(c) 1/8 times the mass of e^-

(d) 16 times the mass of e

Ans. (c) $V_p = 4V_e$

$$\frac{\lambda_p}{\lambda_e} = \frac{2}{1}$$

$$\lambda_e = \frac{h}{m_e v_e},\ \lambda_p = \frac{h}{m_p v_p}$$

$$\frac{\lambda_p}{\lambda_e} = \frac{m_e v_e}{m_p v_p} \Rightarrow \frac{2}{1} = \frac{m_e}{m_p}\ \frac{1}{4}$$

$$\Rightarrow \frac{m_e}{m_p} = 8$$

Q.20. An electron (of mass m) and a photon have the same energy E in the range of a few eV. The ratio of the de-Broglie wavelength associated with the electron and the wavelength of the photon is (c = speed of light in vacuum)

(a) $\left(\frac{E}{2m}\right)^{1/2}$ (b) $\frac{1}{c}\left(\frac{E}{2m}\right)^{1/2}$

(c) $c(2mE)^{1/2}$ (d) $\frac{1}{c}\left(\frac{2E}{m}\right)^{1/2}$

Ans. (b) $\frac{\lambda_e}{\lambda_p} = \frac{\frac{h}{\sqrt{2mE}}}{\frac{hc}{E}} = \frac{E}{c\sqrt{2Em}} = \frac{1}{c}\sqrt{\frac{E}{2m}}$

Q.21. An electron, a doubly ionized helium ion (He^{++}) and a proton are having the same kinetic energy. The relation between their respective de-Broglie wavelengths λ_e, $\lambda_{He^{++}}$ and λ_p, is

(a) $\lambda_e > \lambda_{He^{++}} > \lambda_p$ (b) $\lambda_e < \lambda_p < \lambda_{He^{++}}$

(c) $\lambda_e > \lambda_p > \lambda_{He^{++}}$ (d) $\lambda_e < \lambda_{He^{++}} = \lambda_p$

Ans. (c) $\lambda \propto \frac{1}{\sqrt{m}}$

Since,

$m_{He^{++}} > m_p > m_e$

Therefore (c) is correct

Q.22. Two particle move at right angle to each other. Their de-Broglie wavelengths are λ_1 and λ_2 respectively. The particles suffer perfectly inelastic collision. The de-Broglie wavelength λ, of the final particle, is given by

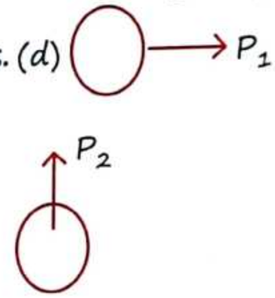
(a) $\lambda = \frac{\lambda_1 + \lambda_2}{2}$

(b) $\frac{2}{\lambda} = \frac{2}{\lambda_1} + \frac{1}{\lambda_2}$

(c) $\lambda = \sqrt{\lambda_1 \lambda_2}$

(d) $\frac{1}{\lambda^2} = \frac{2}{\lambda_1^2} + \frac{1}{\lambda_2^2}$

Ans. (d)



$P_1\hat{i} + P_2\hat{j} = P, \sqrt{P_1^2 + P_1^2} = P$

$\frac{\cancel{h^2}}{\lambda_1^2} + \frac{\cancel{h^2}}{\lambda_2^2} = \frac{\cancel{h^2}}{\lambda_2}$

$\frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$

MR*

"Kismat sath de ya na de lekin kabiliyat jarur sath deti hai."

33 Atoms

1 Postulates of Thomson's atomic model

Postulate 1: An atom consists of a positively charged sphere with electrons embedded in it

Postulate 2: An atom as a whole is electrically neutral because the negative and positive charges are equal in magnitude

Thomson atomic model is compared to watermelon. Where he considered:

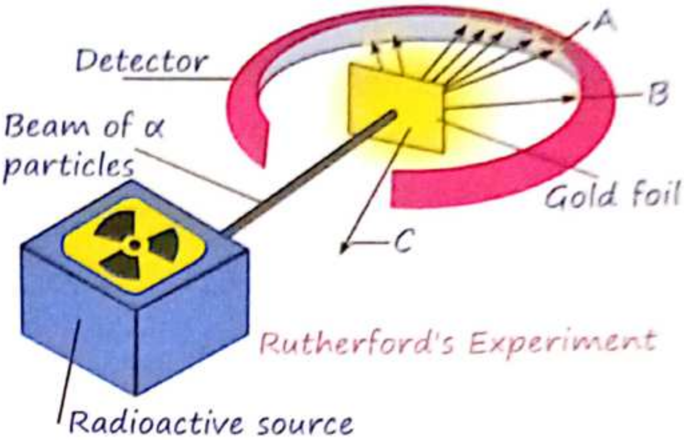
- Watermelon seeds as negatively charged particles
- The red part of the watermelon as positively charged

Limitations of Thomson's atomic model :-

- It failed to explain the stability of an atom because his model of atom failed to explain how a positive charge holds the negatively charged electrons in an atom. Therefore, This theory also failed to account for the position of the nucleus in an atom
- Thomson's model failed to explain the scattering of alpha particles by thin metal foils
- No experimental evidence in its support

2 Rutherford Alpha Particle Scattering Exp

Rutherford and his team took a thin gold foil having a thickness of 2.1×10^{-7} m and placed it in the centre of a rotatable'. detector made of zinc sulfide and a microscope. Then, they directed a beam of 5.5MeV alpha particles emitted from a radioactive source at the foil. Lead bricks collimated these alpha particles as they passed through them.



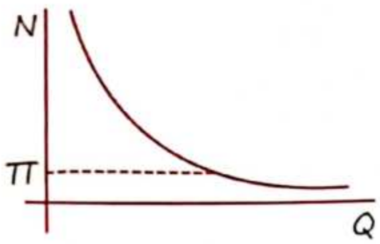
Rutherford's Experiment

After hitting the foil, the scattering of these alpha particles could be studied by the brief flashes on the screen. They expected to learn more about the structure of the atom from the results of this experiment:

- Most of the positively charged alpha particles went undeflected through the foil. This shows that most of the space in an atom is empty.
- Few positively charged alpha particles deflected through small and large angles. This shows that there is presence of positive center in the atom. This positive center is known as nucleus.
- Very few positively charged alpha particles bounced back. This is because the nucleus is very dense and does not allow the alpha particles to pass through it.
- The volume occupied by the nucleus is negligible compared to the total volume of the atom. This shows that radius of atom is much higher than that of the nucleus.

No. of Scattered α-Particle:

$$N \propto \frac{1}{\sin^4(\theta/2)}$$



Distance of closest approach (r_0) :-

$$\underset{\text{bombarding Particle}}{\frac{1}{2}mv^2} = \frac{Kq_1q_2}{r_0} = \frac{2KZe^2}{r_0}$$

α-particle

Q.1 What are the drawbacks of the rutherford atomic model?

Ans. Rutherford's atomic model failed to explain the stability of electrons in a circular path. He stated that electrons revolve around the nucleus in a circular path, but particles in motion would undergo acceleration and cause energy radiation. Eventually, electrons should lose energy and fall into the nucleus. But it never happens.

3 Bohr's Atomic Model

- Bohr proposed that in an atom electron revolves around the nucleus in a definite circular path called orbits or shells.
- These orbits are called 'stationary orbits' and each orbit or shell possesses fixed energy.
- The energy levels are represented by an integer called quantum numbers where these integers indicate principal quantum numbers i.e. orbits that can be assigned as K, L, M, N, corrosponding integer no. n = 1, 2, 3, 4
- The change in energy occur when the electron jumps from one orbit to another orbit.'
- The electron gains energy when it jumps from a lower energy level to a higher energy level and the electron loses its energy as it jumps from the higher energy to a lower energy level.
- The electron's angular momentum is quantized in its orbitals. So, electrons can move only those permissible orbits that should be an integral multiple of $\frac{h}{2\pi}$, where h is planks constant

(i.e., $L=\frac{nh}{2\pi}$)

Postulate 1 : $\frac{mv^2}{r} = \frac{KZe^2}{r^2}$

Postulate 2 : $mvr = \frac{nh}{2\pi}$

Radius of N^{th} Orbit :

$$r = \frac{\varepsilon_0 h^2}{\pi m e^2 z} n^2 = 0.53\frac{n^2}{z}$$

Velocity of N^{th} Orbit :

$$V_n = \frac{nh}{2\pi m r_n} = \frac{e^2}{2\varepsilon_0 h}\frac{z}{n}$$

$$V_n = 2.18\times10^6\frac{z}{n}\ m/s$$

Energies of N^{th} orbit :

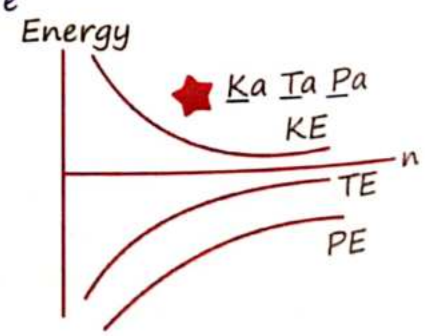
$$KE = \frac{1}{2}mV_n^2 = \frac{me^4z^2}{8\varepsilon_0^2h^2n^2}$$

Potential energy: Kinetic energy : Total energy

$= -1 : \frac{1}{2} : -\frac{1}{2}$

$= -2 : 1 : -1$

- T.E. = $E_n = -13.6\frac{z^2}{n^2}$ eV
- $V_1 = \frac{E_n}{e}$ = Volt



Special Relations :-

- **Time Period :**

$$T = \frac{2\pi r}{V} \qquad T \propto \frac{n^3}{z^2}$$

- **Acceleration :**

$$a_c = \frac{v^2}{r} \qquad a_c \propto \frac{z^3}{n^4}$$

- **Angular Velocity :**

$\omega = 2\pi/T$

$\omega \propto z2/n3$

- **Angular freq^n :**

$$w = 2\pi f \qquad f \propto \frac{z^2}{n^3}$$

- **Current :**

$$i = \frac{q}{t} \qquad i \propto \frac{z^2}{n^3}$$

- **Magnetic dipole Moment :**

$$M = i.A = \frac{eV}{2\pi r} . \pi r2 = \frac{eVr}{2}$$

$M \propto n'$

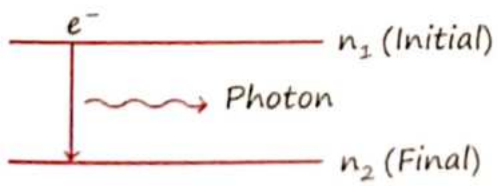
Relation B/N Magnetic Moment & Angular Momentum :-

Magnetic Field $B \propto \frac{1}{n^5}$

$$M = \frac{eVr}{2} = \frac{eL}{2m} \ (L = mvr)$$

$$\vec{M} = \frac{-e}{2m}\vec{L}$$

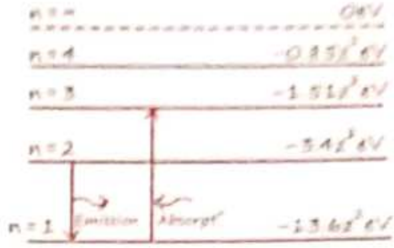
Radiation Energy :-



$$\bar{\nu} = \frac{1}{\lambda} = \frac{13.6\, z^2}{hc}\left[\frac{1}{n_2^2} - \frac{1}{n_1^2}\right] \qquad \overset{13.6}{\curvearrowright} R = 1/hc$$

- $$R = \frac{me^4}{8\varepsilon_0^2 h^3 c} = 1.097 \times 10^7/m$$

Energy Level Diagram :-



* No. of Spectral Line :

$$N = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2} \quad \overset{n_2}{\underset{n_1}{\curvearrowleft}}$$

$$N = \frac{n(n-1)}{2} \quad \overset{n_2}{\underset{G.S.}{\curvearrowleft}}$$

4 Recoiling of an atom

When photon of momentum P emitted from stationary atom then, recoil momentum and energy of atom can be given as follow:

- Recoil Momentum

$$P = \frac{h}{\lambda} = Rz^2h\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \quad \left(\text{Momentum conserves here}\right)$$

- Recoil Energy :

$$E_{atom.} = \frac{P^2}{2m} = \frac{h^2}{2m\lambda^2} \qquad m : \text{mass of Recoil}$$

5 H-Spectrum

For lyman series (K-series)

$\lambda_{min} =$

E_{max} $\overset{n_2 = \infty}{\underset{n_1 = 1}{\curvearrowleft}}$ [Limit Line: last line of K-series]

$\lambda_{max} = E_{min}$ $\overset{n_2 = 2}{\underset{n_1 = 1}{\curvearrowleft}}$ [1st Line of K-series]

Lyman Series : [UV]

$$\lambda_{min} = \boxed{\frac{1}{R} = 912\ A^\circ}$$

$$\lambda_{max} = \frac{4}{3R} = 1216\ A^\circ$$

Balmer Series : [Visible]

$$\lambda_{min} = \frac{4}{R} = 3648\ A^\circ$$

$$\lambda_{max} = \frac{36}{5R} = 6565\ A^\circ$$

Paschen Series : [IR]

$$\lambda_{min} = \frac{9}{R} = 8208\ A^\circ$$

$$\lambda_{max} = \frac{144}{7R} = 18761.1\ A^\circ$$

Brackelt Series : [IR]

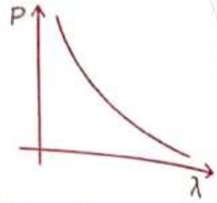
$$\lambda_{min} = \frac{16}{R} = 14592\ A^\circ$$

$$\lambda_{max} = \frac{400}{9R} = 40533\ A^\circ$$

Bohr's Quantum Condition from Debroglie Hypothesis :

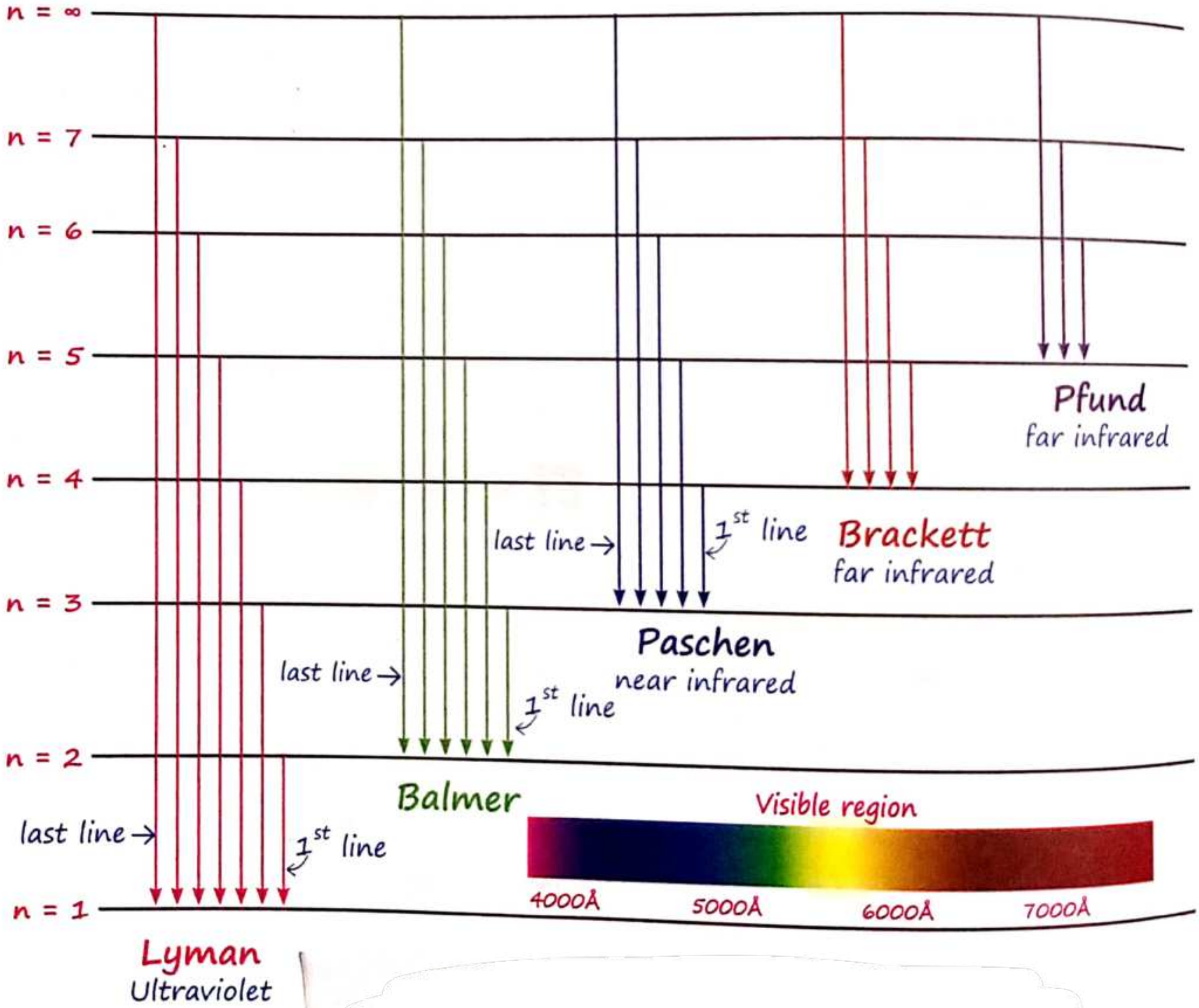
$$2\pi r = n\lambda = \frac{nh}{mv}$$

$$L = \frac{nh}{2\pi}$$



- When H-atom is raised from the ground state to 3rd excited state then → potential energy increases and K.E. decreases.

Spectral Series of Atom



Q.2 The radius of electron's second stationary orbit in Bohr's atom is R. The radius of 3rd orbit will be

(a) R/3 (b) 2.25 R

(c) 3R (d) 9R

Sol. (b) (n = 2, R), (n = 3, R1)

Since, $R \propto n^2$, therefore

$$\frac{R \propto (2)^2}{R^1 \propto (3)^2} \Rightarrow \frac{R}{R^1} = \frac{4}{9}$$

$$R^1 = \frac{9}{4}R = 2.25R$$

Q.3 A small particle of mass m moves in such a way that its potential energy $U = \frac{1}{2}m\omega^2r^2$ where ω is constant and r is the distance of the particle from origin. Assuming Bohr's quantization of momentum and circular orbit the radius of n^{th} orbit will be proportional to:

(a) $\sqrt{n}$ (b) n

(c) n^2 (d) 1/n

Sol. (a) $U = \frac{1}{2}m\omega^2r^2$

$$F = -\frac{dU}{dr} = -m\omega^2r$$

$$\frac{mv^2}{r} = -m\omega^2r$$

$$v^2 \propto r^2$$

$$v \propto r$$

$$mvr = \frac{nh}{2\pi}$$

$$mr \times r = \frac{nh}{2\pi}$$

$$r^2 \propto n$$

$$r \propto \sqrt{n}$$

Q.4 The radius of 2^{nd} orbit of He+ of Bohr's model is r_1 and that of fourth orbit of Be^{3+} is represented as r_2. Now the ratio r_2/r_1 is x : 1. The value of x is ______.

Sol. $r \propto \frac{n^2}{2}$, $\frac{r_2}{r_1} = \frac{4^2}{4} \times \frac{2}{2^2}$

$$\frac{r_2}{r_1} = \frac{2}{1} = \frac{x}{1} \Rightarrow x = 2$$

Q.5 The ratio for the speed of the electron in the 3^{rd} orbit of He^+ to the speed of the electron in the 3^{rd} orbit of hydrogen atom will be:

(a) 1 : 1 (b) 1 : 2

(c) 4 : 1 (d) 2 : 1

Sol. (d) $v \propto \frac{z}{n} \Rightarrow v \propto z$

$$\frac{V_{He+}}{H_{H+}} = \frac{2}{1}$$

Q.6 A hydrogen atom in its ground state absorbs 10.2 eV of energy. The angular momentum of electron of the hydrogen atom will increase by the value of:

(Given, Planck's constant $= 6.6 \times 10^{-34}$ Js).

(a) 2.10×10^{-34} Js

(b) 1.05×10^{-34} Js

(c) 3.15×10^{-34} Js

(d) 4.2×10^{-34} Js

Sol. (b) Change in angular momentum

$$\Delta L = \frac{2h}{2\pi} - \frac{h}{2\pi} = \frac{h}{2\pi}$$

$$= \frac{6.6\times10^{-34}}{2\times3.14} = 1.05 \times 10^{-34} J.s$$

Q.7 The momentum of an electron revolving in nth orbit is given by: (Symbols have their usual meanings)

(a) $\frac{nh}{2\pi r}$ (b) $\frac{nh}{2r}$

(c) $\frac{nh}{2\pi}$ (d) $\frac{2\pi r}{nh}$

Sol. (a) $L = mvr = \frac{nh}{2\pi}$

$$P = \frac{nh}{2\pi r}$$

Q.8 In Bohr's atomic model of hydrogen, let K, P and E are the kinetic energy, potential energy and total energy of the electron respectively. Choose the correct option when the electron undergoes transitions to a higher level:

(a) All K, P and E increase

(b) K decreases, P and E increase.

(c) P decreases, K and E increase.

(d) K increases, P and E decrease.

Sol. (b)

K.E ↓
U ↑
T.E ↑

Q.9 Imagine that the electron in a hydrogen atom is replaced by a muon (μ). The mass of muon particle is 207 times that of an electron and charge is equal to the charge of an electron. The ionization potential of this hydrogen atom will be

(a) 13.6 eV (b) 331.2 eV

(c) 2815.2 eV (d) 27.2 eV

Sol. (c) I.P = T.E = $|K.E|$

$$= \frac{me^4z^2}{8\varepsilon_0^2h^2n^2} = 13.6 \text{ eV} \times 207$$

Q.10 Which level of the single ionized carbon has the same energy as the ground state energy of hydrogen atom?

(a) 6 (b) 8

(c) 4 (d) 1

Sol. (a) For Hydrogen atom,

$$E = -13.6\frac{Z^2}{n^2} = -13.6\frac{(1)^2}{(1)^2} = -13.6$$

For Carbon atom, $z = 6$

$$E = -13.6\frac{z^2}{n^2}$$

$$= -13.6 = -13.6\frac{z^2}{n^2} \Rightarrow n = Z = 6$$

Q.11 The time period of revolution of electron in its ground state orbit in a hydrogen atom is 1.6×10^{-16} s. The frequency of revolution of the electron in its first excited state (in s^{-1}) is:

(a) 6.2×10^{15} (b) 5.6×10^{12}

(c) 7.8×10^{14} (d) 1.6×10^{14}

Sol. (c) $T = T_0 = 1.6 \times 10^{-16} = \frac{n^3}{2^2}$

$$f = f_0\frac{z^2}{n^3} = \frac{2}{3} \times 10^{16} \times \frac{1}{(2)^3}$$

$$= \frac{2 \times 10^{16}}{3 \times 8}$$

$$= \left(\frac{200}{24}\right) \times 10^{14} \approx 7.8 \times 10^{14} \text{ s}^{-1}$$

Q.12 A hydrogen atom, initially in the ground state is excited by absorbing a photon of wavelength 980 Å. The radius of the atom in the excited state, in terms of Bohr radius a_0, will be (h_c = 12500 eV – Å)

(a) $25a_0$ (b) $9a_0$

(c) $16a_0$ (d) $4a_0$

Sol. (c) $E_n - E_1 = \frac{12500}{980}$

$$-13.6\left[\frac{1}{n^2} - \frac{1}{1^2}\right] = 12.75$$

$$n = 4$$

New radius, $r = n^2a_0 = 16a_0$

Q.13 A He+ ion is in its first excited state. Its ionization energy is:

(a) 6.04 eV (b) 13.60 eV

(c) 54.40 eV (d) 48.36 eV

Sol. (d) $|T.E| = \left(\frac{z^2}{n^2}\right) = 13.6$ eV

Q.14 Speed of an electron in Bohr's 7^{th} orbit for Hydrogen atom is 3.6×10^6 m/s. The corresponding speed of the electron in 3^{rd} orbit, in m/s is:]

(a) (1.8×10^6)

(b) (7.5×10^6)

(c) (3.6×10^6)

(d) (8.4×10^6)

Sol. (d) $V \propto \frac{1}{n}$

$$\frac{3.6 \times 10^6}{V_0} = \frac{3}{7} \Rightarrow V_0 = 8.4 \times 10^6$$

Q.15 If λ_1 and λ_2 are the wavelength of the third member of Lyman and first member of the Paschen series respectively, then the value of $\lambda_1 : \lambda_2$ is:

(a) 1 : 3

(b) 7 : 108

(c) 1 : 9

(d) 7 : 135

Sol. (d)

$n_1 = 4$, $n_2 = 1$	$n_1 = 4$, $n_2 = 3$
$\frac{1}{\lambda_2} \; R\left(1 - \frac{1}{16}\right)$	$\frac{1}{\lambda_2} = R\left(\frac{1}{9} - \frac{1}{16}\right)$
$\lambda_1 = \frac{16}{15R}$	$\lambda_2 = \frac{144}{7R}$

$$\lambda_1 : \lambda_2 = 7 : 135$$

Q.16 The electron in a hydrogen atom first jumps from the third excited state to the second excited state and subsequently to the first excited state. The ratio of the respective wavelengths, λ_1/λ_2, of the photons emitted in this process is

(a) 20/7 (b) 7/5

(c) 9/7 (d) 27/5

Sol. (a) Since, $\frac{1}{\lambda} \propto \left(\frac{1}{n_1^2} \;\; \frac{1}{n_2^2}\right)$

Therefore,

$$\frac{\lambda_1}{\lambda_2} = \frac{\left(\frac{1}{2^2} - \frac{1}{3^2}\right)}{\left(\frac{1}{3^2} - \frac{1}{4^2}\right)} = \frac{20}{7}$$

MR*

"Kaam karo aisa ki pehchan ban jaye...chalo to aisa ki nishaan ban jaye..., are zindagi to har koi kaat leta h yaha...., agar dam hai to jiyo aise ki misaal ban jaye."

1 Nucleus

The nucleus is the tiny center of an atom where almost all the positive charge and mass are located.

Composition:

- The nucleus is made up of protons and neutrons. Together, these particles are known as nucleons.
- **Protons:** These carry a positive charge.
- **Neutrons:** These are neutral, meaning they have no charge.

Mass of Proton :-

$$m_p = 1.67 \times 10^{-27}\ Kg = 1.0073\ amu$$

Mass of Neutron :-

$$m_n = 1.67 \times 10^{-27}\ Kg = 1.0087\ amu$$

Mass of an electron :-

$$m_e = 9.1 \times 10^{-31}\ Kg$$

Mass of comparison :-

$$m_\alpha > m_n > m_p > m_e$$

2 Important points related to nucleus

Atomic Number (Z)

- This is the number of protons in the nucleus of an atom. It defines the identity of the element.

Mass Number (A)

- This is the total number of protons and neutrons in the nucleus of an atom. It represents the [illegible]

Rest Mass Energy :

Energy stored in the body due to their Mass

$$E = mc^2$$

Q.1 Find rest mass energy of object having mass 0.5 kg.

Sol. $E = mc^2$

$$= 0.5\,(3 \times 10^8)^2 = 4.5 \times 10^{16}\ J$$

Atomic Mass Unit :

$$1\ amu = \frac{931.5\ MeV}{C^2} = \frac{m_6C^{12}}{12}$$

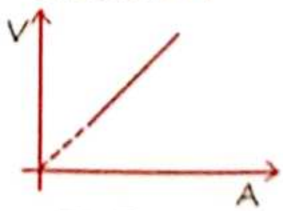
Nuclear Size :

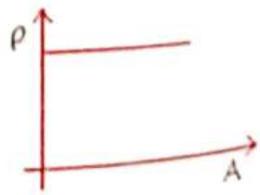
$$R = R_0 A^{1/3} \qquad R_0 = 1.2 f_m\ (1\ f_m = 10^{-15} m)$$

Nuclear Density :

$$\rho = 10^{17}\ Kg/m^3$$ Constant because it does not depend on the mass number.

Volume α A





3 Nuclear Binding Energy (Mass Defect)

$$\text{Mass defect} = \begin{pmatrix}\text{Mass of all the} \\ \text{nucleons of nucleus}\end{pmatrix} - \begin{pmatrix}\text{Mass of} \\ \text{nucleus}\end{pmatrix}$$

$$\Delta m = [ZM_P + (A-Z)M_n - M_{nucleus}]$$

$$BE = \Delta mc^2$$

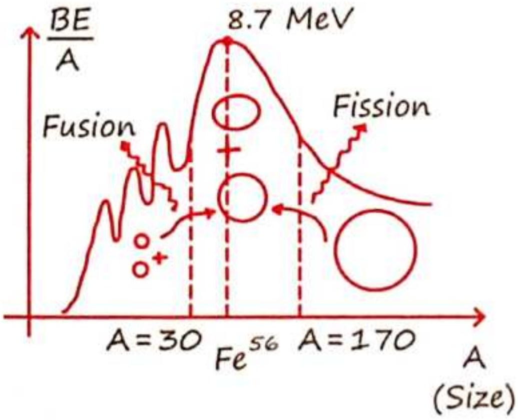
Stability $\propto \frac{B.E.}{A}$ (Binding energy per mass number)

Q.2 Which is more stable ${}_4X^6$ and ${}_6Y^{12}$ having B.E. 24eV and 36eV respectively.

Sol. $\left(\frac{B.E.}{A}\right)_X = \frac{24}{6} = 4ev$

$\left(\frac{B.E.}{A}\right)_Y = \frac{36}{12} = 3ev$ (X is more stable than Y)

Binding Energy Curve:-



- As mass number increases then Ist stability increase then decreases.
- Nucleons of lower mass number fuse for stability and release energy.
- Nucleons of higher mass number break (fission) for stability.

4 Nuclear Force

→ Short range, Non-central, Non conservative

- Does not follow inverse square law
- Charge independent
- Saturation dependent,
- $F_{nuclear} = 10^{39} F_{gravity}$
- Weak nuclear force → shortest range (10^{-16}m), repulsive, Mediated by Boson
- Strong nuclear force → Range 10^{-15}m, attractive, mediated by Meson

Always attractive.

$F_{NN} = F_{PP} = F_{NP}$

Range = 1 Fm

$F_{Nuclear} = 100\ E_{electrostatics}$!

5 Q-Value

$Q = [BE_P - BE_R]$

Q = +ve (Exo)   Q = −ve(Endo)

$BE_P \uparrow$ = Energy Release

$Q = [M_{Reactant} - M_{Product}]\ c^2$

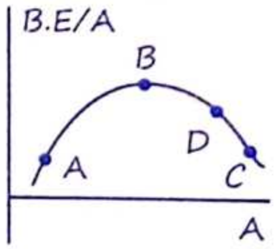
Nuclear reaction coresponding given graph

- $A \longrightarrow B+D$

  Energy released
- $C \longrightarrow D+B$

  Energy released
- $2B \longrightarrow C$

  Energy absorbed



Q-Value of the reaction

$A+B \longrightarrow C+D+Q$

$Q = [(m_A+m_B) - (m_C+m_D)] \times c^2$

$Q = [B.E(C) + B.E(D)] - [(B.E(A) + B.E.(B)]$

- If $[m_A+m_B] > (m_C+m_D)$
- If $[B.E.(C+D)] > B.E.(A+B)$

$\longrightarrow$ energy released

- If $m_A+m_B < m_C+m_D$
- If $B.E.(C+D) < B.E.(A+B)$

$\longrightarrow$ energy absorbed

6 Law of Radioactive Decay

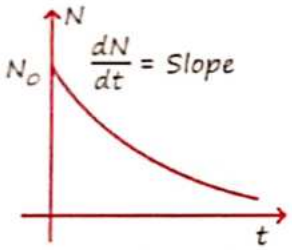
Number of nucleon becomes less and less but always some left.

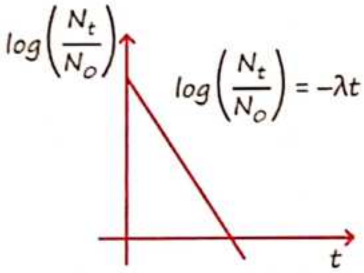
Rate of decay α remaining no. of nucleons

$$\frac{-dN}{dt} = \lambda N$$

λ = decay const^n → Material properties

$$N_t = N_0 e^{-\lambda t}$$

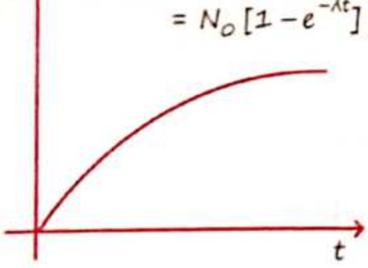




No. of nucleons decayed :

$$N_0 - N_t = N_0 - N_0 e^{-\lambda t} = N_0[1 - e^{-\lambda t}]$$

$$N_t = \frac{N_0}{(2)^n}$$



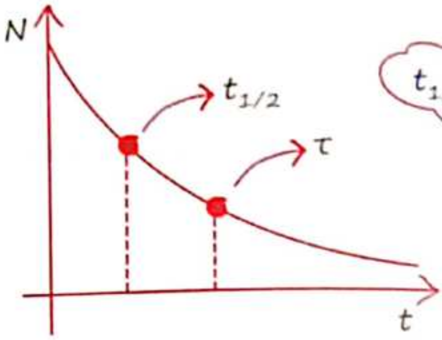
Mean Life :-

$\tau = \frac{1}{\lambda}$ No. of nucleons becomes 37% of initial.

Half-Life :-

No. of Nucleons $\longrightarrow \frac{1}{2} N_0$;

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$



$t_{1/2} = \ln 2\ \tau$

7 Activity

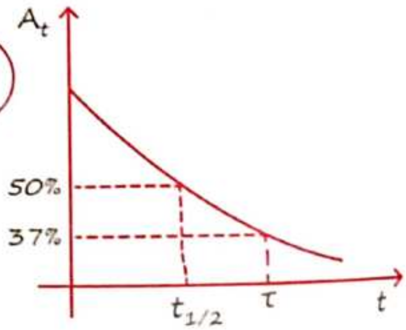
(Rate of disintegrat^n of a sample).

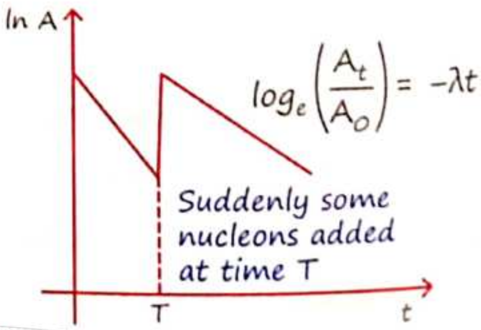
1Bq = 1 decay/s

1 Rutherford = 10^6 dps

1 Curie = 3.7×10^{10} Bq

$$A_t = \lambda N_t$$
$$A_t = A_0 e^{-\lambda t}$$





Time	No. of nuclei at "t" N_t (Remaining)	No. of dis-integrated Nuclei $(N_0 - N_t)$	$\frac{N_t}{N_0 - N_t}$ $\left[\frac{\text{Remain}}{\text{decay}}\right]$	$\frac{N_0}{N_t}$ $\left[\frac{\text{initial}}{\text{remain}}\right]$	$\frac{N_0}{N_0 - N_t}$ $\left[\frac{\text{initial}}{\text{decay}}\right]$
$t = 0$	N_0	0	–	–	–
$t = 1_{t/2}$	$N_0/2$	$N_0/2$	1 : 1	2 : 1	2 : 1
$t = 2_{t/2}$	$N_0/4$	$3N_0/4$	1 : 3	4 : 1	4 : 3
$t = 3_{t/2}$	$N_0/8$	$7N_0/8$	1 : 7	8 : 1	8 : 7
$t = 4_{t/2}$	$N_0/16$	$15N_0/16$	1 : 15	16 : 1	16 : 15

$$= N_0 - \frac{N_0}{2^n} \qquad (1 : 2^n - 1) \quad (2^n : 1)$$

$$nt_{1/2} - \frac{N_0}{2^n} = \frac{N_0}{2^{t/T}}$$

α-decay :

- Recoil velocity of daughter Nuclei :

  Q = Released energy in α-decay

  A = Mass no. of original nuclei

$$\vec{V_D} = \frac{-4\vec{V_\alpha}}{(A-4)}$$

- $KE_\alpha = \left[\frac{A-4}{A}\right]Q$
- $KE_D = \left[\frac{4Q}{A}\right]$

β-decay :

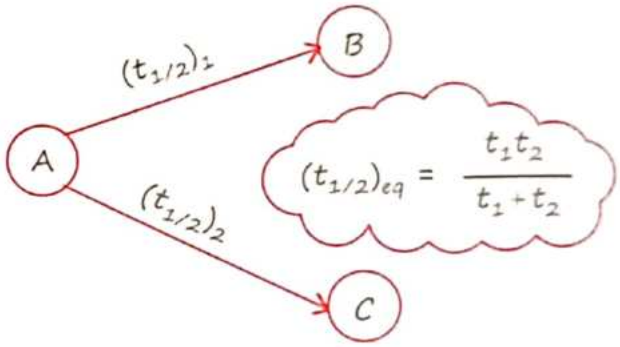
- Q-value in terms of Atomic Mass :
- $\beta^- : Q = [M_R - M_P]c^2$ } same!
- $\beta^+ : Q = [M_R - M_P - 2M_e]\,c^2$

Parallel Disintegration :

In parallel disintegration

$$\lambda_{eq} = \lambda_1 + \lambda_2$$

$$(ty_2) = \frac{t_1 t_2}{t_1 + t_2}$$



$$(t_{1/2})_{eq} = \frac{t_1 t_2}{t_1 + t_2}$$

–: Nuclear Reactor :–

n = 2.5 nucleons/fission

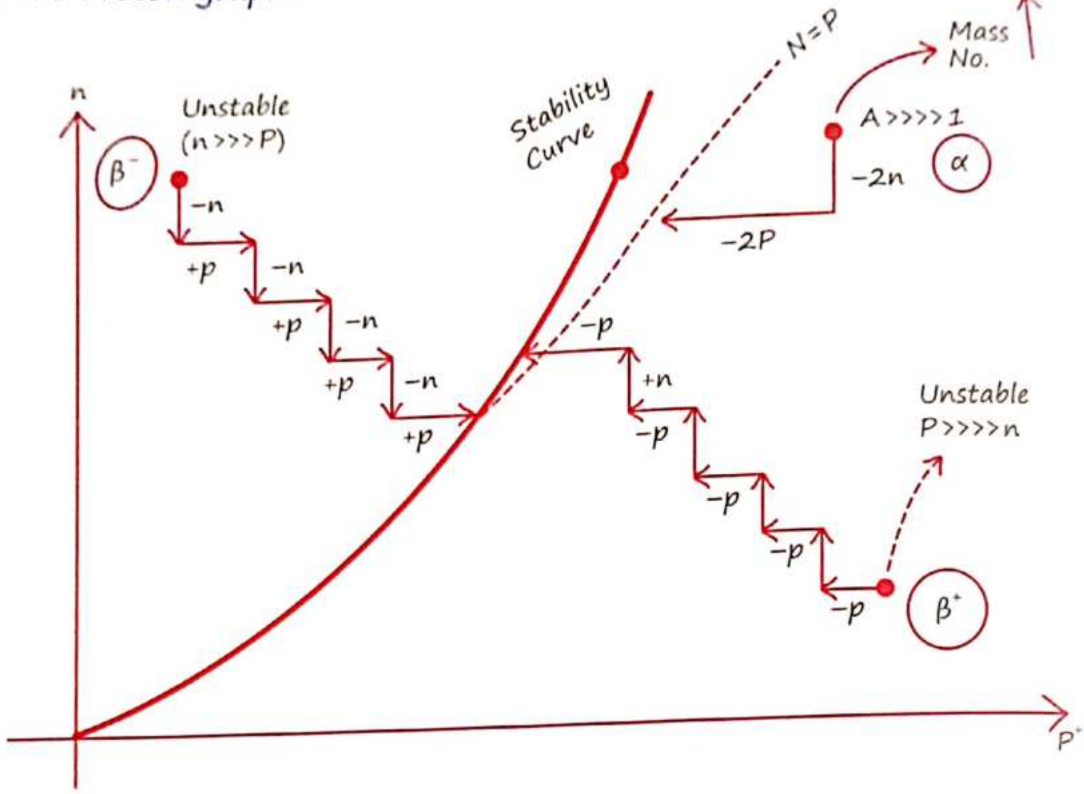
$$K = \frac{\text{Rate of Prod}^n \text{ of nucleon}}{\text{Rate of loss of neutron}}$$

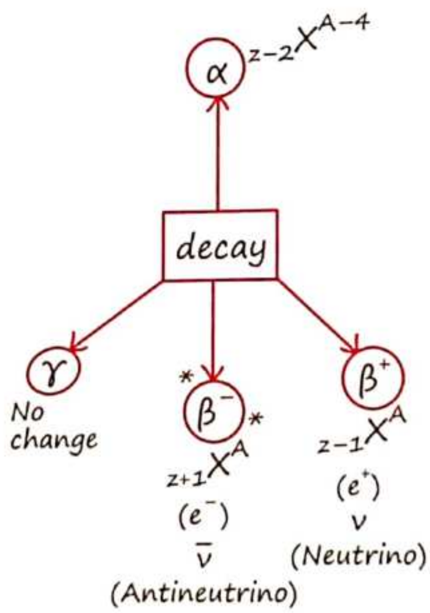
$K = 1$	Critical
$K < 1$	Controlled
$K > 1$	Uncontrolled

–: Power of Reactor :–

$$P = \frac{nE}{t} = \frac{nMc^2}{t}$$

- Neutron v/s Proton graph :

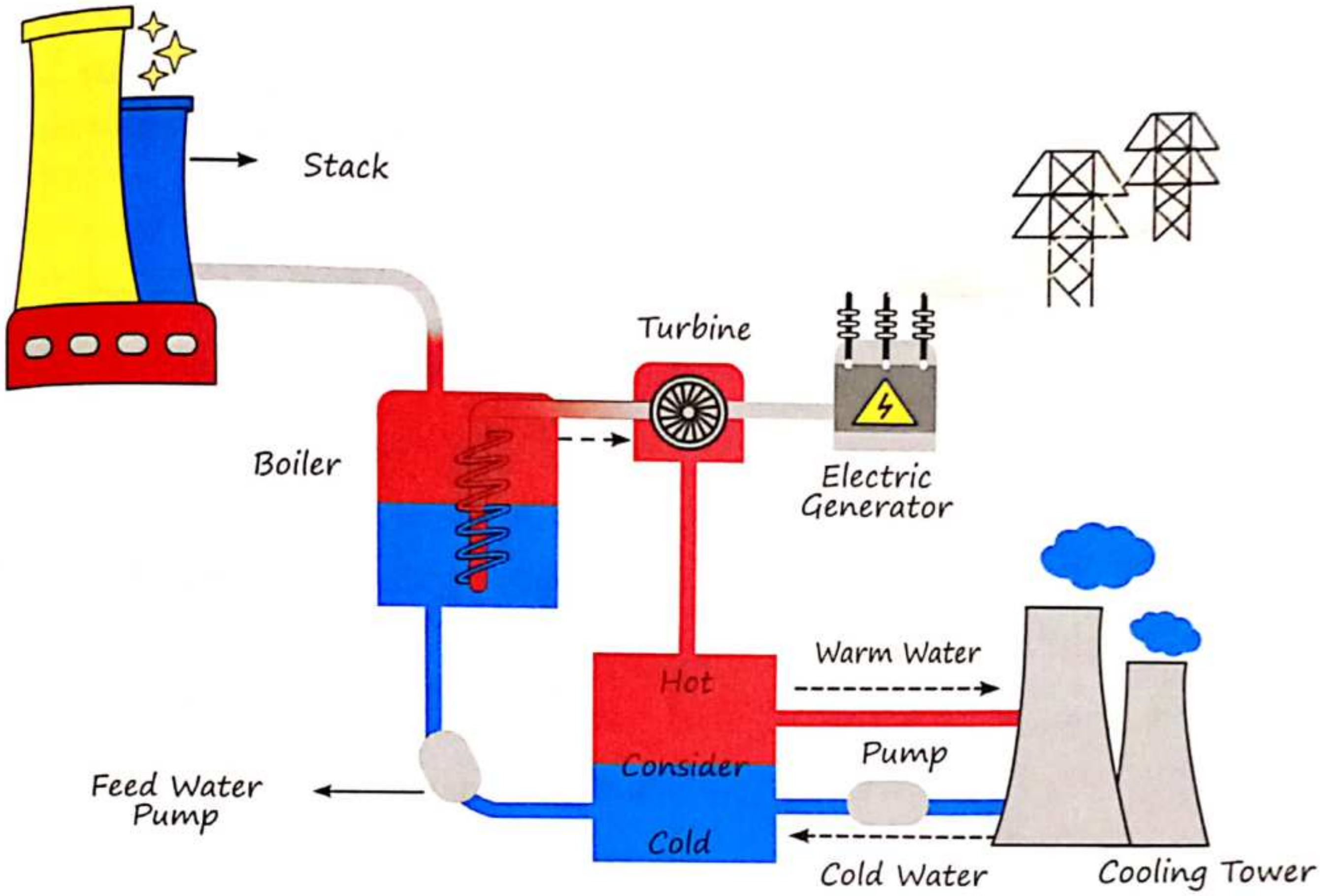




Condition of fusion

1. High temperature = 10^7K
2. High Pressure

Thermal Power Plant



–: Nuclear Bomb :–

H-bomb	Atom Bomb
(Fusion)	(Fission)
→E = 26.7 MeV	E = 200 MeV
Uncontrolled	Controlled!
Chain rx^n	

IMP PYQ

Q.1 Radioactive material A has decay constant 8λ and material B has decay constant λ. Initially, they have same number of nuclei. After what time, the ratio of number of nuclei of material B to that A will be $\frac{1}{e}$?

A $\longrightarrow$ N_o $\quad N' = N_O e^{-8\lambda t}$

B $\longrightarrow$ No $\quad N' = N_O e^{-\lambda t}$

$e = e^{-8\lambda t + \lambda t}$

$e = e^{-7\lambda t}$

$1 = -7\lambda t$

$t = \frac{1}{7\lambda}$

Q.2 A radioactive nucleus of mass M emits a photon of frequency ν and the nucleus recoils. The recoil energy will be

(a) $h^2\nu^2/2\ Mc^2$

(b) zero

(c) $h\nu$

(d) $Mc^2 - h\nu$

Ans. (a)

Momentum of a photon

$p = \frac{h\nu}{c}$

Hence, recoil energy, $E = \frac{p^2}{2M}$

$\therefore \quad E = \frac{\left(\frac{h\nu}{c}\right)^2}{2M}$ or $E = \frac{h^2\nu^2}{2Mc^2}$

Q.3 The activity of a radioactive sample is measured as N_O counts per minute at $t = 0$ and N_O/e counts per minute at $t = 5$ min. The time (in minute) at which the activity reduces to half its value is

(a) $\log_e 2/5$ (b) $\frac{5}{\log_e 2}$

(c) $5 \log_{10} 2$ (d) $5 \log_e 2$

Ans. (d)

Fraction remains after n half lives

$\frac{N}{N_O} = \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^{t/T}$

Given $N = \frac{N_O}{e} \Rightarrow \frac{N_O}{eN_O} = \left(\frac{1}{2}\right)^{5/T}$

or $\frac{1}{e} = \left(\frac{1}{2}\right)^{5/T}$

Taking log on both sides, we get

$\log 1 - \log e = \frac{5}{T}\log\frac{1}{2}$;

$-1 = \frac{5}{T}(-\log 2)$

$\Rightarrow T = 5 \log_e 2$

Now, let t' be the time after which activity reduces to half

$\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^{t'/5\log_e 2} \Rightarrow t' = 5 \log_e 2$

Q.4 Two radioactive materials X_1 and X_2 have decay constants 5λ and λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of X_1 to that of X_2 will be $\frac{1}{e}$ after a time

Save Time

Trick

$t = \frac{\text{Power of}}{\Delta\lambda}$

(a) λ (b) $\frac{1}{2}\lambda$

(c) $\frac{1}{4\lambda}$ (d) $\frac{e}{\lambda}$

Q.5 In compound $X(n, \alpha) \rightarrow {}_3Li^7$, the element X is

(a) ${}_2He^4$ (b) ${}_5B^{10}$

(c) ${}_5B^9$ (d) ${}_4Be^{11}$

Ans. (b)

The given nuclear reaction can be written as

${}_ZX^A + {}_0n^1 \longrightarrow {}_3Li^7 + {}_2He^4$

Conservation of mass number gives,

$A + 1 = 7 + 4 \Rightarrow A = 10$

Conservation of charge number/Atomic No. gives,

$Z + 0 = 2 + 3 \Rightarrow Z = 5$

Hence, $Z = 5$, $A = 10$ corresponds to boron $({}_5B^{10})$.

Q.6 The activity of a radioactive sample is measured as 9750 counts/min at t = 0 and as 975 counts/min at t = 5 min. The decay constant is approximately

(a) 0.922/min (b) 0.691.min

(c) 0.461/min (d) 0.230/min

Ans. (c)

According to law of radioactivity

$$\frac{N}{N_0} = e^{-\lambda t} \quad ...(i)$$

$$\Rightarrow \frac{N_0}{N} = e^{\lambda t}$$

$$\begin{bmatrix} N = \text{final concentration} \\ N_0 = \text{initial concentration} \\ \lambda = \text{decay constant} \end{bmatrix}$$

Taking logarithm on both sides of Eq. (i), we have

$$\log_e\left(\frac{N_0}{N}\right) = \log_e (e^{\lambda t}) = \lambda t \log_e e = \lambda t$$

As we know that,

$\log_e x = 2.3026 \log_{10} x$

Making substitution, we get

$$\lambda = \frac{2.3026 \log_{10}\left(\frac{9750}{975}\right)}{5}$$

$[\because N_0 = 9750$ counts/min and $N = 975$ counts/min$]$

$$= \frac{2.3026}{5} \log_{10} 10 = \frac{2.3026}{5} \text{ min}^{-1}$$

$$= 0.461 \text{ min}^{-1}$$

Q.7 Nuclear fission can be explained by

(a) proton-proton cycle

(b) liquid drop model of nucleus

(c) independent of nuclear particle model

(d) nuclear shell model

Ans. (b)

Q.8 Which of the following is used as a moderator in nuclear reactors ?

(a) Plutonium (b) Cadmium

(c) Heavy water (d) Uranium

Ans. (c)

Q.9 Energy released in the fission of a single ${}_{92}U^{235}$ nucleus is 200 MeV. The fission rate of a ${}_{92}U^{235}$ filled reactor operating at a power level of 5 W is

(a) $1.56 \times 10^{-10}\ s^{-1}$

(b) $1.56 \times 10^{11}\ s^{-1}$

(c) $1.56 \times 10^{-16}\ s^{-1}$

(d) $1.56 \times 10^{-17}\ s^{-1}$

Ans. (b)

$$P = nE/t \left(\frac{n}{t} = P/E\right)$$

$$\text{Fission rate} = \frac{\text{total nuclear power}}{\text{energy produced/fission}}$$

Here, total nuclear power = 5W

Energy released per fission = 200 MeV

$$\therefore \text{ Fission rate} = \frac{5}{200 \text{ MeV}}$$

$$= \frac{5}{200 \times 1.6 \times 10^{-13}} \quad [\because 1 \text{ MeV J}]$$

$$= 1.56 \times 10^{11}\ s^{-1}$$

Q.10 Read the following statements:

A. Volume of the nucleus is directly proportional to the mass number.

B. Volume of the nucleus is independent of mass number.

C. Density of the nucleus is directly proportional to the mass number.

D. Density of nucleus is directly proportional to the cube root of the mass number.

E. Density of the nucleus is independent of the mass number.

Choose the correct option from the following options.

(a) (A) and (D) only.

(b) (A) and (E) only.

(c) (B) and (E) only.

(d) (A) and (C) only

Ans. (b) Volume, $V \propto R^3 \propto A$

Q.11 The ratio of the density of oxygen nucleus ($^{16}_{8}O$) and helium nucleus ($^{4}_{2}He$) is

(a) 4:1 (b) 8:1

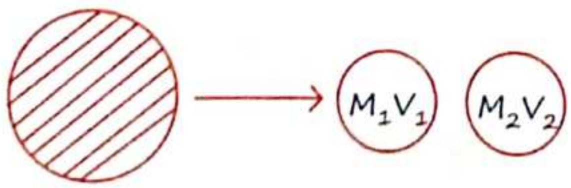
(c) 1:1 (d) 2:1

Ans. (c) Density is independent of mass number (A), Therefore ratio would be 1 : 1

Q.12 A nucleus disintegrates into two smaller parts, which have their velocities in the ratio 3 : 2. The ratio of their nuclear sizes will be $\left(\frac{x}{3}\right)^{1/3}$.

The value of 'x' is ________.

Ans. $\frac{V_1}{V_2} = \frac{3}{2}$



$M_1V_1 = M_2V_2$

$\frac{m_1}{m_2} = \frac{V_1}{V_2} = \frac{2}{3}$

From the formula of Nuclear mass density, $\frac{m}{R^3}$ = constant

Therefore,

$\frac{R_1}{R_2} = \left(\frac{2}{3}\right)^{1/3} = \left(\frac{x}{3}\right)^{1/3}$

$\boxed{x = 2}$

Q.13 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A: The binding energy per nucleon is practically independent of the atomic number for nuclei of mass number in the range 30 to 170.

Reason R: Nuclear force is short ranged.

In the light of the above statements, choose the correct answer from the options given below

(a) Both A and R are true but R is not the correct explanation of A

(b) A is true but R is false

(c) A is false but R is true

(d) Both A and R are true and R is the correct explanation of A

Ans. (d)

Q.14 If the Net force between two protons, two neutrons and between a proton and a neutron is denoted by F_{pp}, F_{nn} and F_{pn}, respectively, then:

(a) $F_{pp} \approx F_{nn} \approx F_{pn}$

(b) $F_{pp} \neq F_{nn}$ and $F_{pn} = F_{nn}$

(c) $F_{pp} = F_{nn} = F_{pn}$

(d) $F_{pp} \neq F_{nn} \neq F_{pn}$

Ans. (b)

Q.15 Energy released in the fission of a single $_{92}U^{235}$ nucleus is 200 MeV. The fission rate of a $_{92}U^{235}$ filled reactor operating at a power level of 5 W is

(a) $1.56 \times 10^{-10}\ s^{-1}$

(b) $1.56 \times 10^{11}\ s^{-1}$

(c) $1.56 \times 10^{-16}\ s^{-1}$

(d) $1.56 \times 10^{-17}\ s^{-1}$

Ans. (b) $P = \left(\frac{n}{t}\right) E$

$$\frac{n}{t} = \frac{P}{E} = \frac{5}{200 \times 10^{6} \times 1.6 \times 10^{-13}}$$

$$= 1.56 \times 10^{11}\ s^{-1}$$

Q.16 The power obtained in a reactor using U^{235} disintegration is 1000 kW. The mass decay of U^{235} per hour is:

(a) 1 microgram (b) 10microgram

(c) 20 microgram (d) 40microgram

Ans. (d) $P = 10^{3} \times 10^{3} = \frac{mc^2}{t}$

$$m = \frac{10^{6} \times t}{c^2}$$

$$= \frac{10^{6} \times 60 \times 60}{(3 \times 10^{8})^2}$$

$$m = 4 \times 10^{-8}\ kg$$

$$m = 4 \times 10^{-2}\ kg \times 10^{-6}$$

$$= 40 \times 10^{-3}\ v_0\ 10^{-6} = 40\ \mu g$$

Q.17 If in nuclear reactor using U^{235} as fuel, the power output is 4.8 MW, the number of fissions per second is:

(Energy released per fission of U^{235} = 200 MeV watts, $1\ eV = 1.6 \times 10^{-19}\ J$)

(a) 1.5×10^{17} (b) 3×10^{19}

(c) 1.5×10^{25} (d) 3×10^{25}

Ans. (a) $4.8 \times 10^{6} = \left(\frac{n}{t}\right) 200 \times 1.6 \times 10^{-13}$.

Q.18 Fusion reaction takes place at a higher temperature because:

(a) atoms get ionized at high temperatures.

(b) Kinetic energy is high enough to overcome the Coulomb repulsion between nuclei.

(c) molecules break up at a high temperature.

(d) nuclei break up at a high temperature.

Ans. (b)

Q.19 For the nuclear reaction:

$$_{92}U^{235} + {}_{56}Ba^{144} + \ldots + 3\,{}_{0}n^{1}$$

The blank space can be filled by:

(a) $_{26}Kr^{89}$ (b) $_{36}Kr^{89}$

(a) $_{26}Sr^{90}$ (a) $_{38}Sr^{89}$

Ans. (a)

236	92
−147	−56
+ 89	26

Q.20 How many alpha and beta particles are emitted when Uranium $_{92}U^{238}$ decays to lead $_{82}Pb^{206}$?

(a) 3 alpha particles and 5 beta particles

(b) 6 alpha particles and 4 beta particles

(c) 4 alpha particles and 5 beta particles

(d) 8 alpha particles and 6 beta particles

Ans. ${}_{92}U^{238} \rightarrow {}_{82}Pb^{206}$

238	No. of alpha-particle
-206	emitted $= \frac{32}{4} = 8$
____	No. of Beta-particle
32	emitted $= 82 - 76 = 6$
No	

Q.21 In the following nuclear reaction

$D \xrightarrow{\alpha} D_1 \xrightarrow{\beta^-} D_2 \xrightarrow{\alpha} D_3 \xrightarrow{\gamma} D_4$

Mass number of D is 182 and atomic number is 74. Mass number and atomic number of D_4 respectively will be

(a) 174 and 71 (b) 174 and 69

(c) 172 and 69 (d) 172 and 71

Ans. (a) ${}^{182}_{74}D \xrightarrow{\alpha} {}^{178}_{72}D_1 \xrightarrow{\beta^-} {}^{178}_{73}D_2 \xrightarrow{\alpha} {}^{174}_{71}D_3 \xrightarrow{\gamma} {}^{174}_{71}D_4$

Q.22 Consider the following radioactive decay process

${}^{218}_{84}A \xrightarrow{\alpha} A_1 \xrightarrow{\beta^-} A_2 \xrightarrow{\gamma} A_3 \xrightarrow{\alpha} A_4 \xrightarrow{\beta^+} A_5 \xrightarrow{\gamma} A_6$

The mass number and the atomic number A_6 are given by:

(a) 210 and 82 (b) 210 and 84

(c) 210 and 80 (d) 211 and 80

Ans. (c)

Q.23 The Q-value of a nuclear reaction and kinetic energy of the projectile particle, K_p are related as:

(a) $Q = K_p$ (b) $(K_p + Q) < 0$

(c) $Q < K_p$ (d) $(K_p + Q) > 0$

Ans. (a) $K + Q = 0$

$\boxed{Q = K.E_1 + K.E_2}$

Q.24 Two lighter nuclei combine to form a comparatively heavier nucleus by the relation given below:

${}^2_1X + {}^2_1X = {}^4_2Y$

The binding energies per nucleon for ${}^2_1X$ and ${}^4_2Y$ are 1.1 MeV and 7.6 MeV respectively. The energy released in this process is _______ MeV.

Ans. $Q = 4(7.6)\text{MeV} - 4(1.1)\text{MeV}$

$= (30.4 - 4.4) = 26$ Mev

Q.25 The mass of proton, neutron and helium nucleus are respectively 1.0073 u, 1.0087 u and 4.0015 u. The binding energy of helium nucleus is:

(a) 14.2 MeV (b) 28.4 MeV

(c) 56.8 MeVX (d) 7.1 MeV

Ans. $BE = [2m_p + 2m_p - m({}^4_2He) \times c^2$

$= (2 \times 1.0073 + 2 \times 1.0087 - 4.0015)$

$= 2.0146$ 4.0320

2.0174 4.0015

____

0.0305

$B.E = 0.0205 \times 931.5$ MeV

0.305×93.15

$= 28.4$ MeV

MR

"Parinde ruk mat, tujhme jaan baki hai,
manjil dur hai abhi, bahut udan baki hai"

35 Semiconductor

1 Types of Material

Conductor :

- Materials that allow electricity to flow easily through them.
- CB & VB Overlap: The energy levels of the electrons (called the Conduction Band and the Valence Band) overlap, so electrons can move freely.
- Conductivity, $\sigma = 10^2$ to 10^8 Sm^{-1} (Siemens per meter)

Semi-Conductor :

Materials that can conduct electricity under certain conditions but not as well as conductors.

Examples: Silicon (Si) and Germanium (Ge).

- Energy gap, $E_g < 3eV$ Si & Ge
- $\sigma = 10^5$ to 10^{-6} Sm^{-1}
- $E_g = (E_{CB})_{min} - (E_{VB})_{max}$
- At 0 K current can flow through conductor but can not flow through semi-conductor.

Insulators :

Materials that do not allow electricity to flow through them easily.

Examples: Rubber, glass.

- Energy gap, $E_g \geq 3eV$
- $\sigma = 10^{-11}$ to 10^{-19} Sm^{-1}

Intrinsic Semi-Conduct :

Pure semiconductors without any added impurities.

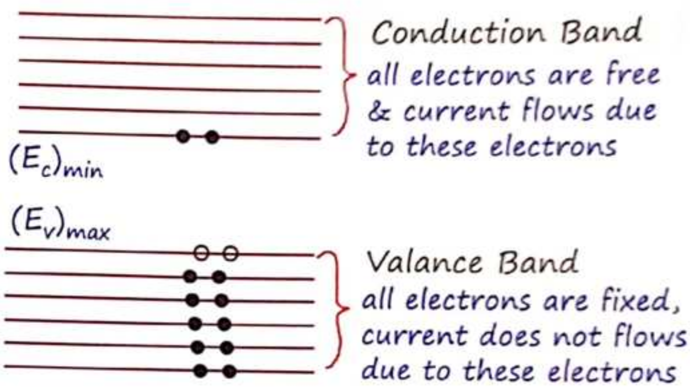
Examples: Pure Silicon (Si) and Germanium (Ge).

[Pure] $\longrightarrow$ Si & Ge

At T = 0 K insulator.

Band diagram

$E_g = (E_c)_{min} - (E_v)_{max}$



Electric Current in intrinsic Semi-conductor

$n_e = n_h = n$ = no. of intrinsic charge density

$\vec{J} = \sigma\vec{E}$

$J = en(\mu_e + \mu_h)E$; $i = i_e + i_h$

$\sigma = en(\mu_e + \mu_h)$

$i_e > i_h$ $\quad \mu_e > \mu_h$

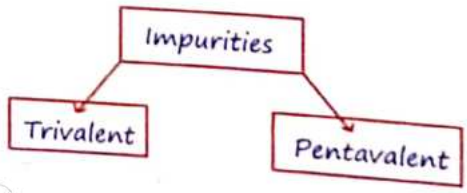
$\mu = \frac{e\tau}{m}$ $\quad (M_{eff})_{hole} > (M_{eff})_{e-}$

* Condition for intrinsic Semi-Conductor :

$n_i = n_e = n_h$; $\sigma = en_i(\mu_e + \mu_h)$

Extrinsic Semi-Conductor :

intrinsic S.C. + impurities = E.S.C



P-type (+)	N-type (−)
$n_h >>> n_e$	$n_e >>> n_h$
"B-Family"	"N-Family"
Neutral	Neutral
$n_h \geq N_A$	$n_{e^-} \geq N_D$
N_A = no. of acceptor ion density	N_D = no. of donor ion density
Fixed −ve ion	Fixed +ve ion
Acceptor $1e^-$ accept udhar lega Ge/Si se phir bond bnayga	Donor $1e^-$ donate kar ke Ge/Si se bond bna lega

2 Law of Mass Action

$$n_i^2 = n_e . n_h$$

N-Type

$n_e \approx N_D$ (Donar)

$$n_h = \frac{n_i^2}{n_e} = \frac{n_i^2}{N_D}$$

P-Type

$n_h \simeq N_A$ (Acceptor)

$$n_e = \frac{n_i^2}{n_h} = \frac{n_i^2}{N_A}$$

Q.1 In a semiconductor, the number density of intrinsic charge carriers at 27°C is $1.5 \times 10^{16}\ m^{-3}$. If the semiconductor is doped with impurity atom, the hole density increases to $4.5 \times 10^{22}\ m^{-3}$. The electron density in the doped semiconductor is

Ans. $n_e n_h = n_i^2$

$$\Rightarrow n_e = \frac{n_i^2}{n_h} = \frac{(1.5 \times 10^{16})^2}{4.5 \times 10^{22}}$$

$$\Rightarrow n_e = \frac{1.5 \times 1.5 \times 10^{32}}{4.5 \times 10^{22}}$$

$$\Rightarrow n_e = 5 \times 10^9\ m^{-3}$$

Q.2 Statement-I: By doping silicon semiconductor with pentavalent material, the electrons density increases.

Statement-II: Then-typesemiconductor has net negative charge. In the light of the above statements, choose the most appropriate answer from the options given below:

(a) Statement-I is false but statement-II is true

(b) Statement-I is true but statement-II is false

(c) Both statement-I and statement-II are true

(d) Both statement-I and statement-II is false

Ans. (b)

(I) Doping Silicon with pentavalent impurity result in an extra electron, therefore electron density increases.

(II) The net charge on either type of semiconductor is zero.

Q.3 The effect of increase in temperature on the number of electrons in conduction band (n_e) and resistance of a semiconductor will be as:

(a) Both n_e and resistance decrease

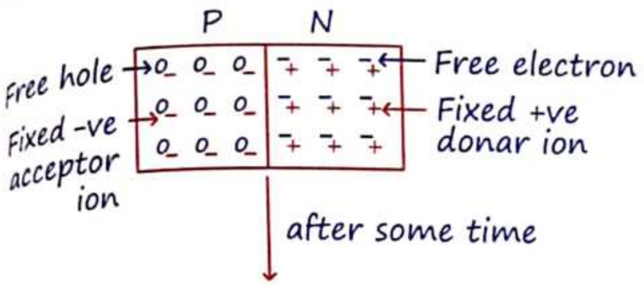
(b) Both n_e and resistance increase

(c) n_e increases, resistance decrease

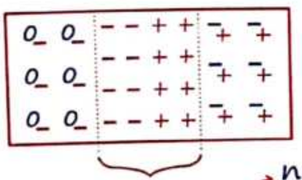
(d) n_e decreases, resistance increases

Ans. (c) When the temperature rises, electrons in the valence band move to the conduction band, increasing the number of electrons there. As a result, the resistivity of a semiconductor decreases, leading to lower resistance as the temperature goes up.

3 P-N Junction Diode

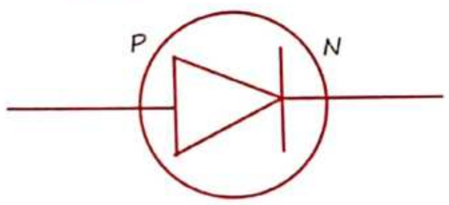


Diffusion current from P to N due to {some electron diffuse from N to P side & some hole diffuse from P to N side}



Depletion region → no free carriers
→ Contains fixed ions creating an electric field and a potential barrier.

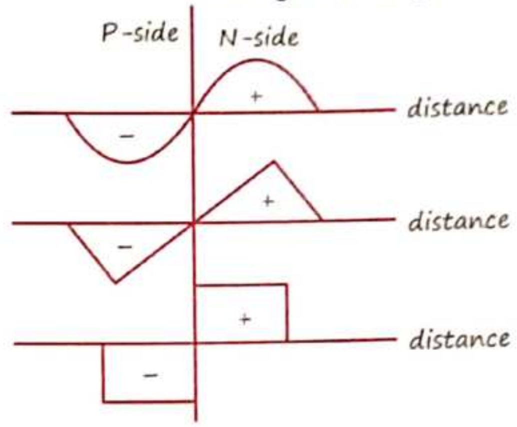
Potential barrier → Formed due to immobile fixed acceptor and donor ion



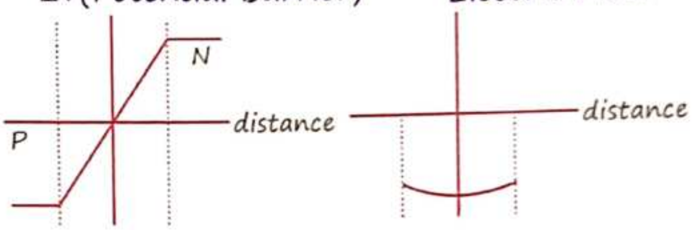
- Electric Field formed from N to P side within Depletion region :
- Width of depletion $\propto \dfrac{1}{\text{doping level}} \propto \text{Temp}^r$
- Drift current now from N to P due to electric field
- At equilibrium $i_{Drift} = i_{Diffusion}$

Graphs :

Charge density



ΔV(Potential Barrier)   Electric Field



Knee Voltage : (V_B)

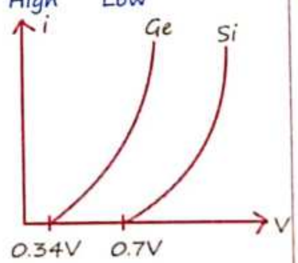
- V_{min} to flow e^- & hole pairs
- "V" above wiz. current rises rapidly.

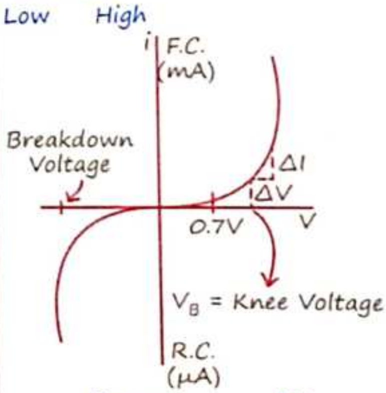
Si = 0.7 V = 0.7 eV

Ge = 0.34 V = 0.34 eV

Biasing :

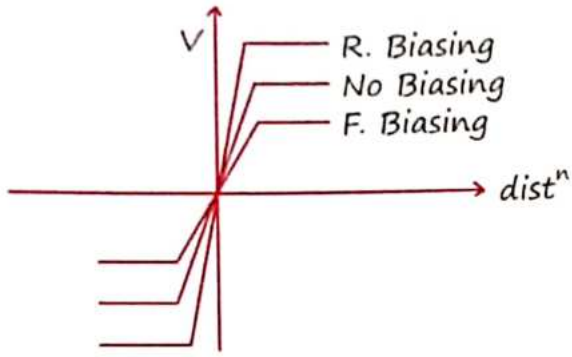
Forward	Reverse
i = ✓	i = 0
P ⟶ N	P ⟶ N
High Low	Low High





Dynamic Resistance $R_B = \dfrac{\Delta V}{\Delta I}$

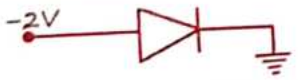
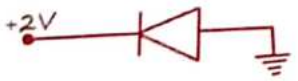
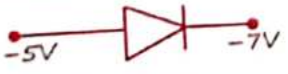
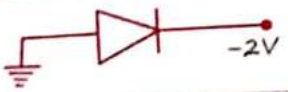
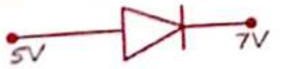
Depl^n layer width "0.5μm to 1μm"



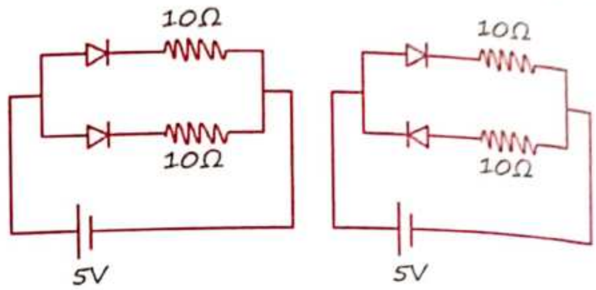
- **MR Feel Table :**

Biasing	Electric Field	Potential diffⁿ	Width of Depletion	$i_{Diffusion}$	i_{Drift}
Forward	↓	↓	↓	↑	↓
Reverse	↑	↑	↑	↓	↑

Zener Diode	**Avalanche Breakdown**
In high doped semi-C	In low doped semi-C
In reverse bias as V↑ the e^-s hole become free due to breaking of co-valent bond	In revese bias at vary high voltage
Reversible	**Not reversible**

Diode	Biasing
	R.B.
	R.B.
	F.B.
	F.B.
	R.B.

Q.4 Find the current through the battery in each of the circuits shown in figure.



(i) (ii)

Ans. In fig. (i) Both diodes are forward biased. Thus the net diode resistance is 0.

$$i = \frac{5}{(10 \times 10)/10 \times 10} = \frac{5}{5} = 1A$$

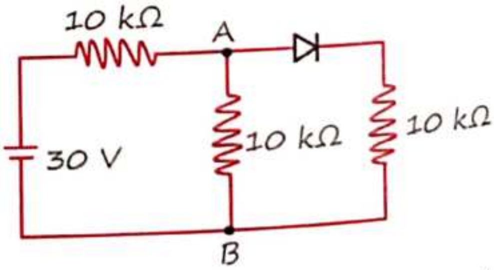
In fig (ii) One diode is forward biased and other is reverse biased. Current passes through the forward biased diode only.

$$i = \frac{V}{R_{net}} = \frac{5}{10+0} = 0.5A$$

Q.5 In the figure, potential difference between A and B is

(a) 5 V (b) 10 V

(c) zero (d) 15 V



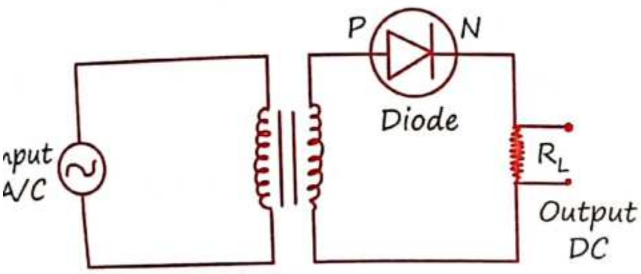
Ans. IHere, diode is in forward biased, therefore, it behaves like a conductive wire, potential difference between A and B:

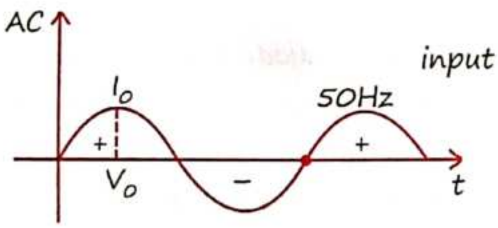
$$V_A - V_B = V \times \frac{R_{AB}}{R_{total}}$$

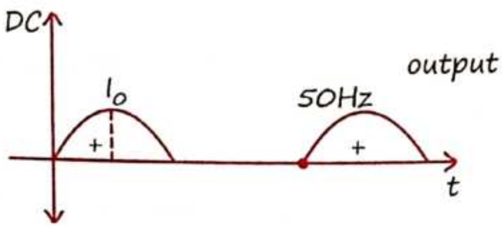
$$= 30 \times \frac{\dfrac{10 \times 10}{10 + 10}}{\left(\dfrac{10 \times 10}{10 + 10} + 10\right)} = 10V$$

Rectifier

Half-wave rectifier : (1 Diode use)







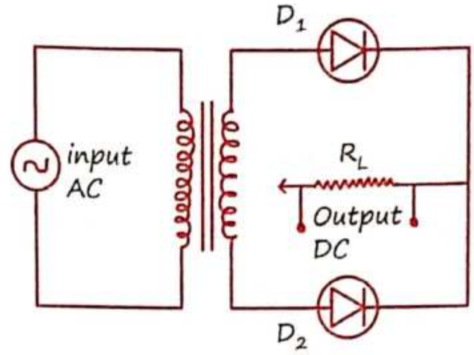
$$i_{DC} = \frac{i_o}{\pi} \qquad V_{DC} = \frac{V_o}{\pi}$$

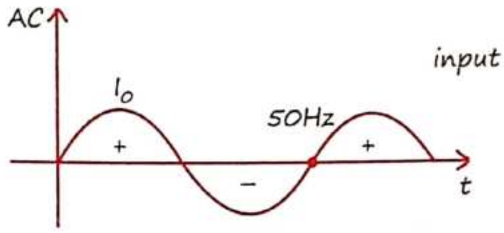
$$i_{rms} = \frac{i_o}{2} \qquad V_{rms} = \frac{V_o}{2}$$

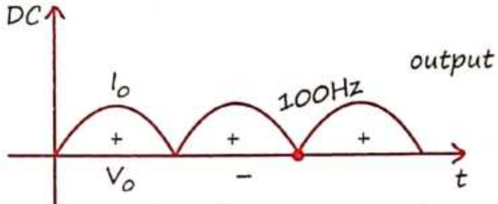
$$\eta = \frac{P_o}{P_i} \times 100 \qquad P = iV = \frac{V^2}{R}$$

Full wave rectifier :

- **Center tapped rectifier : (2-Diode use)**







Ripple frequency = $2 \times f_{input}$

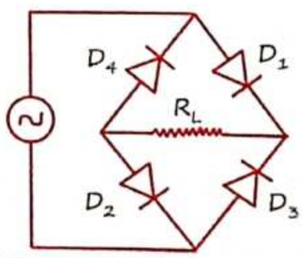
$$\eta = P_o/P_i \times 100$$

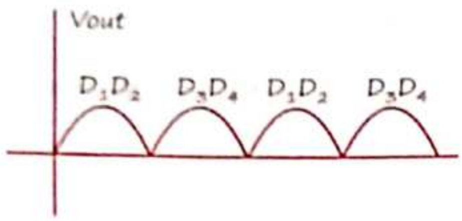
$$i_{avg} = \frac{2i_o}{\pi} \qquad V_{avg} = \frac{2V_o}{\pi}$$

$$i_{rms} = \frac{i_o}{\sqrt{2}} \qquad V_{rms} = \frac{V_o}{\sqrt{2}}$$

- When capacitor is connected parallel with load resistance then output voltage remains constant.

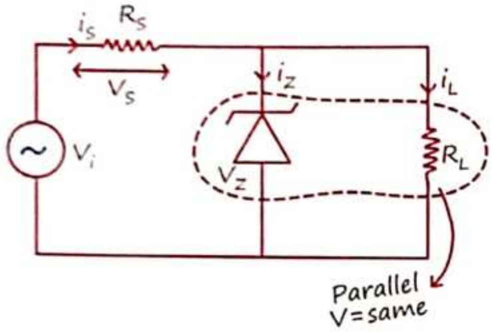
Bridge rectifier : (4-Diode use)





Zener Diode as a Regulator :

- Works in reverse baised condition
- Acts as a voltage requlator.



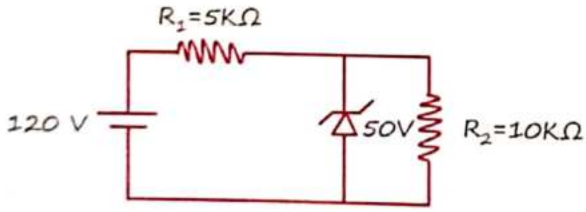
$i_S = i_Z + i_L$ $\qquad V_Z = i_L R_L$

$i_S = \frac{V_S}{R_S}$ $\qquad V_i = V_S + V_Z$

$V_{out} = V_Z = i_L R_L = \text{Constant}$

When diode is working.

Q.6 For the circuit shown below, the current through the zener diode is:



Ans. Assuming Z.D. does not undergo breakdown, current in circuit $= \frac{120}{15000}$ $= 8\text{mA}$

Voltage drop across diode = 80V > 50 V. The diode undergo breakdown.

Current in $R_1 = \frac{70}{5000} = 14\text{ mA}$

Current in $R_2 = \frac{50}{10000} = 5\text{ mA}$

Current through diode = 9 mA

Q.7 Statement-I: To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect a capacitor across the output parallel to the load RL.

Statement-II: To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect an inductor in series with RL. In the light of the above statements, choose the most appropriate answer from the options given below:

(a) Statement-I is false but statement-II is true

(b) Statement-I is true but statement-II is false

(c) Both statement-I and statement-II are true

(d) Both statement-I and statement-II is false

Ans. (b) A photodiode, when exposed to light and under reverse bias, generates charge carriers. The change in minority carriers is noticeable because the starting current is very low.

Q.8 Choose the correct statement about Zener diode:

(a) It work as a voltage regulator in reverse bias and behaves like simple p-n junction diode in forward bias.

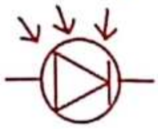
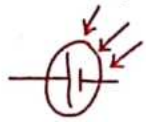
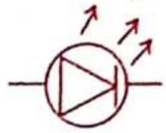
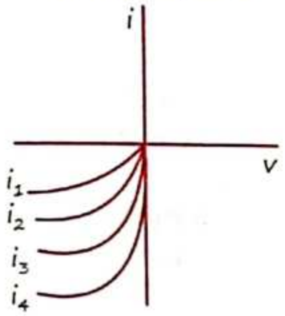
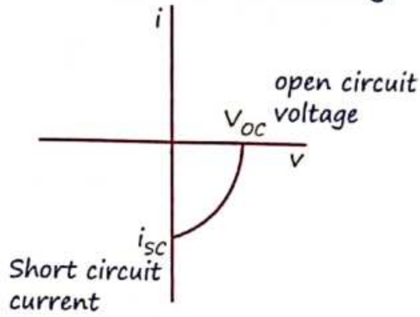
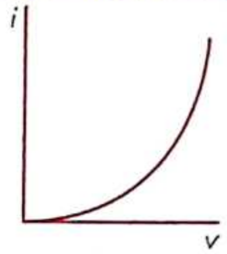
(b) It works as a voltage regulator in both forward and reverse bias.

(c) It works a voltage regulator only in forward bias.

(d) It works as a voltage regulator in forward bias and behaves like simple p-n junction diode in reverse bias.

Ans. (a)

5 Application of Diodes

Photodiode	Solar Cell	LED
$h\nu > E_g$	$h\nu > E_g$	$h\nu \le E_g$
		
Reverse Biased	Conneted in No Biasing	Forward Biasing
		
$i_4 > i_3 > i_2 > i_1$ Intensity ↑ Photo Current ↑ • Act as Light Sensor		Ga – As = Infrared Ga – As – P → Red, Yellow Ga, P → Red, Green

Q.9 Which of the following statement is not correct in the case of light emitting diodes?

A. It is a heavily doped p-n junction.

B. It emits light only when it is forward biased.

C. It emits light only when it is reverse biased.

D. The energy of the light emitted is equal to or slightly less than the energy gap of the semiconductor used.

Choose the correct answer from the options given below:

(a) C and D (b) A only

(c) C only (d) B only

Ans. (c)

Q.10 The photodiode is used to detect the optical signals. These diodes are preferably operated in reverse biased mode because

(a) Fractional change in majority carriers produce higher forward bias current

(b) Fraction change in majority carriers produce higher reverse bias current

(c) Fractional change in majority carriers produce higher forward bias current

(d) Fractional change in minority carriers produce higher reverse bias current

Ans. (d)

Q.11 Assertion(A): Photodiodes are preferably operated in reverse bias condition for light intensity measurement.
Reason (R): The current in the forward bias in more than the current in the reverse bias for a p-n junction diode.
In the light of the above statement, choose the correct answer from the options given below:

(a) A is false bu R is true.

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) Both A and R are true and R is the correct explanation of A

Ans. (b)

6 Logic Gates

Q. 12 MR and Ramlal ne bank me joint account open kiya, dono ko different atm password mila. Atm me dono ka password match hone ke bad paisa milega then atm me Koin sa gate use huaa hai..?

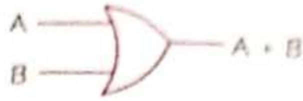
Ans. AND gate

Q. 13 MR audi car se ja raha hai, Ramlal apna truck le ke nikla MR ko takkar marne. Jo aage ya piche khi se takkar mar skta hai, air bag open karne ke liya car me koin sa gate use hoga ....?

Ans. OR gate

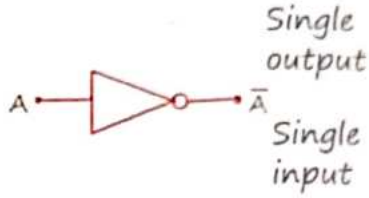
Fundamental Gates :

(1) OR-Gate :



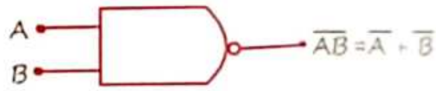
(2) AND-Gate :

(3) NOT-Gate :



Universal Gate :

(1) NAND-Gate :



(2) NOR-Gate :

Exclusive Gates :

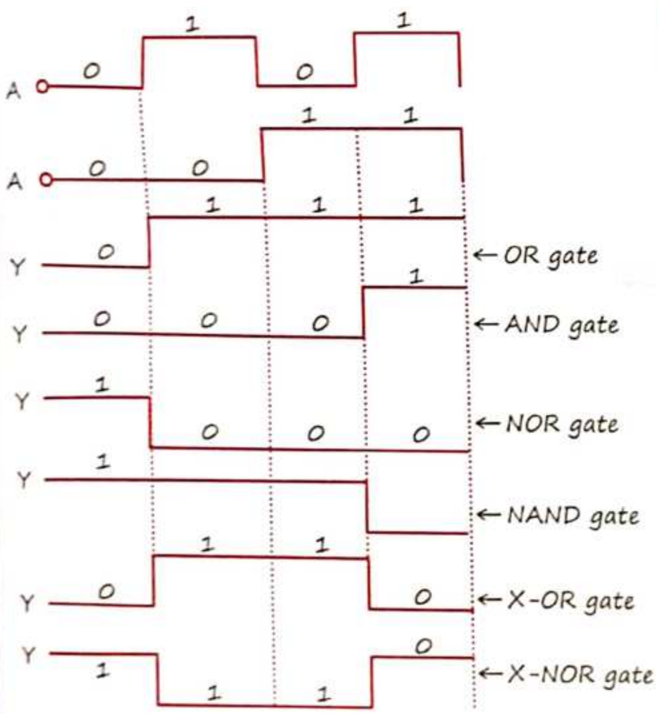
(1) XNOR :

(2) XOR :

Truth table for all gate :-

Input		Output					
A	B	OR Gate	NOR Gate	AND Gate	NAND Gate	X-OR Gate	X-NOR Gate
0	0	0	1	0	1	0	1
1	0	1	0	0	1	1	0
0	1	1	0	0	1	1	0
1	1	1	0	1	0	0	1

Time Scale for different gate :-



7 Basic Boolean Exp

$0 + A = A$	$0.A = 0$	$0.1 = 0$
$1 + A = 1$	$A.A = A$	$1.1 = 1$
$A + A = A$	$1.A = A$	$\overline{1} = 0$
$A + \overline{A} = 1$	$A.\overline{A} = 0$	$\overline{\overline{1}} = 1$
$1 + 0 = 1$		$\overline{0} = 1$
$1 + 1 = 1$	$\overline{\overline{A}} = A$	$\overline{\overline{0}} = 0$

De Morgan Principle :

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

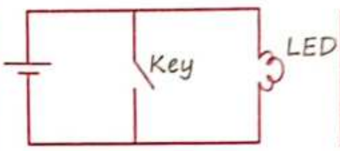
$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

Special Case :

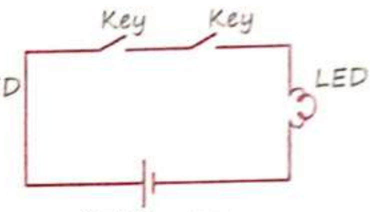
$$\overline{\overline{A} + \overline{B}} = A \cdot B$$

$$\overline{\overline{A}\,\overline{B}} = A + B$$

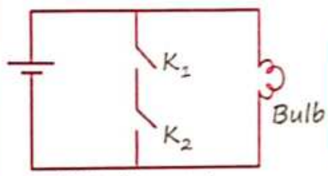
Electrical equivalent circuit :



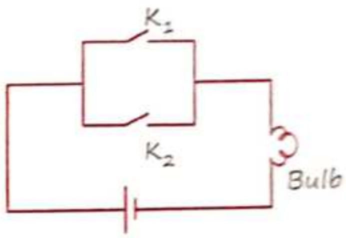
(NOT gate)



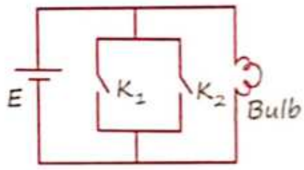
(AND gate)



(NAND gate)



(OR gate)



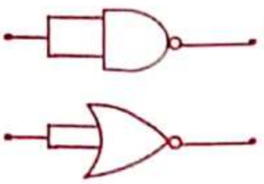
(NOR gate)

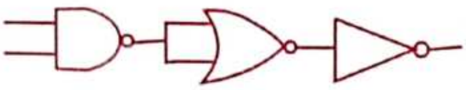
Formation of Different gates using NAND gate:

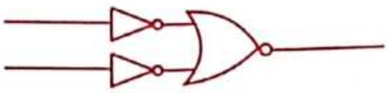
Gate	NOT	AND	OR	NOR
No. of NAND gate required	1	2	3	4

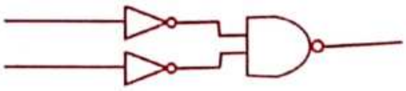
Formation of Different gates using NOR gate:

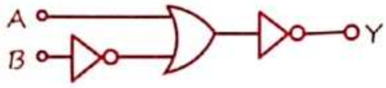
Gate	NOT	AND	OR	NAND
No. of NOR gate required	1	3	2	4

- 
Single input NAND and Single input NOR gate will behave as NOT gate

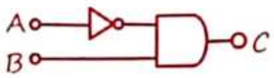
- 
NAND gate

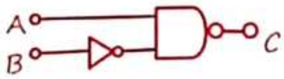
- 
AND gate

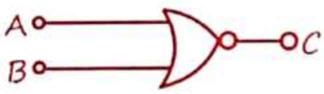
- 
OR gate

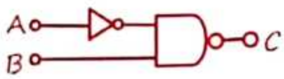
Q.14 

The logic circuit shown above is equivalent to:

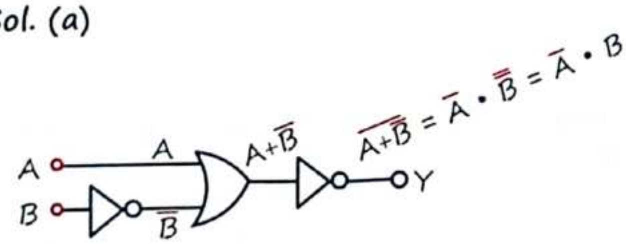
(a) 

(b) 

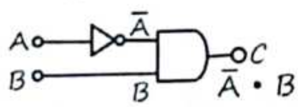
(c) 

(d) 

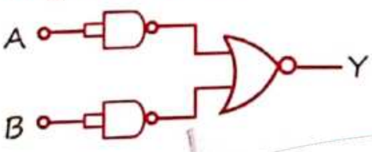
Sol. (a)



Checking with option (a)



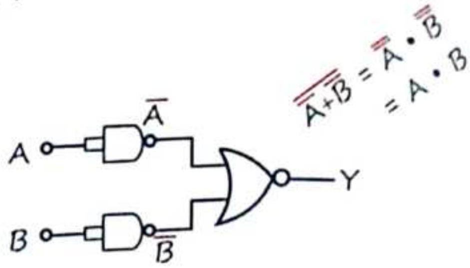
Q.15 Identify the logic operation carried out by the given circuit:



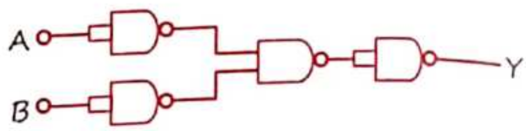
(a) NAND (b) NOR

(c) AND (d) OR

Sol. (c)



Q.16 The following logic gate is equivalent to:



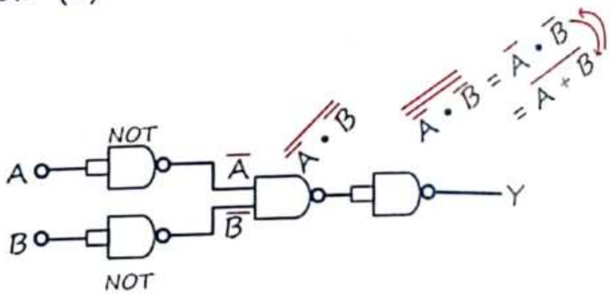
(a) NOR gate

(b) NAND gate

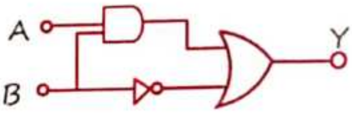
(c) OR gate

(d) AND gate

Sol. (a)



Q.17 Find the truth for the function Y of A and B represented in the following figure:



(a)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	0

(b)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

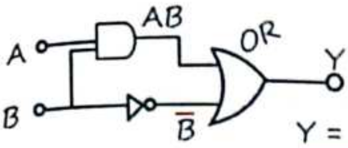
(c)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(d)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

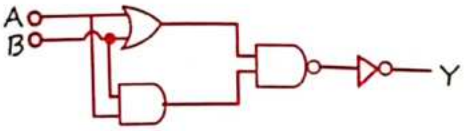
Sol. (b)



$$Y = \overset{0\,1}{AB} + \overline{B}$$

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

Q.18 The logic operations performed by the given digital circuit is equivalent to:



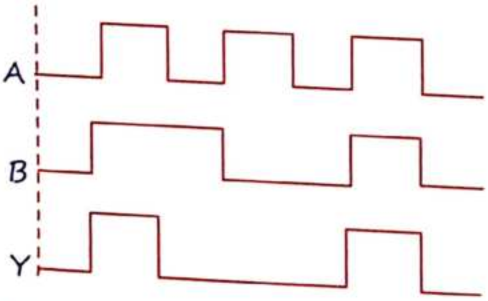
(a) AND (b) NOR

(c) OR (d) NAND

Sol. (a)

$$Y = (A+B)\cdot A\cdot B$$
$$= AAB + ABB = AB + AB$$
$$= AB(1 + 1) = AB$$

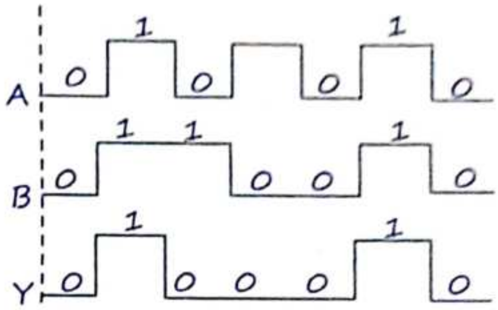
Q.19 A logic gate circuit has two inputs A and B and output Y. The voltage waveforms of A, B and Y are shown below. The logic gate circuit is:



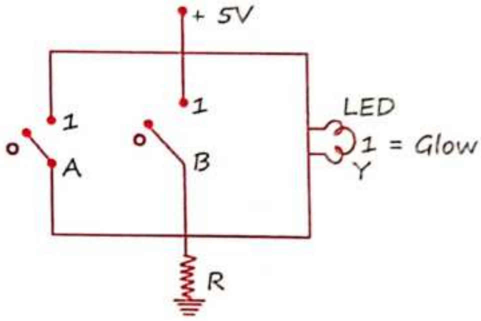
(a) AND gate (b) OR gate

(c) NOR gate (d) NANA gate

Sol. (a)



Q.20 Name the logic gate equivalent to the diagram attached.



(a) OR

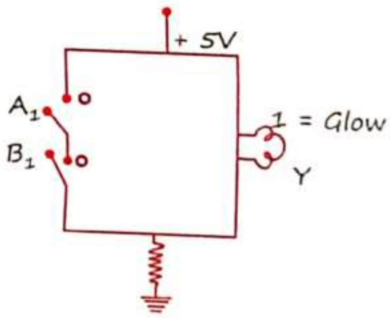
(b) NOR

(c) NAND

(d) AND

Sol. (b)

Q.21 The logic gate equivalent to the given circuit diagram is:



(a) OR

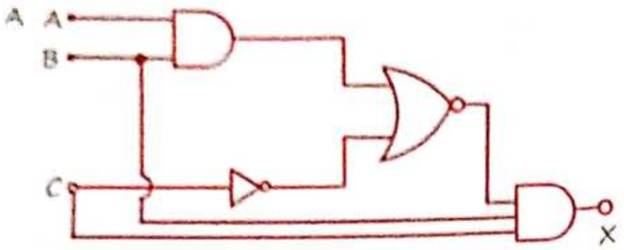
(b) NAND

(c) NOr

(d) AND

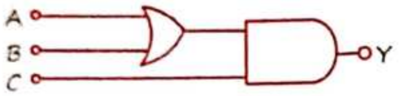
Sol. (b)

Q.22 Find the output boolean function for the logic circuit.



Sol. $X = \bar{A}BC$

Q.23 To get output $Y = 1$ for the following circuit, the correct choice for the input is:



(a) $A = 1, B = 0, C = 0$

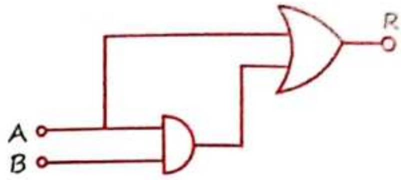
(b) $A = 1, B = 1, C = 0$

(c) $A = 1, B = 0, C = 1$

(d) $A = 0, B = 1, C = 0$

Ans. (c) The boolean expression of the given logic circuit $y = (A + B) \cdot C$.

Q.24 A combination of logic gates is shown in the circuit. If A is at 0 V and B is at 5V, then the potential of R is:



(a) 0 V (b) 5 V

(c) 10 V (d) Any of these

Ans. (a)

MR*

"Manjil Mile na mile yah to kismat ki bat h
hum koshish hi na kare ye to galt bat hai."